4. The magnetic flux through a loop placed in a magnetic field can be changed by changing :

(A) area of the loop only

(B) the value of magnetic field only

(C) orientation of the loop in the magnetic field only

(D) any one or more of the factors given in (A), (B) and (C) CBSE 2026

4. The magnetic flux through a loop placed in a magnetic field can be changed by changing :

(A) area of the loop only

(B) the value of magnetic field only

(C) orientation of the loop in the magnetic field only

(D) any one or more of the factors given in (A), (B) and (C) CBSE 2026

(D) any one or more of the factors given in option (A), (B) and (C)

13. *Assertion (A) :* Induced emf produced in a coil will be more when the magnetic flux linked with the coil is more.

Reason (R) : Induced emf produced is directly proportional to the magnetic flux.

CBSE 2026

13. *Assertion (A) :* Induced emf produced in a coil will be more when the magnetic flux linked with the coil is more.

Reason (R) : Induced emf produced is directly proportional to the magnetic flux.

CBSE 2026

(D) Both Assertion(A) and reason (R) are false.

9. A rectangular loop of size 5 cm × 8 cm is lying in x-y plane in a uniform magnetic field $\vec{B} = (2\cdot0\ \text{T})\ \hat{k}$. The total magnetic flux linked with the loop is :

(A) 80 Wb

(B) 16 Wb

(C) 8×10^{-2} Wb

(D) 8×10^{-3} Wb

CBSE 2026

9. A rectangular loop of size 5 cm × 8 cm is lying in x-y plane in a uniform magnetic field $\vec{B} = (2{\cdot}0\ \mathrm{T})\ \hat{k}$. The total magnetic flux linked with the loop is :

(A) 80 Wb (B) 16 Wb

(C) 8×10^{-2} Wb (D) 8×10^{-3} Wb CBSE 2026

(D) 8×10^{-3} Wb

3. A square loop of side 0·50 m is placed in a uniform magnetic field of 0·4 T perpendicular to the plane of the loop. The loop is rotated through an angle of 60° in 0·2 s. The value of emf induced in the loop will be :

(A) 5 V (B) 3·5 V

(C) 2·5 V (D) Zero V

CBSE 2026

3. A square loop of side 0·50 m is placed in a uniform magnetic field of 0·4 T perpendicular to the plane of the loop. The loop is rotated through an angle of 60° in 0·2 s. The value of emf induced in the loop will be :

(A) 5 V (B) 3·5 V

(C) 2·5 V (D) Zero V

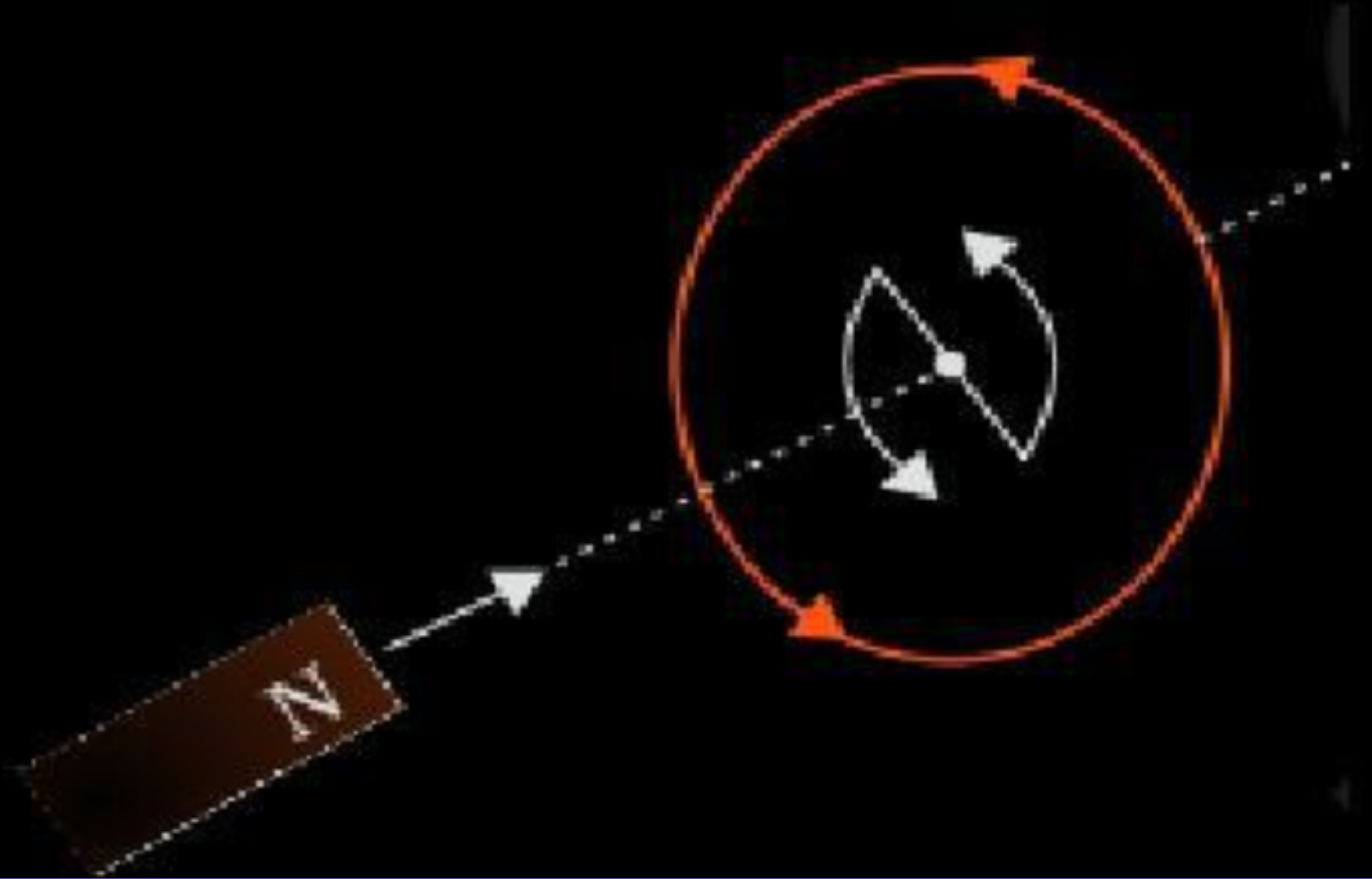
CBSE 2026

Award full mark for attempt

17. A light copper ring is freely suspended by a light string. A bar magnet is held horizontally with its length along the axis of the ring. The magnet is moved towards the ring with its N pole facing the loop. What will happen to the ring and its position ? Explain. CBSE 2026 *2*

17. A light copper ring is freely suspended by a light string. A bar magnet is held horizontally with its length along the axis of the ring. The magnet is moved towards the ring with its N pole facing the loop. What will happen to the ring and its position ? Explain. CBSE 2026 *2*

Current is induced in the ring	½
Reason- Due to electromagnetic induction.	½
The ring will be repelled.	½
Reason- As per Lenz's law, Current is induced in the anticlockwise direction and magnetic moment associated with this current has north polarity.	½
Note: Award 1 mark if the student draws diagram showing current in anticlockwise direction.	



4. The dimensions of the rate of change of magnetic flux are :

(A) $[M\,L\,T^{-3}\,A]$

(B) $[M\,L^2\,T^{-3}\,A^{-1}]$

(C) $[M\,L^2\,T^{-2}\,A^{-1}]$

(D) $[M\,L^2\,T^{-3}\,A^{-2}]$

CBSE 2026

4. The dimensions of the rate of change of magnetic flux are :

(A) $[M\,L\,T^{-3}\,A]$

(B) $[M\,L^{2}\,T^{-3}\,A^{-1}]$

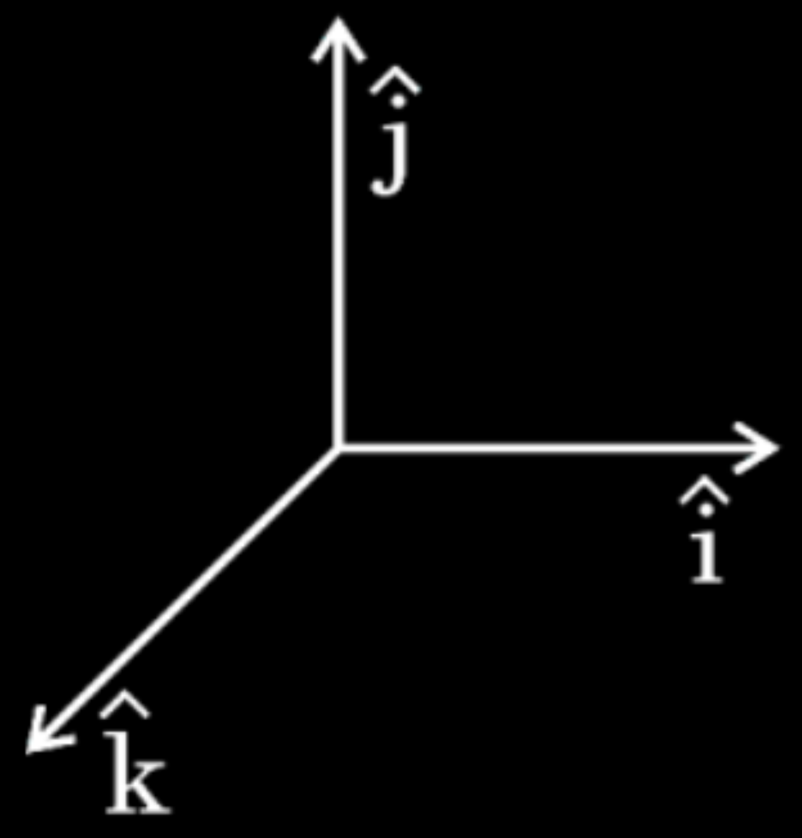
(C) $[M\,L^{2}\,T^{-2}\,A^{-1}]$

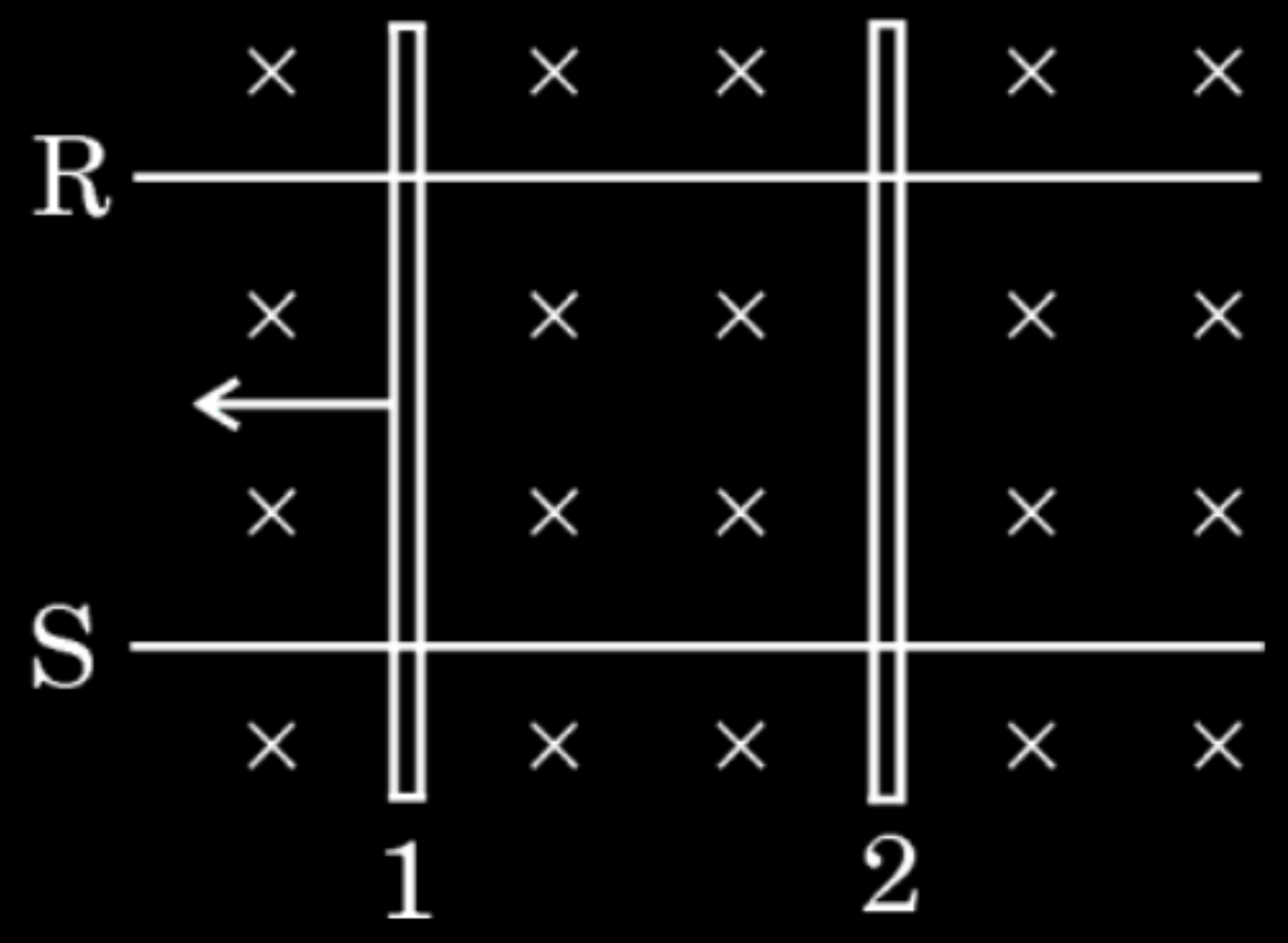
(D) $[M\,L^{2}\,T^{-3}\,A^{-2}]$

CBSE 2026

(B) $\left[ML^{2}T^{-3}A^{-1}\right]$

5. Two identical conductors 1 and 2 are placed on two frictionless conducting rails R and S in a uniform magnetic field directed vertically downward into the plane of the page. If conductor 1 is moved with a constant velocity in the direction as shown in figure, the force on conductor 2 will be along :





(A) $-\hat{i}$

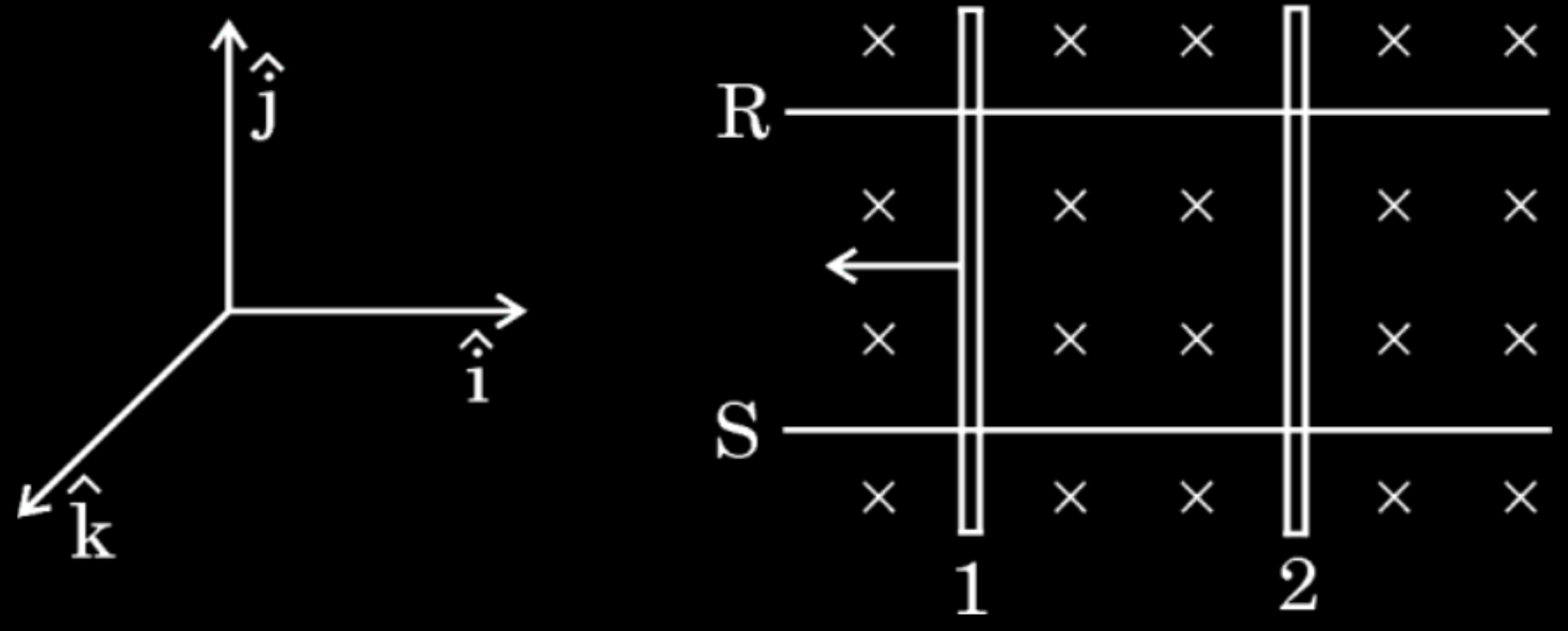
(B) $-\hat{j}$

(C) $\hat{k}$

(D) $\hat{j}$

CBSE 2026

5. Two identical conductors 1 and 2 are placed on two frictionless conducting rails R and S in a uniform magnetic field directed vertically downward into the plane of the page. If conductor 1 is moved with a constant velocity in the direction as shown in figure, the force on conductor 2 will be along :



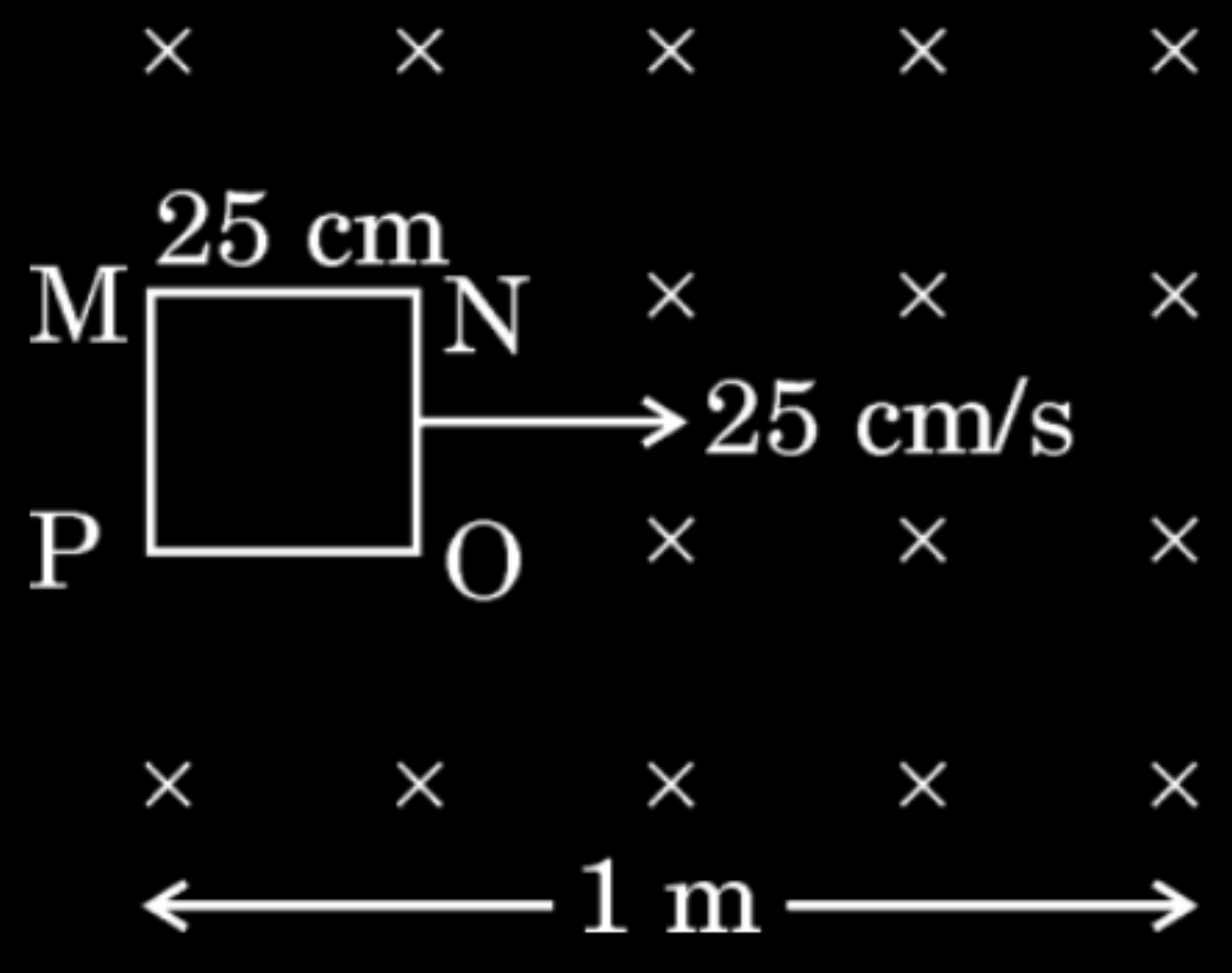
(A) $-\hat{i}$ (B) $-\hat{j}$

(C) $\hat{k}$ (D) $\hat{j}$

CBSE 2026

(A) $-\hat{i}$

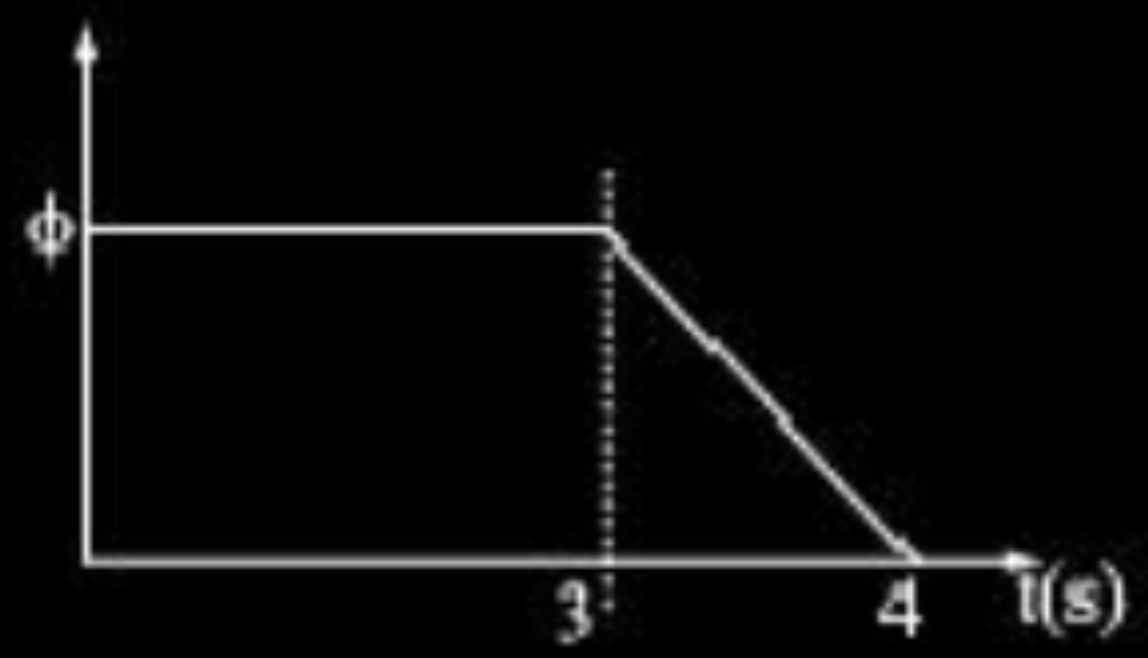
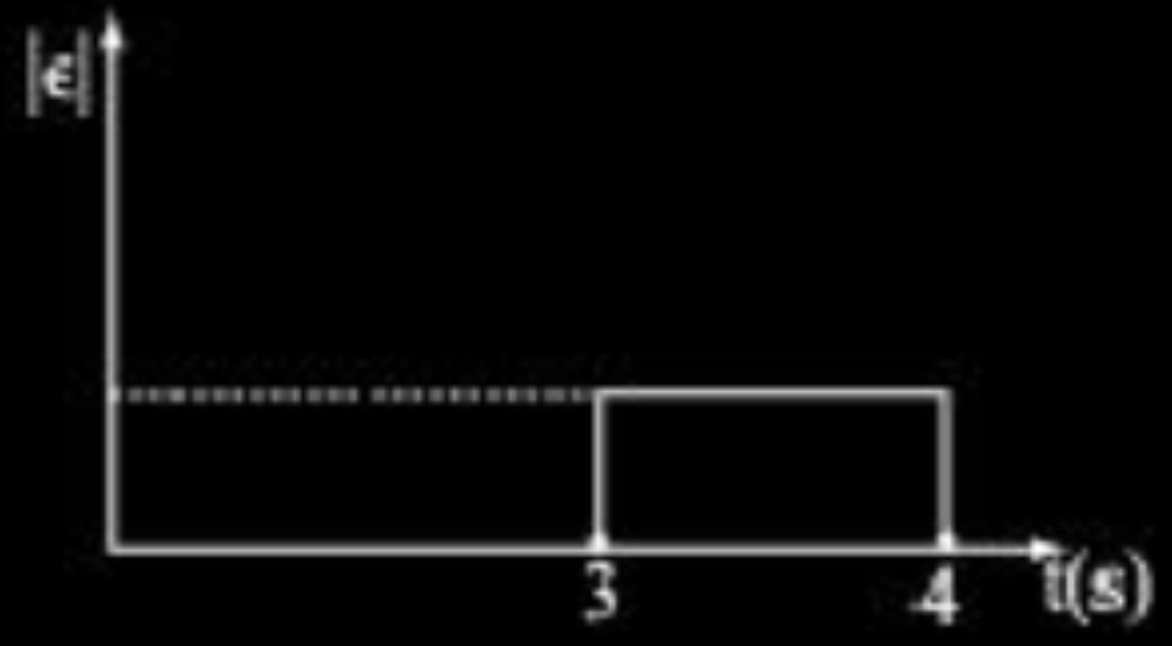
24. The figure given below shows a square-shaped loop MNOP of side 25 cm placed horizontally in a uniform magnetic field $\vec{B}$ directed vertically downward. The position of the loop at t = 0 s is as shown in figure. The loop is pulled with a constant velocity of 25 cm/s till it goes out of the field.



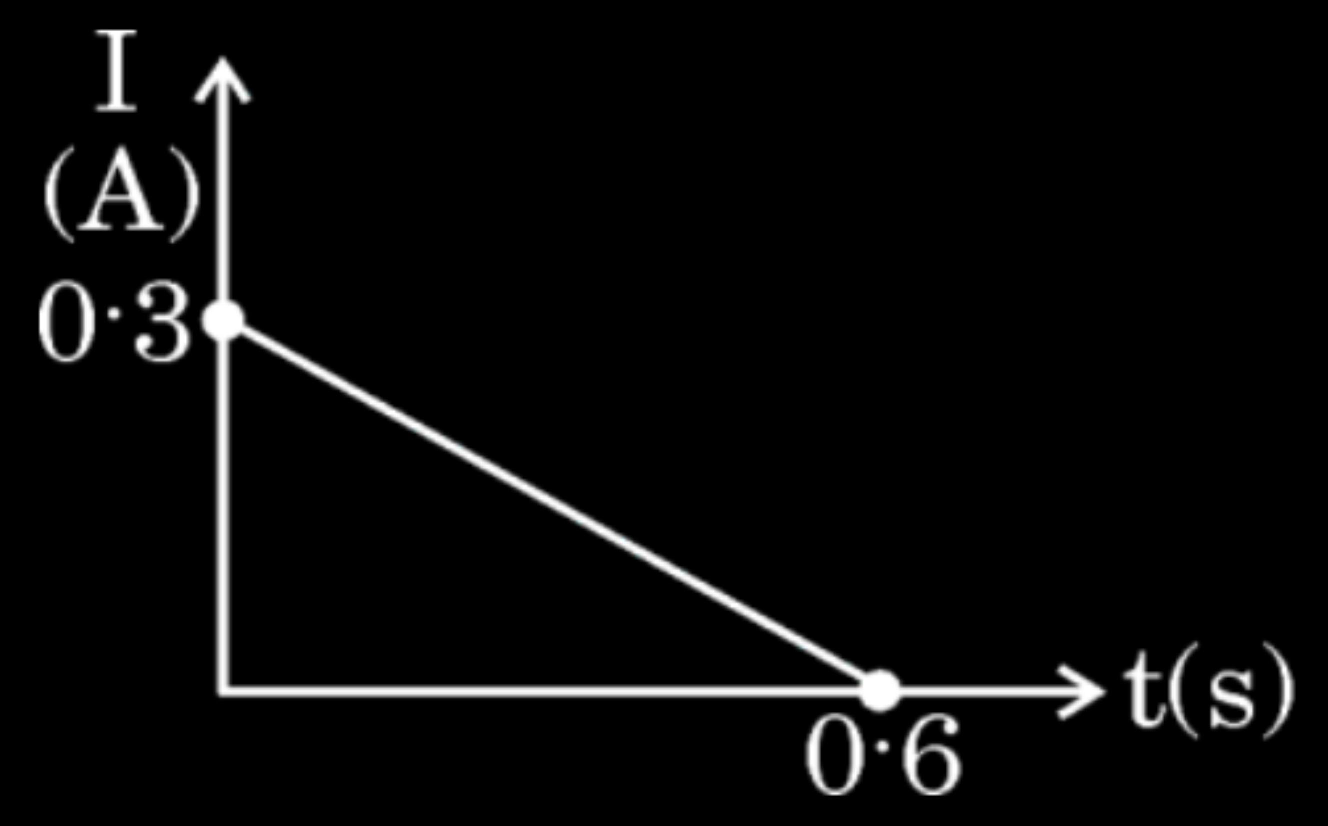
CBSE 2026

(a) What will be the direction of the induced current in the loop as it goes out of the field ? For how long would the current in the loop persist ?

(b) Plot graphs showing the variation of magnetic flux and magnitude of induced emf as a function of time. *3*

(a) As the magnetic flux linked with the loop is decreasing, hence in accordance with Lenz's Law, the induced current will be in the clockwise direction.	1
$t = \dfrac{25\,\text{cm}}{25\,\text{cm/s}} = 1\text{s}$	1
(b)  	½+½

27. A conducting rectangular loop of area 5 cm^2 and resistance 4 Ω is removed from a region of uniform magnetic field, acting normal to the plane of the loop. The value of induced current I in the loop varies with time t, as shown in the figure.



Find : *3*

(a) total charge that passed through the loop

(b) change in magnetic flux through the loop

(c) magnitude of magnetic field in the region

CBSE 2026

(a) Q = Area under (I–t) curve $= \frac{1}{2} \times \text{base} \times \text{height}$ $= \frac{1}{2} \times 0.6 \times 0.3 = 0.09\ \text{C}$	½ ½
(b) Change in magnetic flux, $\Delta\phi = QR$ $= 0.09 \times 4$ $= 0.36\ \text{Wb}$	½ ½
(c) Magnitude of magnetic field $B = \frac{\Delta\phi}{A} = \frac{0.36}{5\times10^{-4}}$ $= 720\ \text{T}$	½ ½

2. A square loop of side 50 cm is placed in a uniform magnetic field of 3·0 T acting perpendicular to the plane of the loop. If the loop is rotated through an angle of 90° in 0·3 s, the value of emf induced in the loop would be :

(A) 0·25 V

(B) 0·50 V

(C) 0·75 V

(D) 1·0 V

CBSE 2026

2. A square loop of side 50 cm is placed in a uniform magnetic field of 3·0 T acting perpendicular to the plane of the loop. If the loop is rotated through an angle of 90° in 0·3 s, the value of emf induced in the loop would be :

(A) 0·25 V

(B) 0·50 V

(C) 0·75 V

(D) 1·0 V

CBSE 2026

No option is correct (Award full credit for attempting)

(b) (i) State Lenz's law and explain that it follows the law of conservation of energy.

(i)	The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it. Consider north pole of a magnet being moved towards a coil. The magnetic flux through the coil increases, hence current is induced in counter clockwise direction making the side of the coil behave as north pole causing repulsion. If this is not the case then bringing magnet towards a coil will construct a perpetual motion machine by a suitable arrangement. Therefore production of induced emf in a direction which opposes the change in magnetic flux follows from law of conservation of energy.	1 1

7. A metallic rod of 1 m length is rotated with a frequency of 40 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field of 0·5 T parallel to the axis is present in the region. The value of emf induced between the centre and the metallic ring is close to :

(A) 20 V (B) 32 V

(C) 40 V (D) 63 V

CBSE 2026

7. A metallic rod of 1 m length is rotated with a frequency of 40 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field of 0·5 T parallel to the axis is present in the region. The value of emf induced between the centre and the metallic ring is close to :

(A) 20 V
(B) 32 V
(C) 40 V
(D) 63 V

CBSE 2026

(D) 63 V

4. A square loop of side L lies in the x-y plane in a magnetic field $\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$, where B_0 is a constant. The magnetic flux through the loop is :

(A) $\sqrt{29}\ B_0 L^2$

(B) $4 B_0 L^2$

CBSE 2026

(C) $3 B_0 L^2$

(D) $2 B_0 L^2$

4. A square loop of side L lies in the x-y plane in a magnetic field $\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$, where B_0 is a constant. The magnetic flux through the loop is :

(A) $\sqrt{29}\, B_0 L^2$ (B) $4\, B_0 L^2$

(C) $3\, B_0 L^2$ (D) $2\, B_0 L^2$

CBSE 2026

(B) $4\, B_0 L^2$

3. An electric current can be induced in a wire loop if – **1**

(A) the wire loop is wound around a bar magnet.

(B) a bar magnet is placed in contact with the wire loop.

(C) a bar magnet is placed near the wire loop.

(D) a bar magnet is moved rapidly towards the wire loop. CBSE 2026

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(D) a bar magnet is moved rapidly toward the wire loop

24. (a) Briefly explain how Lenz's law is consistent with the law of conservation of energy.

24. (a) Briefly explain how Lenz's law is consistent with the law of conservation of energy.

(a) Lenz's law is consistent with the law of conservation of energy because: When a magnet is moved towards or away from a coil, then induced current creates its own magnetic field that opposes the motion of the magnet. Due to this opposition an external force must be applied to keep the magnet moving. The work done by this external force is converted into electrical energy.	1

8. The magnetic flux ϕ (in Wb) linked with a coil is related to time t (in s) as
$\phi = 5\,At^2 + Bt - 2C$
The SI units of A and B are respectively **1**

(A) Wbs^2, Wbs (B) $Wb\,s^{-1}$, Wb

(C) Wbs^{-2}, Wbs^{-1} (D) Wbs^{-1}, Wbs^{-2}

CBSE 2026

8. The magnetic flux ϕ (in Wb) linked with a coil is related to time t (in s) as
$\phi = 5\,At^2 + Bt - 2C$
The SI units of A and B are respectively **1**

(A) Wbs^2, Wbs (B) $Wb\,s^{-1}$, Wb

(C) Wbs^{-2}, Wbs^{-1} (D) Wbs^{-1}, Wbs^{-2}

CBSE 2026

(C) Wbs^{-2}, Wbs^{-1}

10. A magnet held vertically, with its north pole down is dropped along the axis of a closed solenoid placed vertically on a table. If the observer looks down from the top, **1**

(A) the induced current will flow in the anticlockwise direction.

(B) the induced current will flow in the clockwise direction.

(C) no induced current will flow in the solenoid.

(D) the magnet will fall with a constant velocity.

CBSE 2026

10. A magnet held vertically, with its north pole down is dropped along the axis of a closed solenoid placed vertically on a table. If the observer looks down from the top, **1**

(A) the induced current will flow in the anticlockwise direction.

(B) the induced current will flow in the clockwise direction.

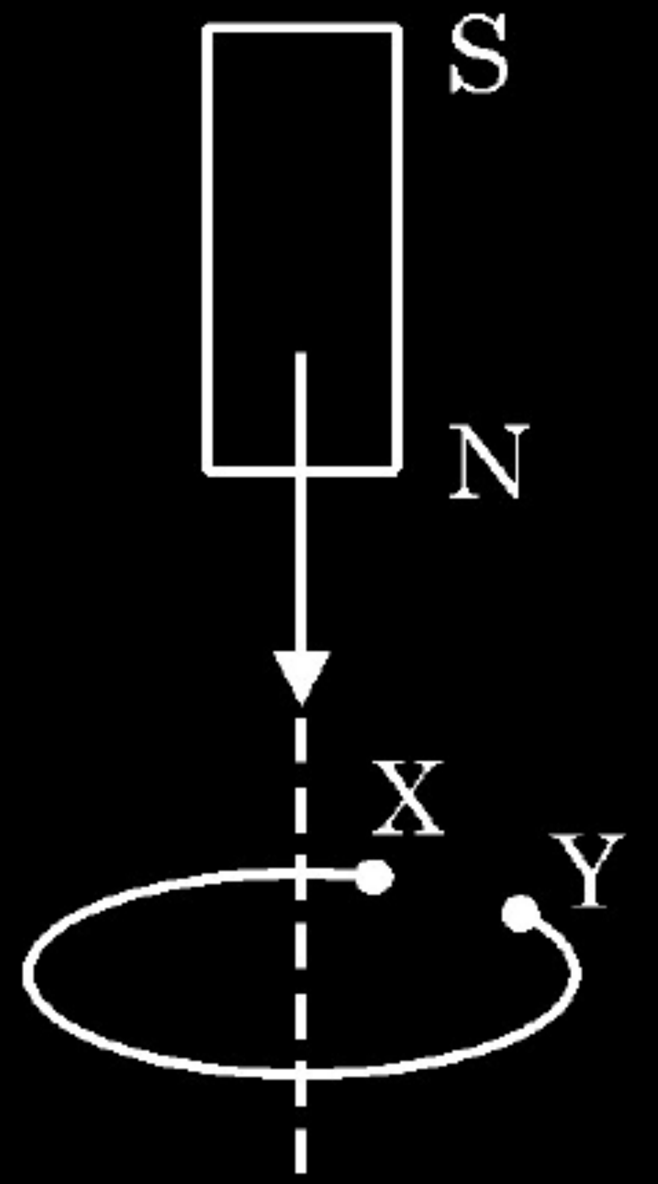
(C) no induced current will flow in the solenoid.

(D) the magnet will fall with a constant velocity.

CBSE 2026

(A) The induced current will flow in the anticlockwise direction

6. Figure shows a magnet dropped through a small loop with a small cut. Which of the following statements is correct ? 1



CBSE 2026

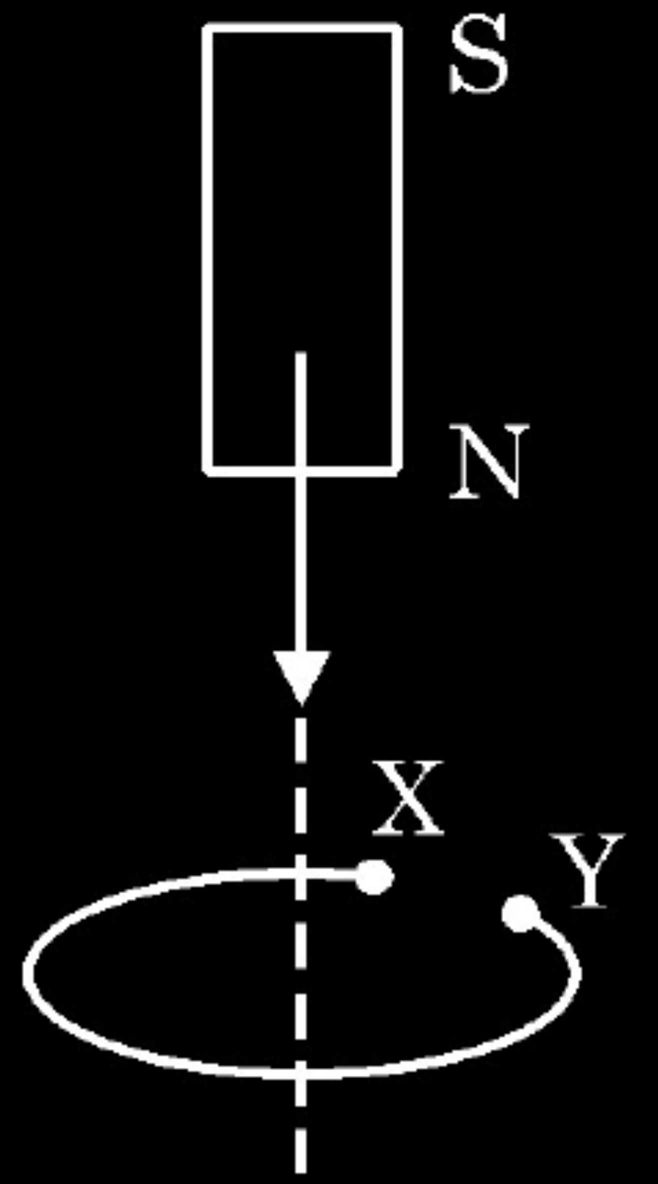
(A) The speed of the falling magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(B) Acceleration of magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(C) Speed of magnet remains uniform as it moves through the loop.

(D) Acceleration of the magnet remains uniform as it moves through the loop.

6. Figure shows a magnet dropped through a small loop with a small cut. Which of the following statements is correct ? **1**



CBSE 2026

(A) The speed of the falling magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(B) Acceleration of magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(C) Speed of magnet remains uniform as it moves through the loop.

(D) Acceleration of the magnet remains uniform as it moves through the loop.

(D) Acceleration of the magnet remains uniform as it moves through the loop.

8. Two identical circular loops A and B of metal wire are arranged on a table close to each other. Loop A is connected to a battery and the current in it increases with time. In response, the loop B **1**

(A) remains stationary

(B) is attracted by loop A

(C) is repelled by loop A

(D) rotates about its centre of mass which is fixed

CBSE 2026

8. Two identical circular loops A and B of metal wire are arranged on a table close to each other. Loop A is connected to a battery and the current in it increases with time. In response, the loop B **1**

(A) remains stationary

(B) is attracted by loop A

(C) is repelled by loop A

(D) rotates about its centre of mass which is fixed

CBSE 2026

(C) is repelled by loop A

33. (a) State Faraday's law of electromagnetic induction. **5**

(c) A conducting rod of length 50 cm, with one end pivoted, is rotated with angular speed of 60 rpm in a uniform magnetic field of 4.0 mT directed perpendicular to the plane of rotation of rod. Find the emf induced in the rod. CBSE 2026

(a) The magnitude of the induced e.m.f in a circuit is equal to the time rate of change of magnetic flux through the circuit. **Alternatively:** $\varepsilon = -\dfrac{d\phi_B}{dt}$	1
(c) Induced e.m.f	
$\varepsilon = \dfrac{1}{2} B l^2 \omega$	½
$= \dfrac{1}{2} \times 4 \times 10^{-3} \times \left(50 \times 10^{-2}\right)^2 \times (2\pi \times 1)$	1
$= 3.14\ \text{mV}$	½

OR

6. A metal rod of length 50 cm is held vertically and moved with a velocity of 10 m/s towards east. The horizontal component of the Earth's magnetic field at the place is 0·4 G. The emf induced across the ends of the rod is :

(A) 0·1 mV

(B) 0·2 mV

(C) 0·8 mV

(D) 1·6 mV

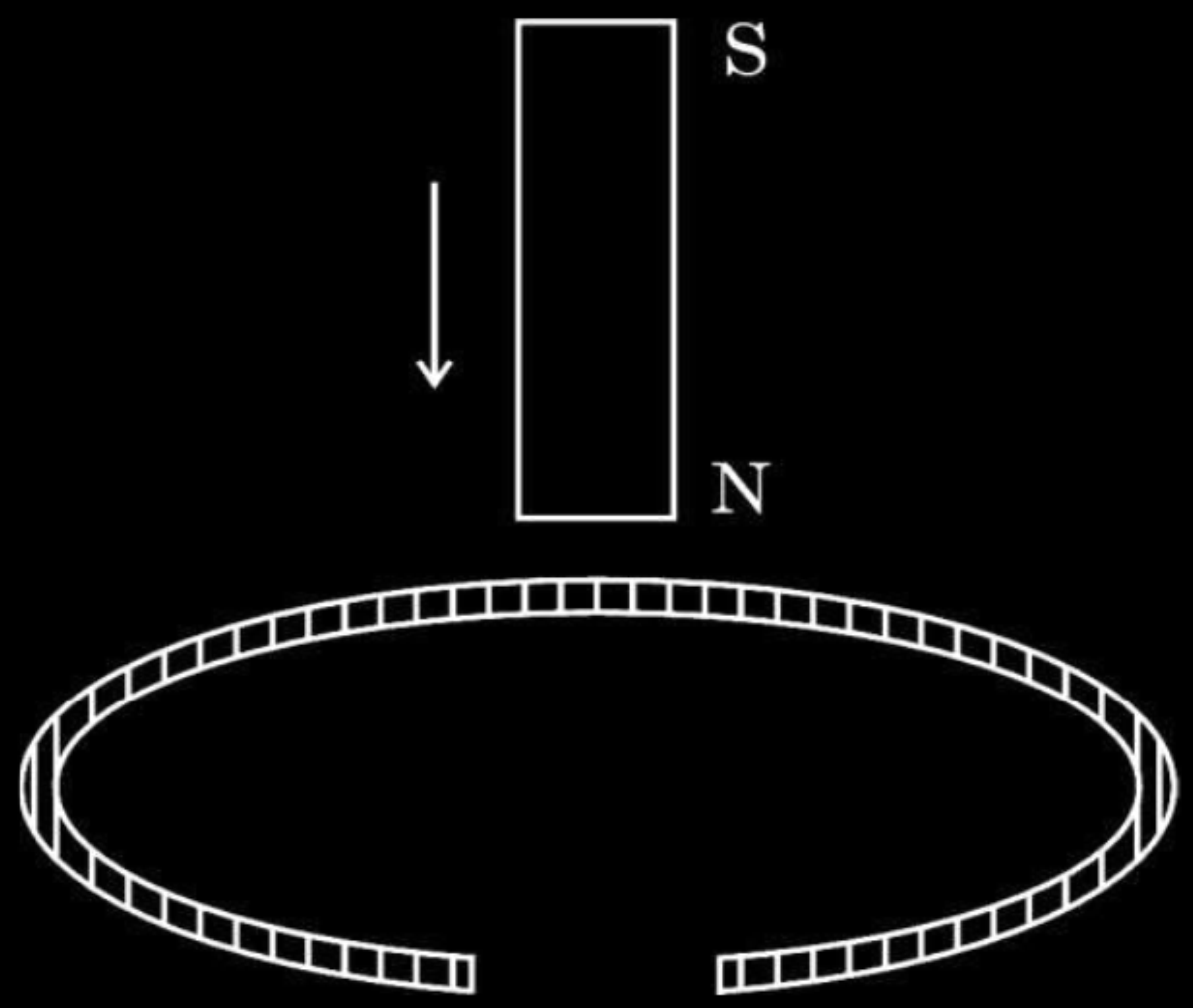
6. A metal rod of length 50 cm is held vertically and moved with a velocity of 10 m/s towards east. The horizontal component of the Earth's magnetic field at the place is 0·4 G. The emf induced across the ends of the rod is :

(A) 0·1 mV   (B) 0·2 mV

(C) 0·8 mV   (D) 1·6 mV

(B) 0.2 mV

4. A vertically held bar magnet is dropped along the axis of a copper ring having a cut as shown in the diagram. The acceleration of the falling magnet is :



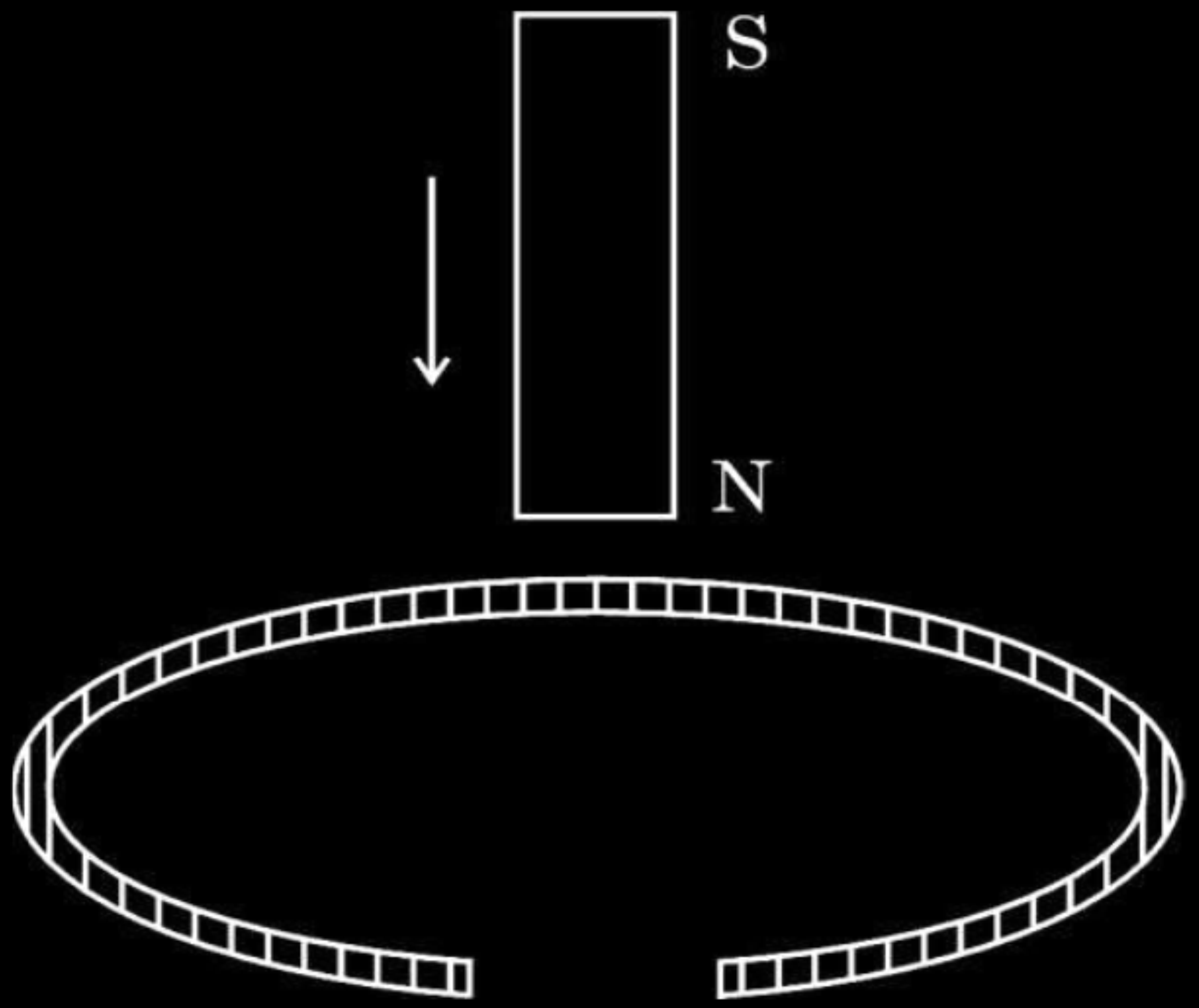
(A) zero

(B) less than g

(C) g

(D) greater than g

4. A vertically held bar magnet is dropped along the axis of a copper ring having a cut as shown in the diagram. The acceleration of the falling magnet is :



(A) zero

(B) less than g

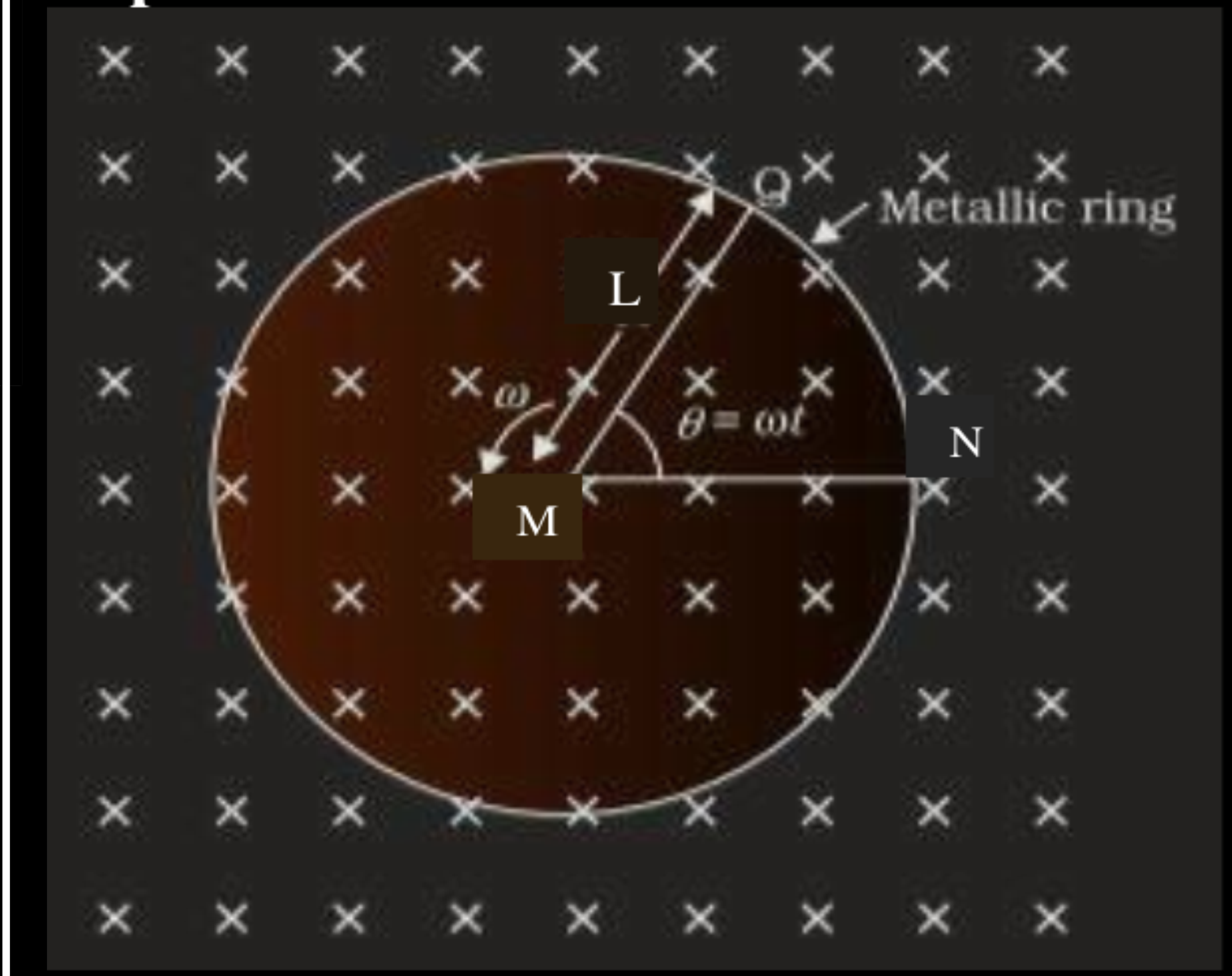
(C) g

(D) greater than g

(C) g

28. (a) State Lenz's law. A rod MN of length L is rotated about an axis passing through its end M perpendicular to its length, with a constant angular velocity ω in a uniform magnetic field $\vec{B}$ parallel to the axis. Obtain an expression for emf induced between its ends. *3*

Lenz's law The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.	1
The magnitude of the emf generated across the length dr of the rod as it moves at right angles to the magnetic field is given by $d\varepsilon = Bv\,dr$	½
$\varepsilon = \int d\varepsilon$ $= \int_0^L Bv\,dr$	½
$\varepsilon = \int_0^L B\omega r\,dr$	½
$\varepsilon = \frac{1}{2}BL^2\omega$	½
Alternatively: Area of the sector (QMN) = $\frac{1}{2}L^2\theta$	½
Induced emf is $\varepsilon = B\times\frac{d}{dt}\left(\frac{1}{2}L^2\theta\right)$	½
$\varepsilon = \frac{1}{2}BL^2\frac{d\theta}{dt}$	½
$\varepsilon = \frac{1}{2}BL^2\omega$	½



6. The magnetic flux through a loop changes from 3.6 Wb to 1.6 Wb in 0.4 s. The magnitude of the induced emf in the loop is – **1**

(A) 5 V (B) 2.5 V

(C) 1.5 V (D) 0.5 V

6. The magnetic flux through a loop changes from 3.6 Wb to 1.6 Wb in 0.4 s. The magnitude of the induced emf in the loop is – **1**

(A) 5 V (B) 2.5 V

(C) 1.5 V (D) 0.5 V

(A) 5V

13. **Assertion (A) :** Only a continuous change of magnetic flux will maintain an induced emf in a coil. **1**

Reason (R) : An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic field that induces the current.

13. **Assertion (A) :** Only a continuous change of magnetic flux will maintain an induced emf in a coil. **1**

Reason (R) : An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic field that induces the current.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is the not the correct explanation of the Assertion (A)

6. The magnetic flux linked with a coil changes with time t as $\phi = (8t^2 + 5t + 7)$, where t is in seconds and ϕ is in Wb. The value of emf induced in the coil at $t = 4$ s is :

(A) 32 V

(B) 37 V

(C) 64 V

(D) 69 V

6. The magnetic flux linked with a coil changes with time t as $\phi = (8t^2 + 5t + 7)$, where t is in seconds and ϕ is in Wb. The value of emf induced in the coil at $t = 4$ s is :

(A) 32 V (B) 37 V

(C) 64 V (D) 69 V

(D) 69 V

6. The magnetic flux linked with a closed coil (in Wb) varies with time t (in s) as $\phi = 5t^2 + 4t - 2$. If the resistance of the circuit is 14 Ω, the magnitude of induced current in the coil at t = 1 s will be :

(A) 0·5 A (B) 1·0 A

(C) 1·5 A (D) 2·0 A

6. The magnetic flux linked with a closed coil (in Wb) varies with time t (in s) as $\phi = 5t^2 + 4t - 2$. If the resistance of the circuit is 14 Ω, the magnitude of induced current in the coil at t = 1 s will be :

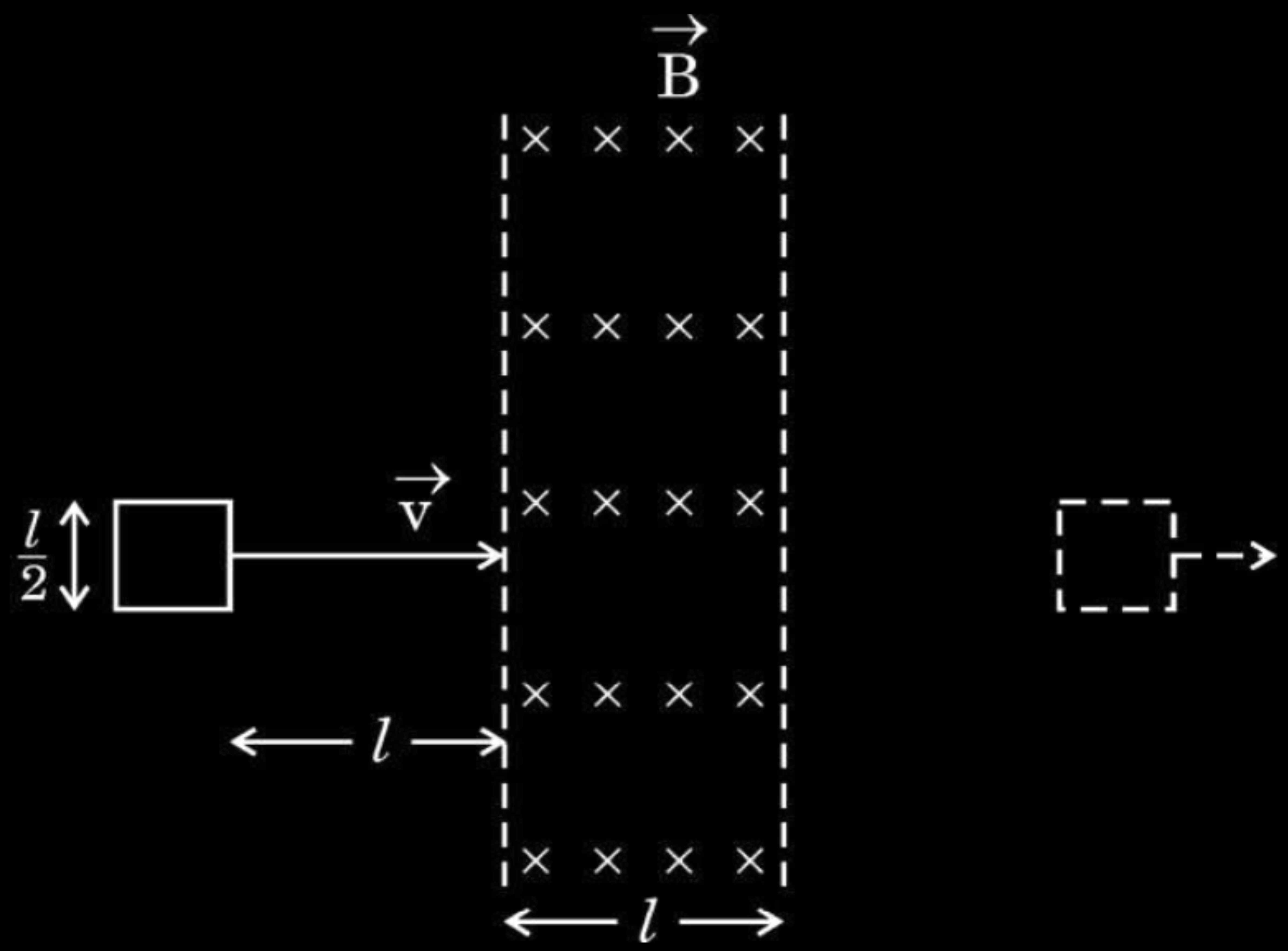
(A) 0·5 A (B) 1·0 A

(C) 1·5 A (D) 2·0 A

(B) 1.0 A

33. (a) (i) State Lenz's law and explain how this law is a consequence of conservation of energy principle.

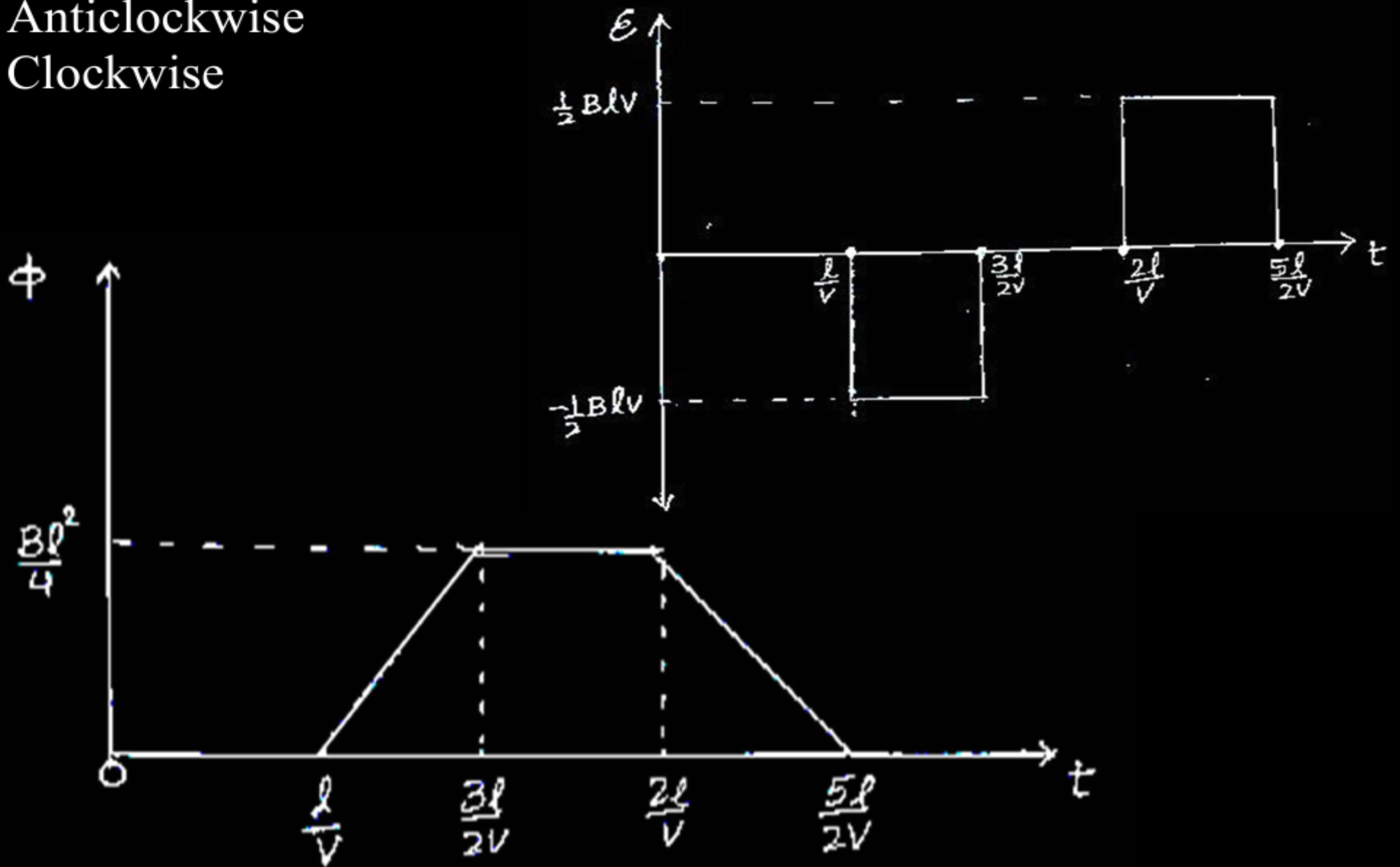
(ii) A square shaped loop of side $\frac{l}{2}$ is initially lying outside a region of uniform magnetic field $\vec{B}$ as shown in the figure. The loop is moved towards right with a constant velocity $\vec{v}$ till it goes out of the region of magnetic field.



(I) What will be the directions of induced current when the loop enters the field and when it leaves the field ?

(II) Draw the plots showing the variation of magnetic flux ϕ linked with the loop with time t and variation of induced emf E with time t. Mark the relevant values of E, ϕ and t on the graphs. 5

Lenz's law – Polarity of the induced emf is such that it tends to produce a current, which opposes the change in magnetic flux that produces it.	½
When magnet is moved closer/ away from the loop, same/ opposite pole is developed on the approaching face of the loop. So mechanical work is required to move a magnet which gets converted into electrical energy which is consistent with the law of conservation of energy.	½
(ii)	½
(I) Anticlockwise Clockwise	½
(II)	1½ 1½



24. (a) Differentiate between magnetic flux through an area and magnetic field at a point.

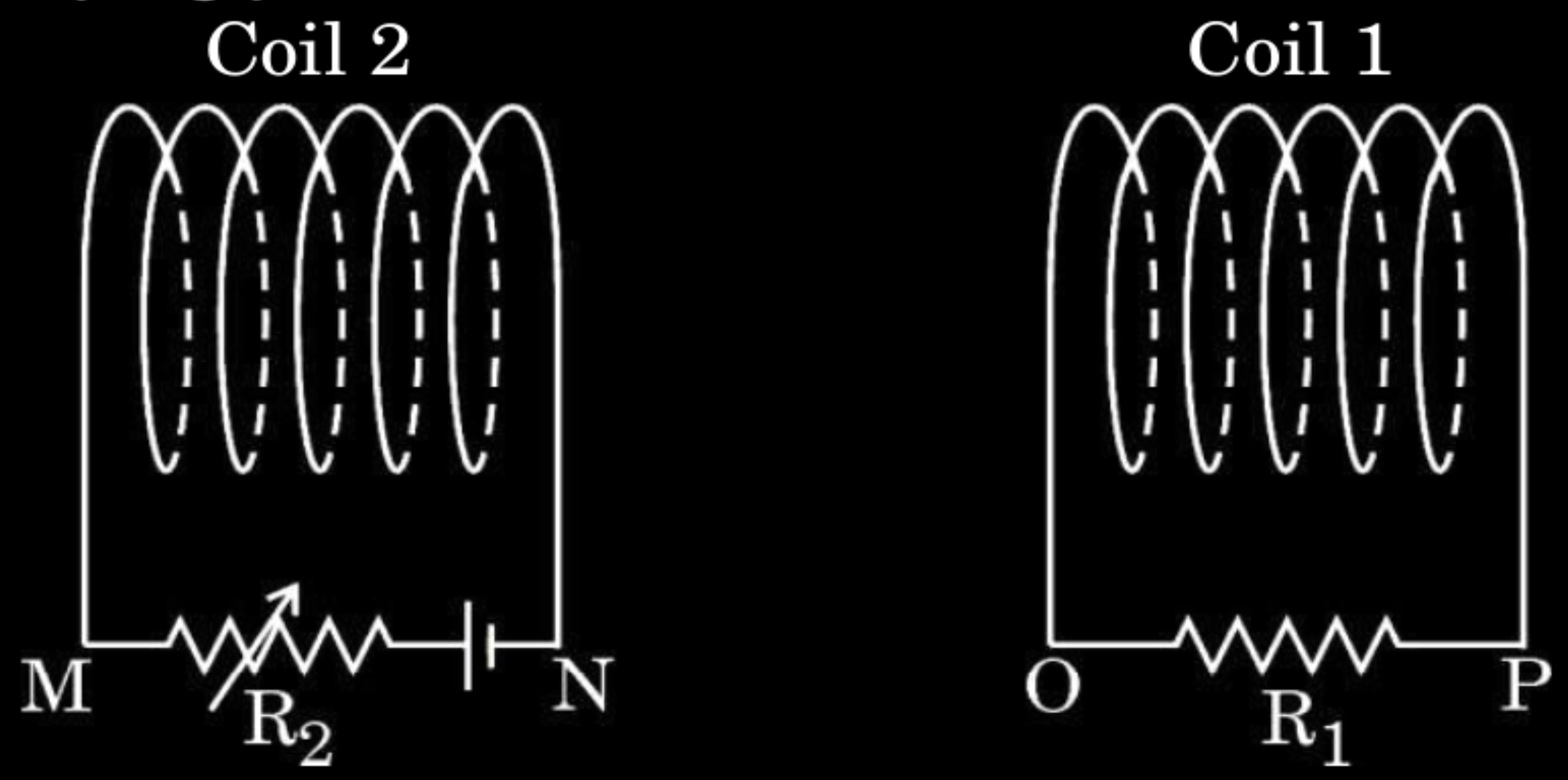
(b) A bar magnet is held with its length along the axis of a closed coil. Initially the south pole of the magnet faces the coil. If the magnet is moved towards the coil, explain how a current is induced in the coil and in what direction. *3*

24. (a) Differentiate between magnetic flux through an area and magnetic field at a point.

(b) A bar magnet is held with its length along the axis of a closed coil. Initially the south pole of the magnet faces the coil. If the magnet is moved towards the coil, explain how a current is induced in the coil and in what direction. *3*

(a) Magnetic flux is equal to total number of magnetic field lines passing normal to given area.	½
Magnetic field at a point in the space around the magnet or moving charge where magnetic force can be experienced.	½
(b) When south pole of the bar magnet moves closer to coil, the magnetic flux through the coil increases. Hence according to Faraday's law induced emf/current generate in the coil.	1
According to Lenz's law near end of the coil become south pole and as a consequence, current flows in the coil is clockwise direction.	1

24. Two coils '1' and '2' are placed close to each other as shown in the figure. Find the direction of induced current in coil '1' in each of the following situations, justifying your answers : *3*



(a) Coil '2' is moving towards coil '1'.

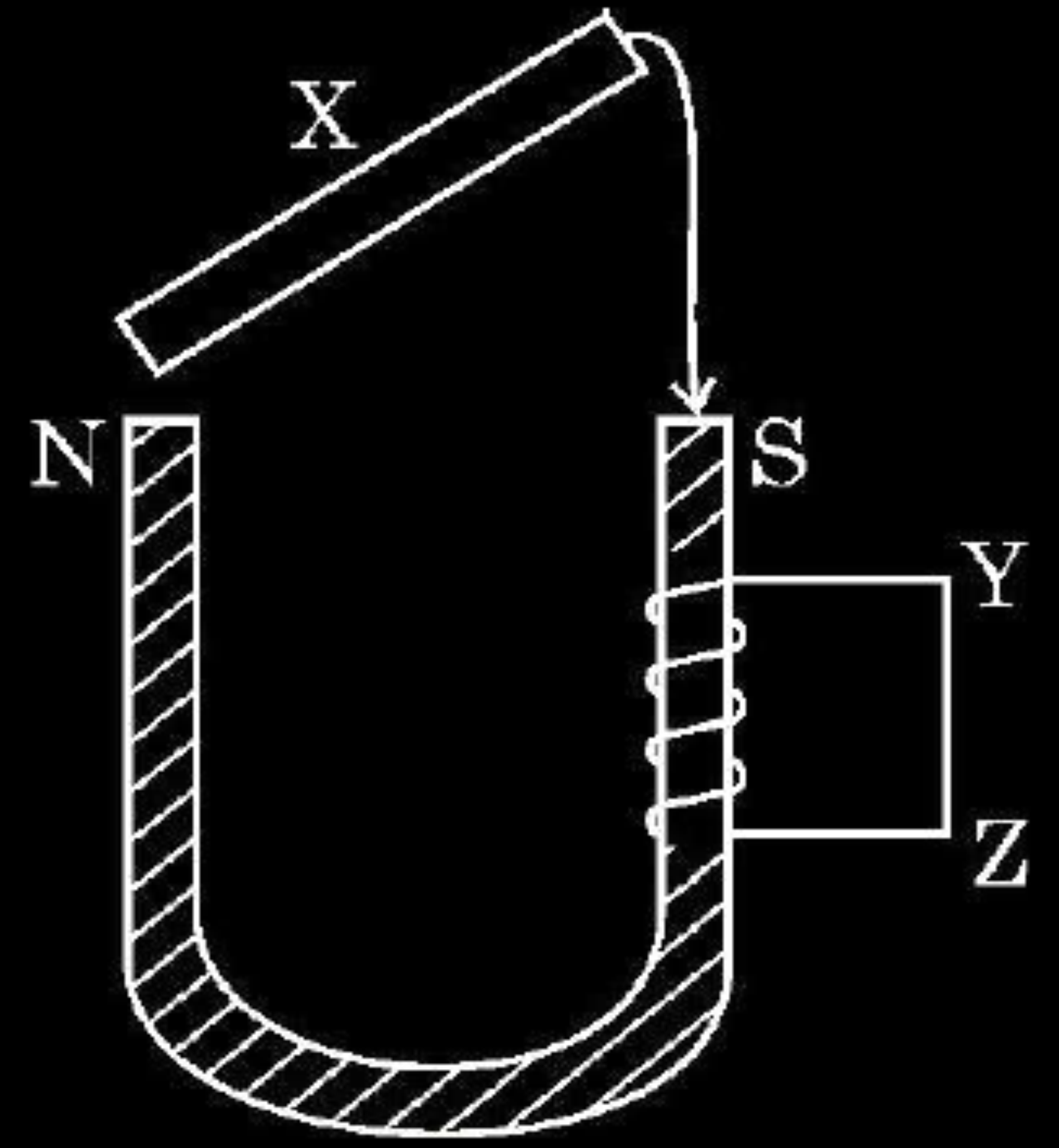
(b) Coil '2' is moving away from coil '1'.

(c) The resistance connected with coil '2' is increased keeping both the coils stationary.

Finding direction of induced current in coil and justification	
(a) coil '2' is moving toward coil '1'	½+½
(b) coil '2' is moving away from coil '1'	½+½
(c) Resistance connected with coil '2' is increased keeping both the coils stationary	½+½

(a) Induced current will be clockwise.	½
Due to the direction flow of current in coil 2, the face approaching to coil 1 behaves like south pole. Induced polarity on coil 1 viewed from the side of coil 2 will be south pole	½
(b) Induced current will be Anticlockwise	½
Due to direction of flow of current in coil 2, face going away from coil 1 behaves like south pole. Induced polarity on coil 1 viewed from the side of coil 2, will be north pole.	½
(c) Induced current will be Anticlockwise	½
Till the resistance in coil 2 is increasing, the magnetic flux associated with coil 1 is decreasing. Hence according to Lenz's law current will induce in coil 1, momentarily and after that no induced current in coil	½

5. A soft iron rod X is allowed to fall on the two poles of a U shaped permanent magnet as shown in figure. A coil is wrapped over one arm of the U shaped magnet. 1



During fall of the rod, the current in the coil will be

(A) clockwise current (B) anticlockwise current

(C) alternating current (D) zero

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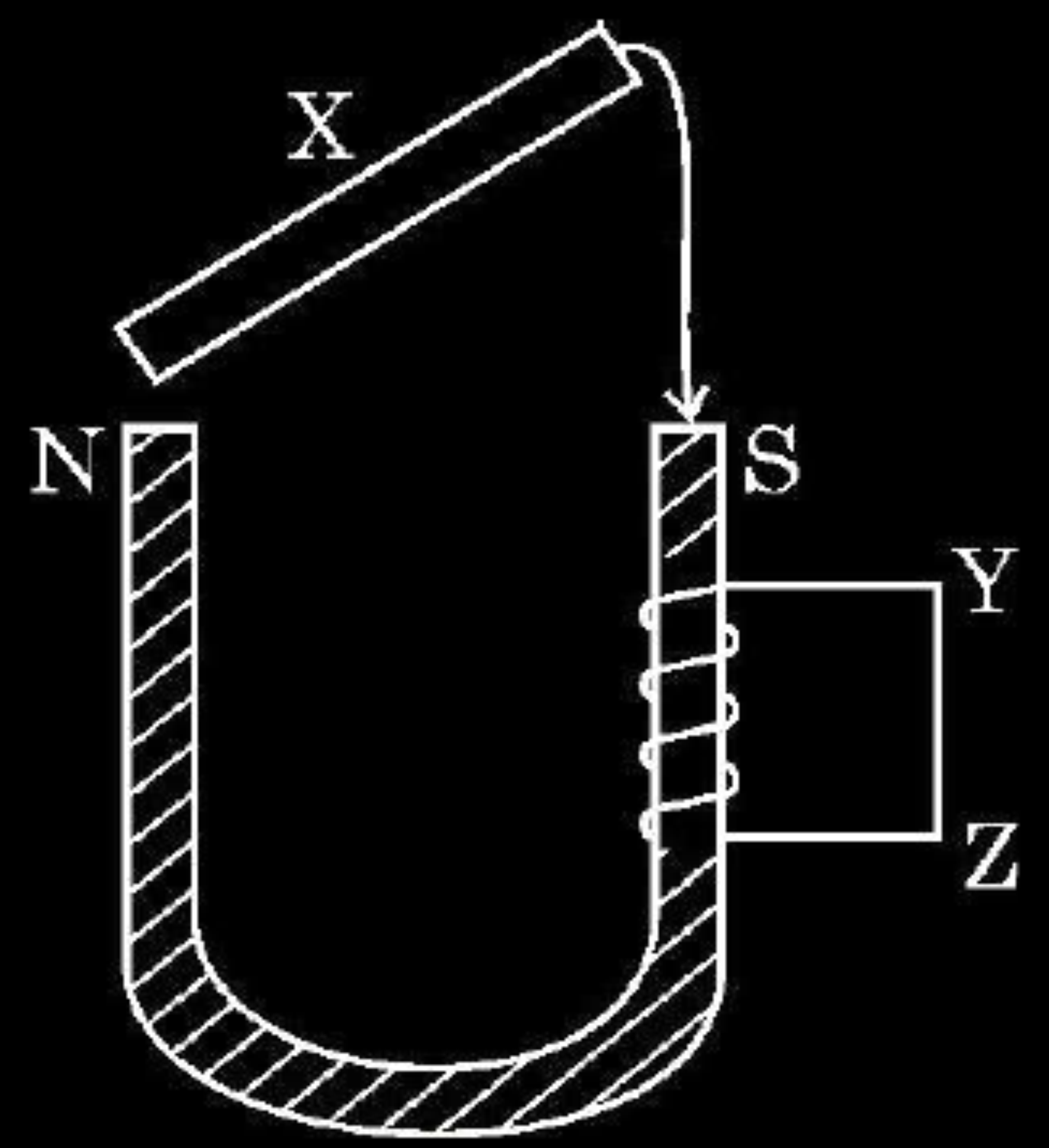


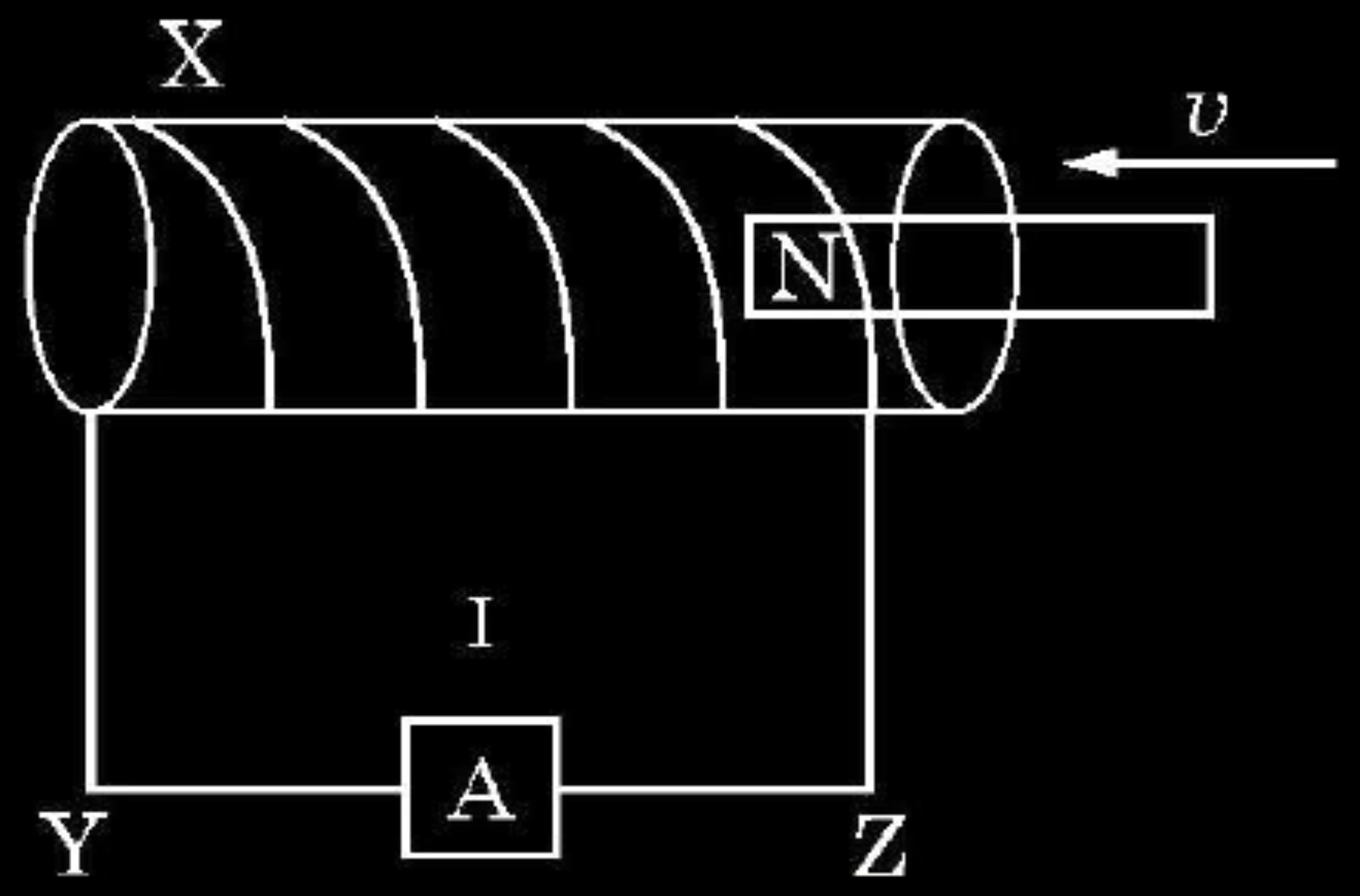
During fall of the rod, the current in the coil will be

(A) clockwise current (B) anticlockwise current

(C) alternating current (D) zero

(B) anticlockwise current

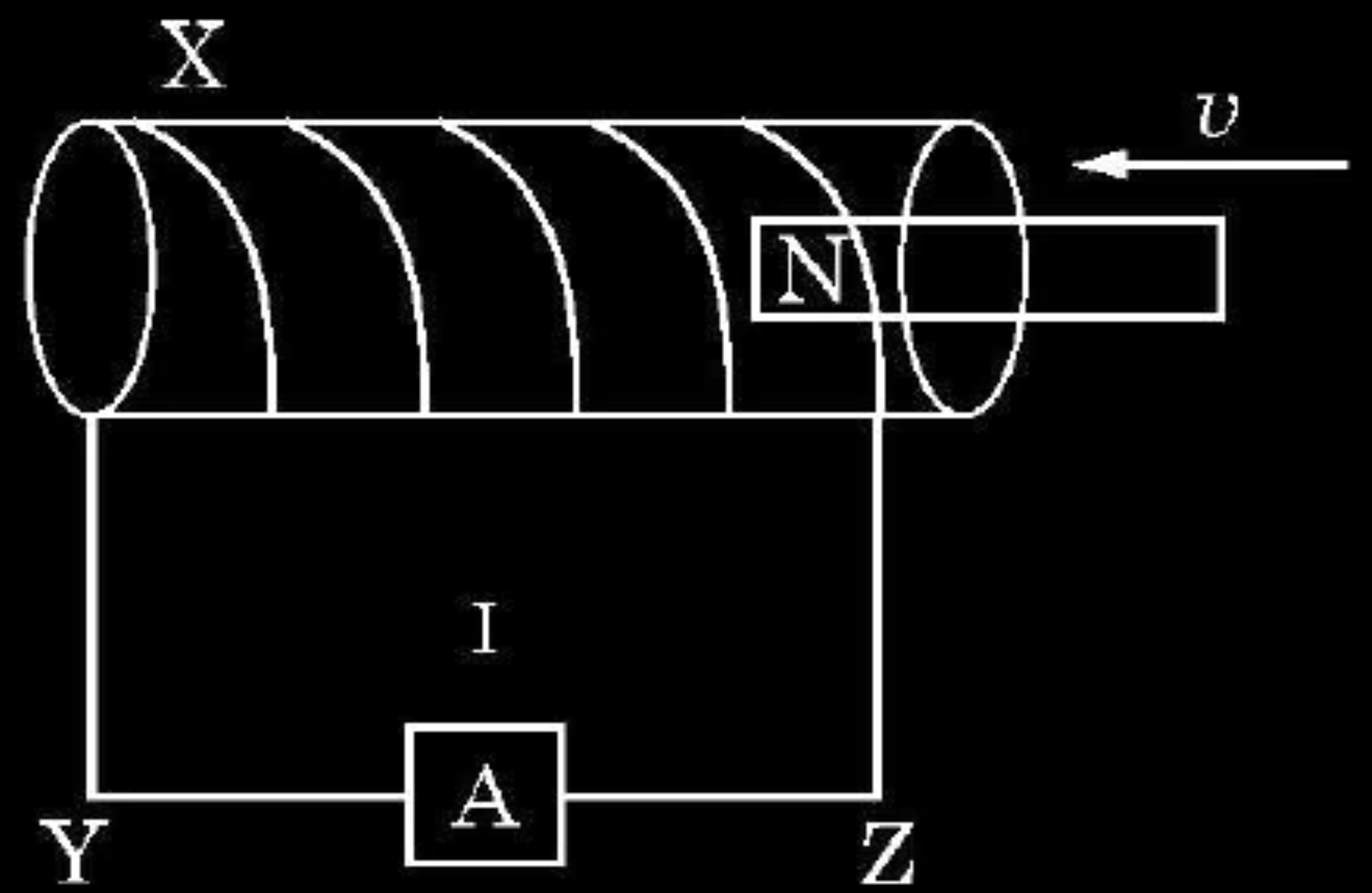
5. In the figure X is a coil wound over a hollow wooden pipe. **1**



A permanent magnet is pushed at a constant speed v from the right into the pipe and it comes out at the left end of the pipe. During the entry and the exit of the magnet, the current in the wire YZ will be from

(A) Y to Z and then Y to Z
(B) Z to Y and then Y to Z
(C) Y to Z and then Z to Y
(D) Z to Y and then Z to Y

5. In the figure X is a coil wound over a hollow wooden pipe. 1



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(A) Y to Z and then Y to Z

(B) Z to Y and then Y to Z

(C) Y to Z and then Z to Y

(D) Z to Y and then Z to Y

(B) Z to Y and then Y to Z

15. **Assertion (A)** : It is difficult to move a magnet into a coil of large number of turns when the circuit of the coil is closed. **1**

Reason (R) : The direction of induced current in a coil with its circuit closed, due to motion of a magnet, is such that it opposes the cause.

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Reason (R) : The direction of induced current in a coil with its circuit closed, due to motion of a magnet, is such that it opposes the cause.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

4. A circular coil of diameter 15 mm having 300 turns is placed in a magnetic field of 30 mT such that the plane of the coil is perpendicular to the direction of magnetic field. The magnetic field is reduced uniformly to zero in 20 ms and again increased uniformly to 30 mT in 40 ms. If the emfs induced in the two time intervals are e_1 and e_2 respectively, then the value of e_1/e_2 is **1**

(A) $\frac{1}{2}$ (B) 1

(C) 2 (D) 4

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(A) $\frac{1}{2}$ (B) 1

(C) 2 (D) 4

(C) 2

4. A coil has 100 turns, each of area 0.05 m^2 and total resistance 1.5 Ω. It is inserted at an instant in a magnetic field of 90 mT, with its axis parallel to the field. The charge induced in the coil at that instant is : **1**

(A) 3.0 mC (B) 0.30 C

(C) 0.45 C (D) 1.5 C

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(A) 3.0 mC
(B) 0.30 C
(C) 0.45 C
(D) 1.5 C

(B)	0.30 C

22. (a) (i) State Lenz's Law. In a closed circuit, the induced current opposes the change in magnetic flux that produced it as per the law of conservation of energy. Justify.

(ii) A metal rod of length 2 m is rotated with a frequency 60 rev/s about an axis passing through its centre and perpendicular to its length. A uniform magnetic field of 2T perpendicular to its plane of rotation is switched-on in the region. Calculate the e.m.f. induced between the centre and the end of the rod.

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(a) (i) The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.	1
In a closed loop, when the polarity of induced emf is such that, the induced current favours the change in magnetic flux then the magnetic flux and consequently the current will go on increasing without any external source of energy. This violates law of conservation of energy.	½
$\varepsilon = \frac{1}{2} B l^2 \omega$	½
$= \frac{1}{2} \times 2 \times (2)^2 \times (2\pi \times 60)$	½
$= 480\pi$ V	½
$= 1.51 \times 10^3$ V	

5. A coil of area of cross-section $0.5\ m^2$ is placed in a magnetic field acting normally to its plane. The field varies as $B = 0.5t^2 + 2t$, where B is in tesla and t in seconds. The emf induced in the coil at $t = 1s$ is

(A) 0.5 V (B) 1.0 V

(C) 1.5 V (D) 3.0 V

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(A) 0.5 V (B) 1.0 V

(C) 1.5 V (D) 3.0 V

(C) 1.5V

24. Prove that induced charge depends on the net change in the magnetic flux and not on the time interval of the flux change. *3*

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Using Faraday's law of electromagnetic induction	
$\lvert\varepsilon\rvert = \frac{\Delta\phi}{\Delta t}$	½
$I = \frac{\lvert\varepsilon\rvert}{R}$	½
$I = \frac{1}{R}\left(\frac{\Delta\phi}{\Delta t}\right)$	½
$\frac{\Delta Q}{\Delta t} = \frac{1}{R}\left(\frac{\Delta\phi}{\Delta t}\right)$	½
$\Delta Q = \frac{\Delta\phi}{R}$	½
Hence induced charge depends on change in magnetic flux, not on the time interval of flux change.	½

(ii) The magnetic flux through a coil of resistance 5 Ω increases with time as :

$$\phi = (2{\cdot}0\ t^3 + 5{\cdot}0\ t^2 + 6{\cdot}0\ t)\ \text{mWb}$$

Find the magnitude of induced current through the coil at $t = 2$ s.

(ii) The magnetic flux through a coil of resistance 5 Ω increases with time as :

$$\phi = (2{\cdot}0\ t^3 + 5{\cdot}0\ t^2 + 6{\cdot}0\ t)\ \text{mWb}$$

Find the magnitude of induced current through the coil at $t = 2$ s.

(ii) $\phi = (2.0t^3 + 5.0t^2 + 6.0t)$ mWb	
$\lvert\varepsilon\rvert = \frac{d\phi}{dt} = 50 \times 10^{-3}$ V	½
$I = \frac{\lvert\varepsilon\rvert}{R}$	½
$I = \frac{50\times10^{-3}}{5}$ A=10 mA	½

15. *Assertion (A) :* Lenz's law is a consequence of the law of conservation of energy.

Reason (R) : There is no power loss in an ideal inductor.

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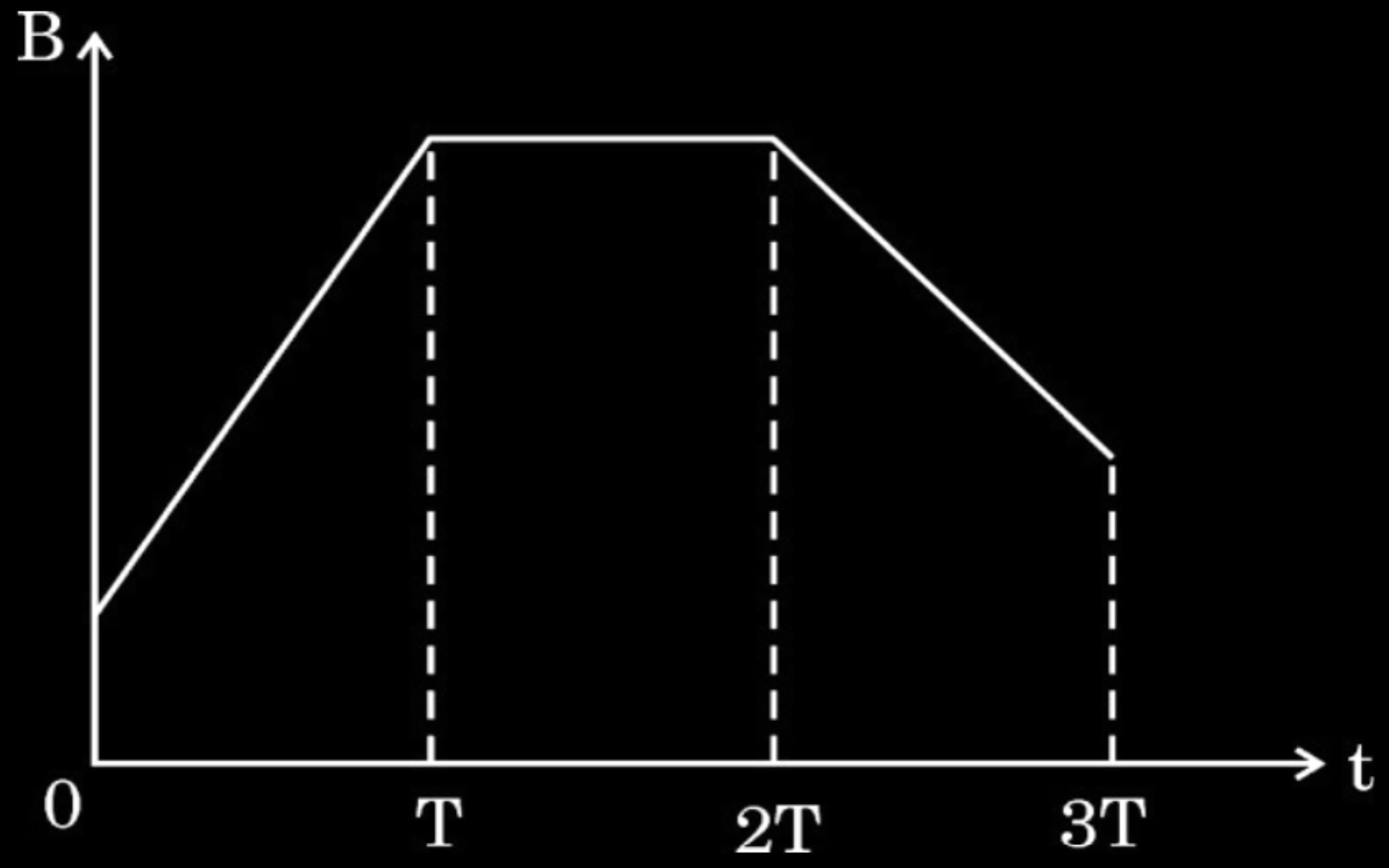
Reason (R) : There is no power loss in an ideal inductor.

(B) Both Assertion(A) and Reason (R) are true but Reason(R) is not the correct explanation of the Assertion (A)

3. A conducting loop is placed in a magnetic field, normal to its plane. The magnitude of the magnetic field varies with time as shown in the figure. If ε_1, ε_2 and ε_3 are magnitudes of induced emfs during periods $0 \le t \le T$, $T \le t \le 2T$ and $2T \le t \le 3T$, then :



(A) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

(B) $\varepsilon_2 > \varepsilon_3 > \varepsilon_1$

(C) $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$

(D) $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

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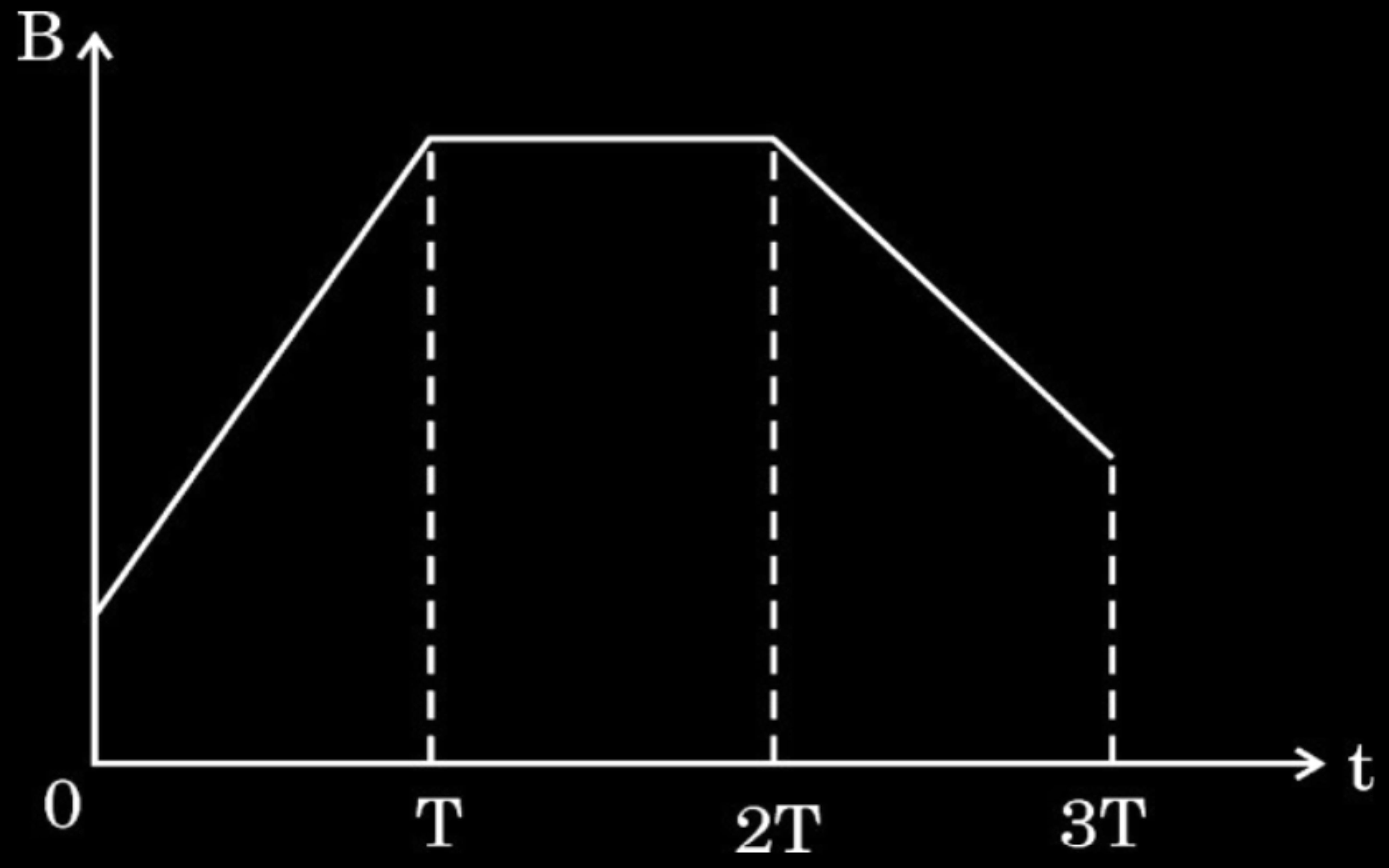

(A) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

(B) $\varepsilon_2 > \varepsilon_3 > \varepsilon_1$

(C) $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$

(D) $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

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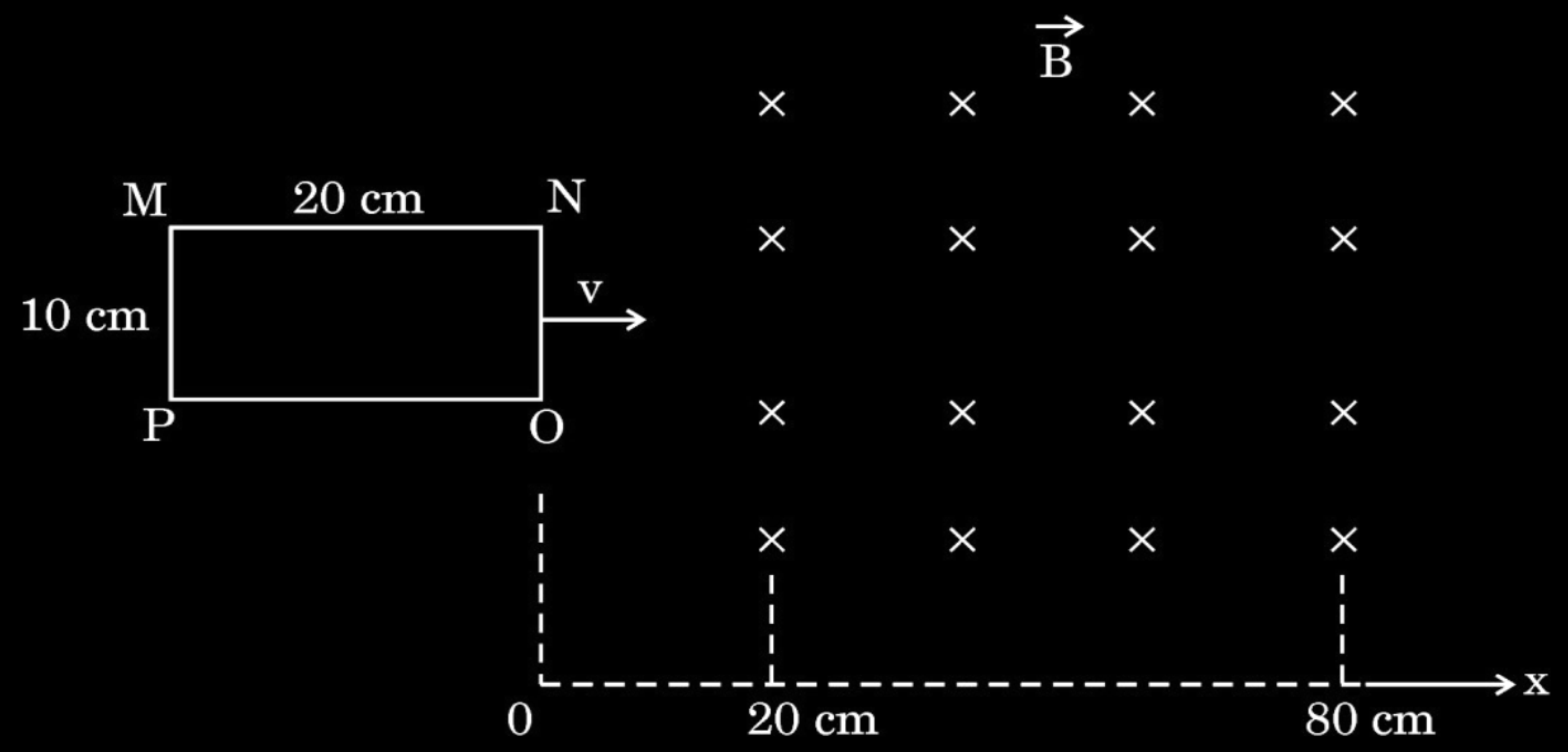
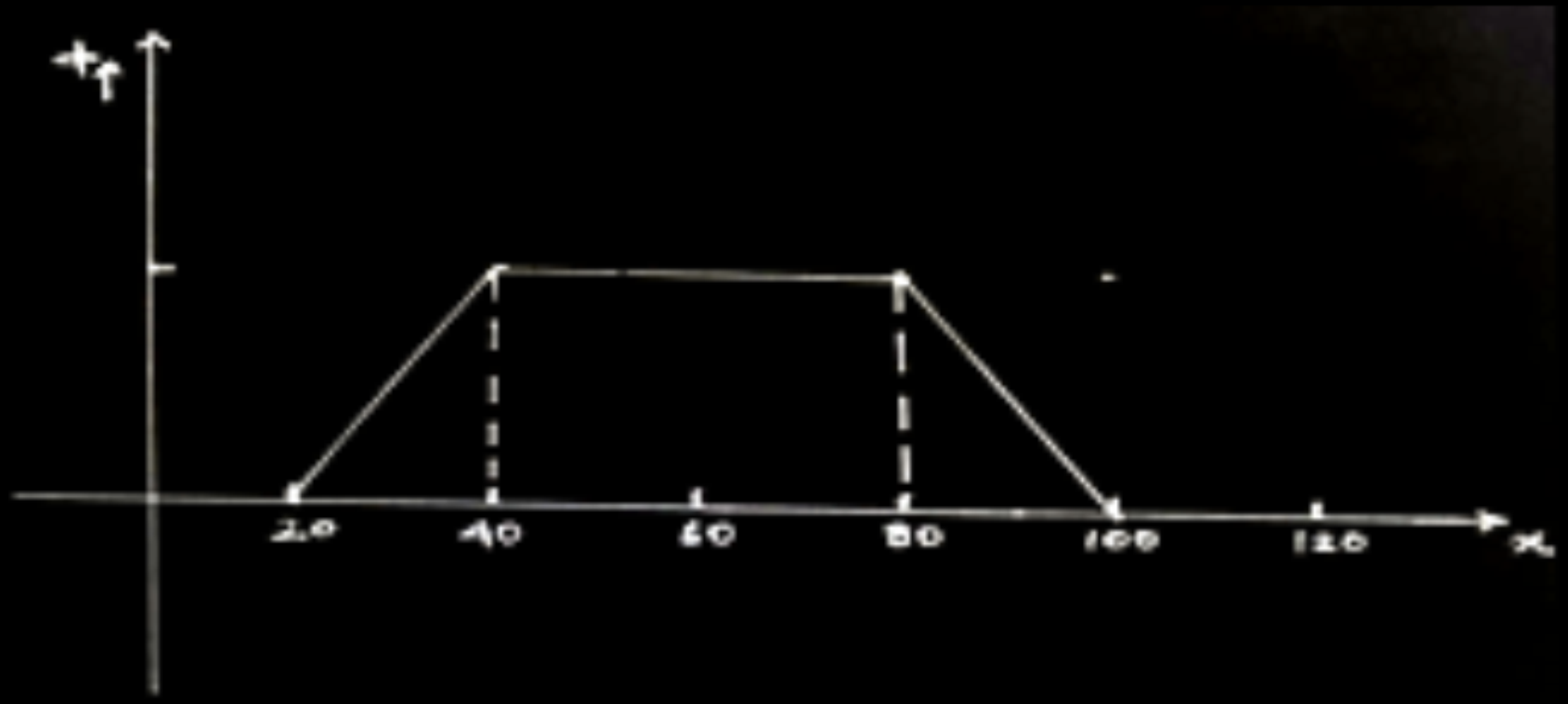
24. A rectangular loop of sides 10 cm × 20 cm is kept outside a region of uniform magnetic field $|\vec{B}| = 5$ mT as shown in the figure. The loop is moved with the velocity of 5 cm/s till it goes completely out of the magnetic field.

(a) Plot a graph showing variation of the magnetic flux ϕ with x ($0 \le x \le 100$ cm).

(b) Find the maximum value of magnetic flux linked with the loop.

(c) Will an external work be required to be done to move the loop through the magnetic field ? *3*



(a) 	1 ½
(b) $\phi = B.A$	½
$= 5\times10^{-3}\times20\times10^{-2}\times10\times10^{-2}$	
$= 10^{-4}\,Wb$	½
(c) Yes, external work is required.	½

24. A 100-turn coil of radius $1 \cdot 6$ cm and resistance $5 \cdot 0\ \Omega$ is co-axial with a solenoid of 250 turns/cm and radius $1 \cdot 8$ cm. The solenoid current drops from $1 \cdot 5$ A to zero in 25 ms. Calculate the current induced in the coil in this duration. (Take $\pi^2 = 10$) *3*

24. A 100-turn coil of radius 1·6 cm and resistance 5·0 Ω is co-axial with a solenoid of 250 turns/cm and radius 1·8 cm. The solenoid current drops from 1·5 A to zero in 25 ms. Calculate the current induced in the coil in this duration. (Take $\pi^2 = 10$) *3*

Induced emf $(\varepsilon) = \frac{-Nd\phi}{dt}$	½
$= \frac{-NAdB}{dt}$	
$= -NA\frac{d}{dt}(\mu_0 nI)$	
$= -N\mu_0 n(\pi r^2)\frac{dI}{dt}$	
$\varepsilon = \frac{100\times 4\pi\times 10^{-7}\times 250\times 10^{2}\times\pi\times(1.6\times 10^{-2})^2\times 1.5}{25\times 10^{-3}}$	1
$=0.1536\text{V}$	½
$I = \frac{\varepsilon}{R}$	½
$= 0.03\text{A}$	½

7. A conducting circular loop is placed in a uniform magnetic field $B = 50$ mT with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of 1 mm s^{-1}. At the instant the radius of the loop is 4 cm, the induced emf in the loop is :

(A) π μV (B) 2π μV

(C) 4π μV (D) 8π μV

7. A conducting circular loop is placed in a uniform magnetic field $B = 50\ mT$ with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of $1\ mm\ s^{-1}$. At the instant the radius of the loop is 4 cm, the induced emf in the loop is :

(A) $\pi\ \mu V$ (B) $2\pi\ \mu V$

(C) $4\pi\ \mu V$ (D) $8\pi\ \mu V$

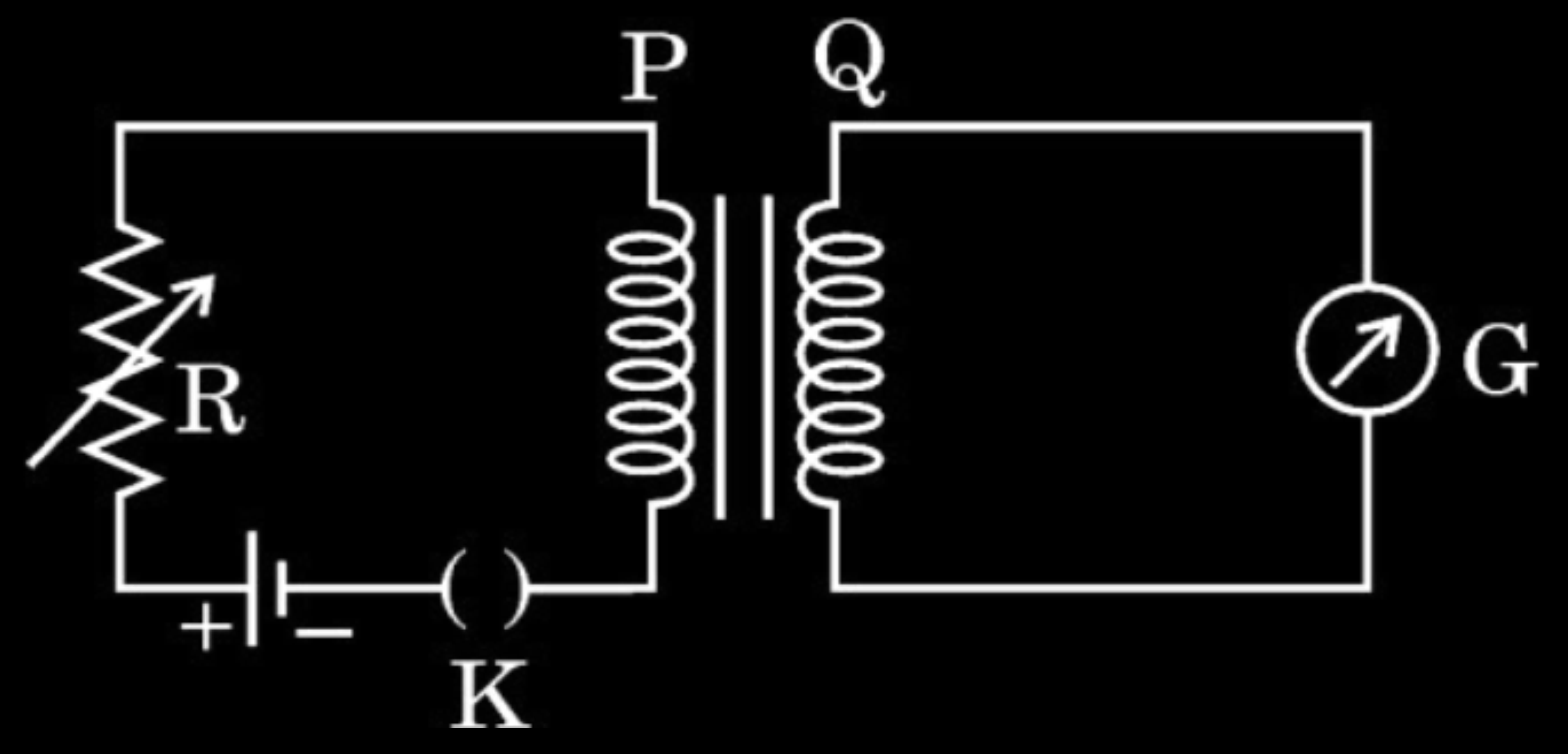
(C) $4\pi\mu V$

25. Consider the arrangement of two coils P and Q shown in the figure. When current in coil P is switched on or switched off, a current flows in coil Q.

(a) Explain the phenomenon involved in it.

(b) Mention two factors on which the current produced in coil Q depends.

(c) Give the direction of current in coil Q when there is a current in the coil P and (i) R is increased, and (ii) R is decreased. *3*



(a) Mutual Induction When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it.	1
(b) Factors on which the current produced in coil Q depends will be: (Any two) (i) Number of turns in coil P and Q (ii) Current flowing through coil P. (iii) Resistance of coil Q. (iv) Mutual Induction between the two coils.	½ + ½
(c) The direction of current through coil Q: (i) Clockwise when R is increased. (ii) Anticlockwise when R is decreased.	½ ½

6. The magnetic flux ϕ (in Wb) through a closed loop of resistance $3 \cdot 0\ \Omega$ varies with time t (s) as $\phi = 6t^2 + 3 \cdot 0t + 5$. The value of the induced current in the loop at $t = 1 \cdot 0$ s is :

(A) 15·0 A (B) 3·0 A

(C) 4·0 A (D) 5·0 A

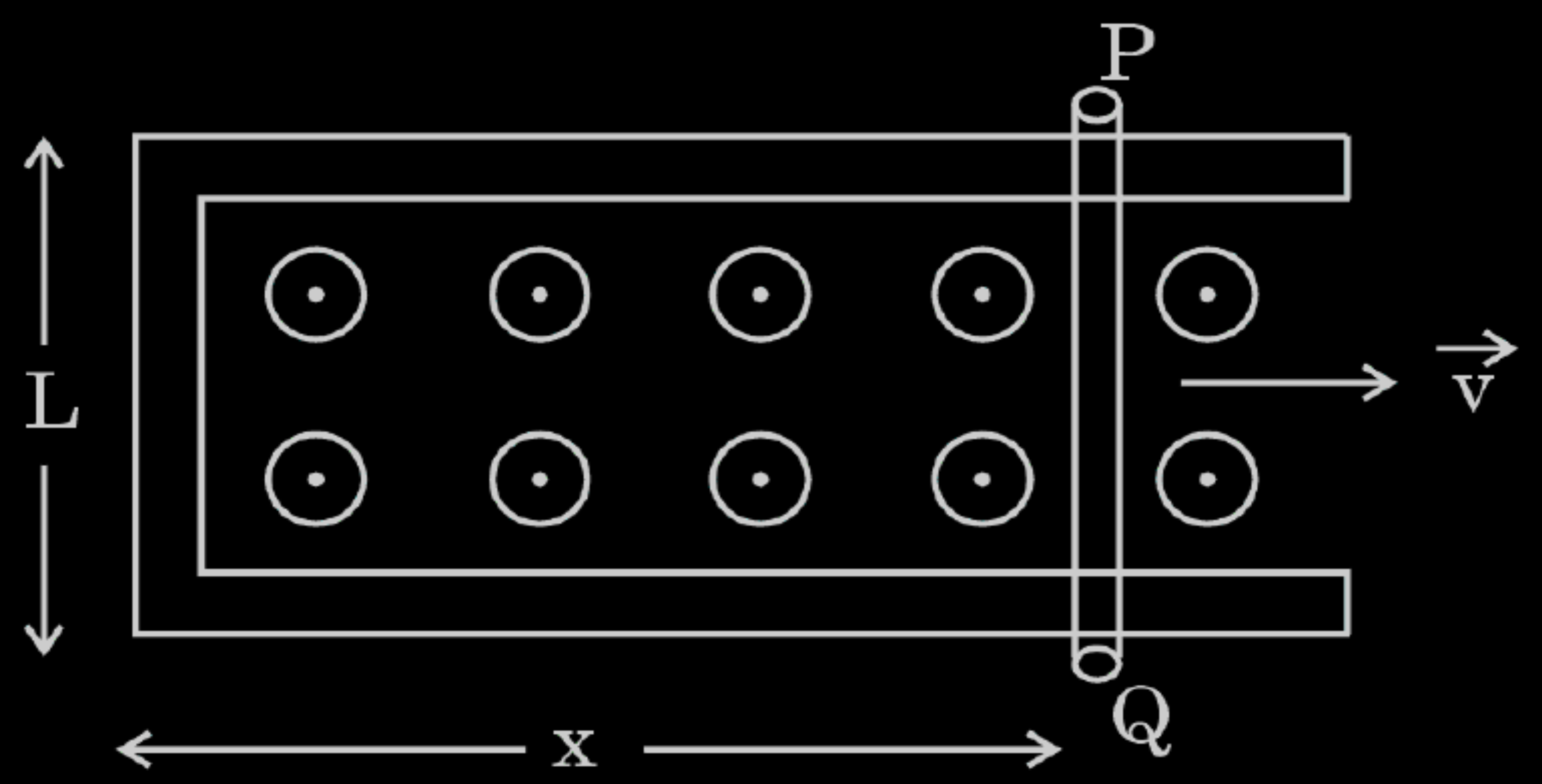
6. The magnetic flux ϕ (in Wb) through a closed loop of resistance $3{\cdot}0\ \Omega$ varies with time t (s) as $\phi = 6t^2 + 3{\cdot}0t + 5$. The value of the induced current in the loop at $t = 1{\cdot}0$ s is :

(A) 15·0 A (B) 3·0 A

(C) 4·0 A (D) 5·0 A

(D) 5.0 A

4. A rod PQ of resistance R lies across frictionless conducting rails in a constant uniform magnetic field $\vec{B}$, as shown in the figure. Assuming that the rails have negligible resistance, the force required to pull the rod to the right at constant velocity $\vec{v}$ is :



(A) 0

(B) Blv

(C) $\dfrac{Blv}{R}$

(D) $\dfrac{B^2 l^2 v}{R}$

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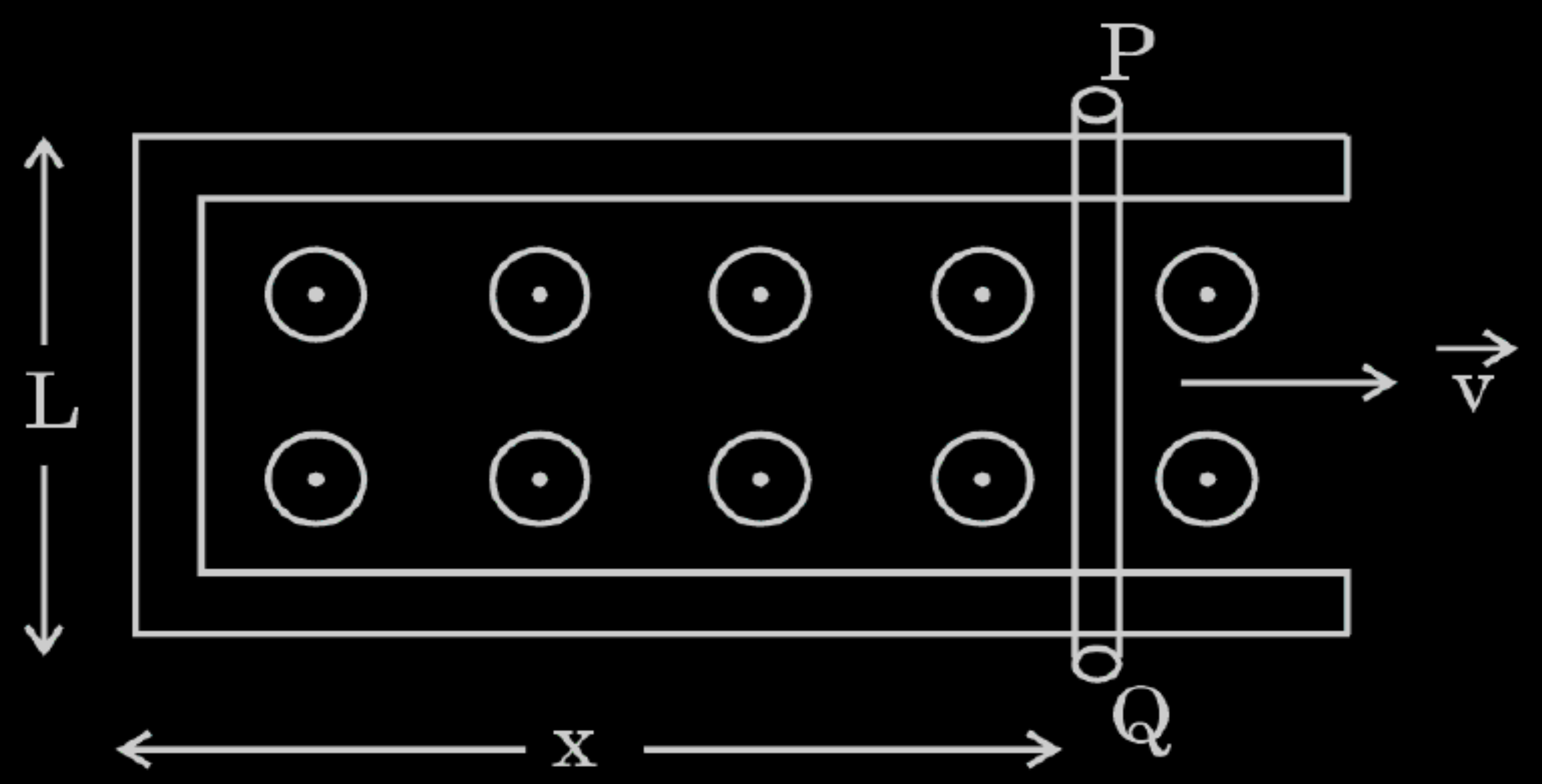

(A) 0

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(C) $\dfrac{Blv}{R}$

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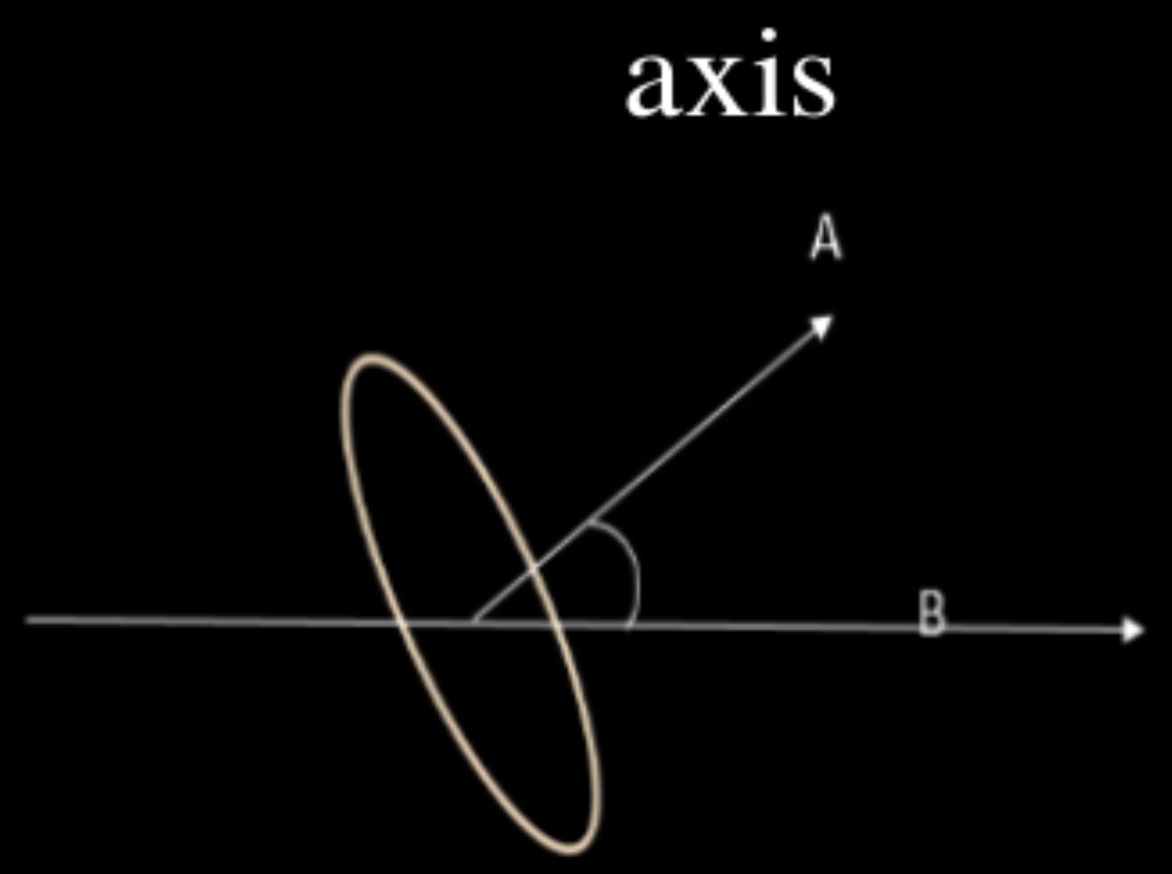
26. A small circular loop of area $\frac{6}{\pi}$ cm^2 is placed inside a long solenoid at its centre such that its axis makes an angle of 60° with the axis of the solenoid. The number of turns per cm is 10 in the solenoid. The current in the solenoid changes uniformly from 5 A to zero in 10 ms. Calculate the emf induced in the loop. *3*

26. A small circular loop of area $\frac{6}{\pi}$ cm^2 is placed inside a long solenoid at its centre such that its axis makes an angle of 60° with the axis of the solenoid. The number of turns per cm is 10 in the solenoid. The current in the solenoid changes uniformly from 5 A to zero in 10 ms. Calculate the emf induced in the loop. *3*

E.m.f induced in the circular loop:



$$|\varepsilon| = \frac{d\varphi}{dt}$$ ½

$$= \frac{d}{dt}(BA\cos\theta)$$

$$= \frac{d}{dt}(\mu_0 n I A\cos\theta)$$ 1

$$|\varepsilon| = \left(\mu_0 n A\cos\theta . \frac{dI}{dt}\right)$$ 1

$$|\varepsilon| = 4\pi \times 10^{-7} \times 1000 \times \frac{6}{\pi} \times 10^{-4} \times \cos 60^0 \times \frac{5}{10^{-2}}$$

$$= 6 \times 10^{-5} \text{ V}$$ ½

6. A bar magnet is thrown from a distance into a closed loop with its north pole towards the loop and passes through it. The induced current in the loop is :

(A) first clockwise and then anticlockwise

(B) first anticlockwise and then clockwise

(C) clockwise only

(D) anticlockwise only

6. A bar magnet is thrown from a distance into a closed loop with its north pole towards the loop and passes through it. The induced current in the loop is :

(A) first clockwise and then anticlockwise

(B) first anticlockwise and then clockwise

(C) clockwise only

(D) anticlockwise only

(B) first anticlockwise and then clockwise

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4. The magnetic flux (in SI units) through a coil varies with time as $\phi = 3t^2 + 4t + 7$. The ratio of emf induced in the coil at $t = 2$ s to that at $t = 1$ s will be :

(a) 2 (b) 0·8

(c) 1·6 (d) 4

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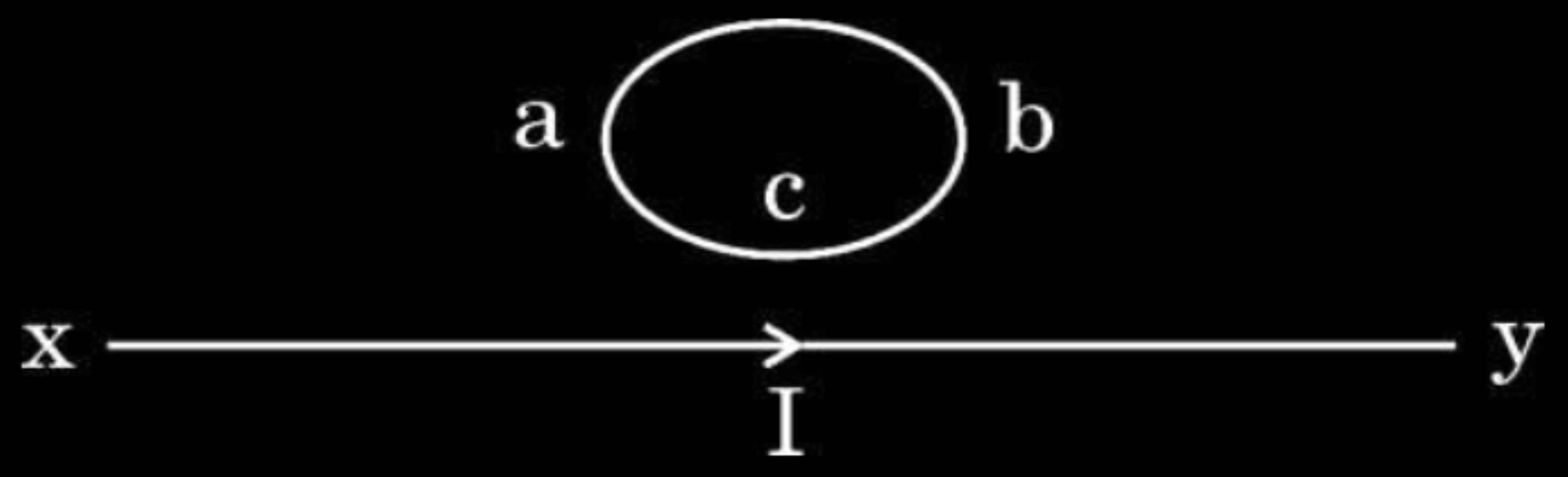
4. The magnetic flux (in SI units) through a coil varies with time as $\phi = 3t^2 + 4t + 7$. The ratio of emf induced in the coil at t = 2 s to that at t = 1 s will be :

(a) 2 (b) 0·8

(c) 1·6 (d) 4

(c) 1.6

11. The direction of induced current in the loop abc is :



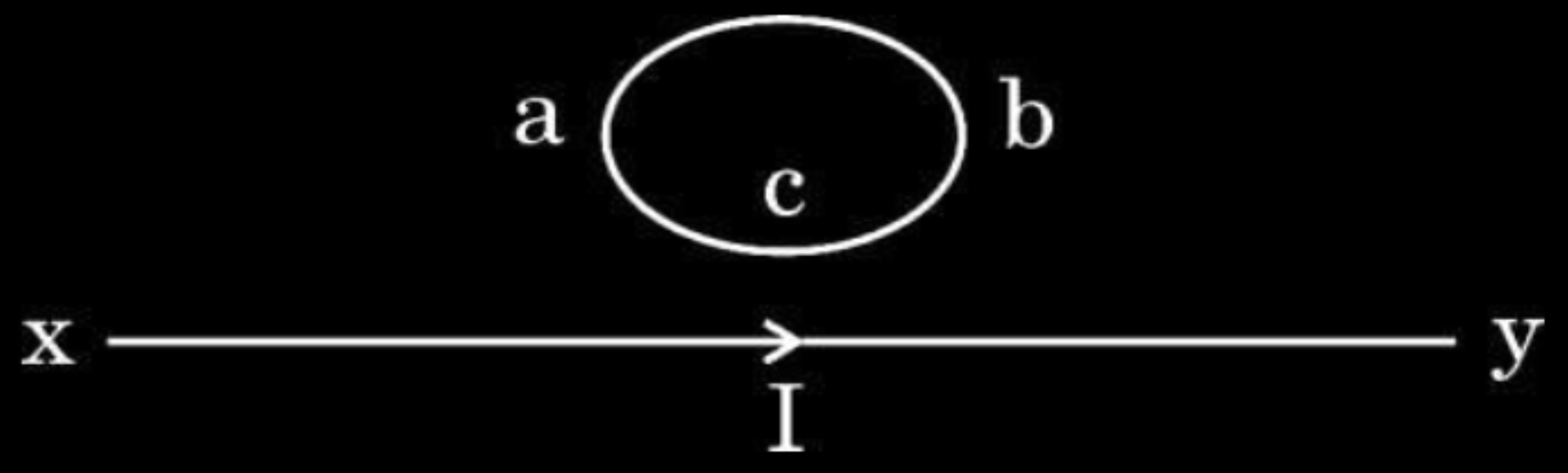
(a) along abc if I decreases

(b) along acb if I increases

(c) along abc if I is constant

(d) along abc if I increases

11. The direction of induced current in the loop abc is :



(a) along abc if I decreases

(b) along acb if I increases

(c) along abc if I is constant

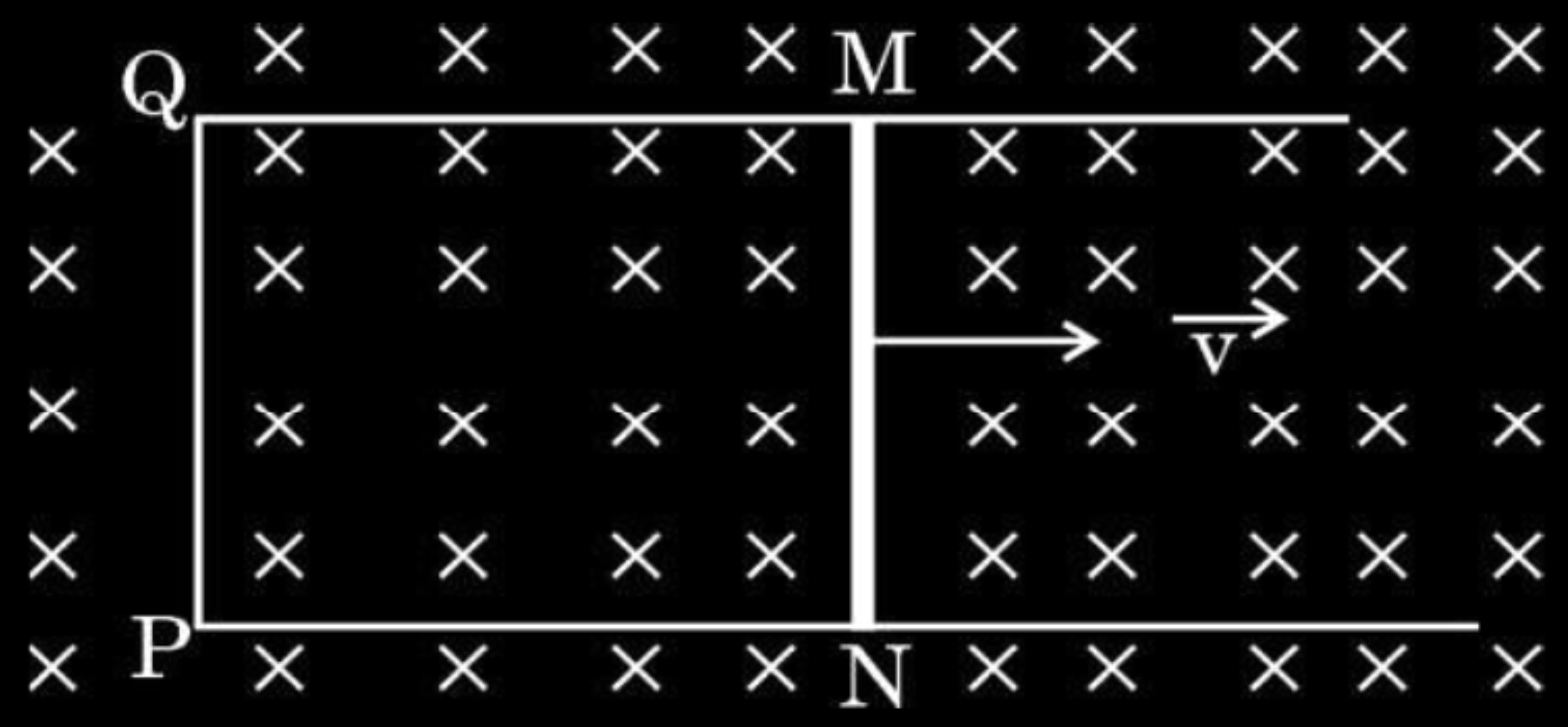
(d) along abc if I increases

(d) along abc if I increases

30. A rectangular conductor MNPQ with a movable arm MN (resistance r) is kept in a uniform magnetic field as shown in the figure. Resistance of arms MQ, QP and PN are negligible. Obtain the expression for the : *3*

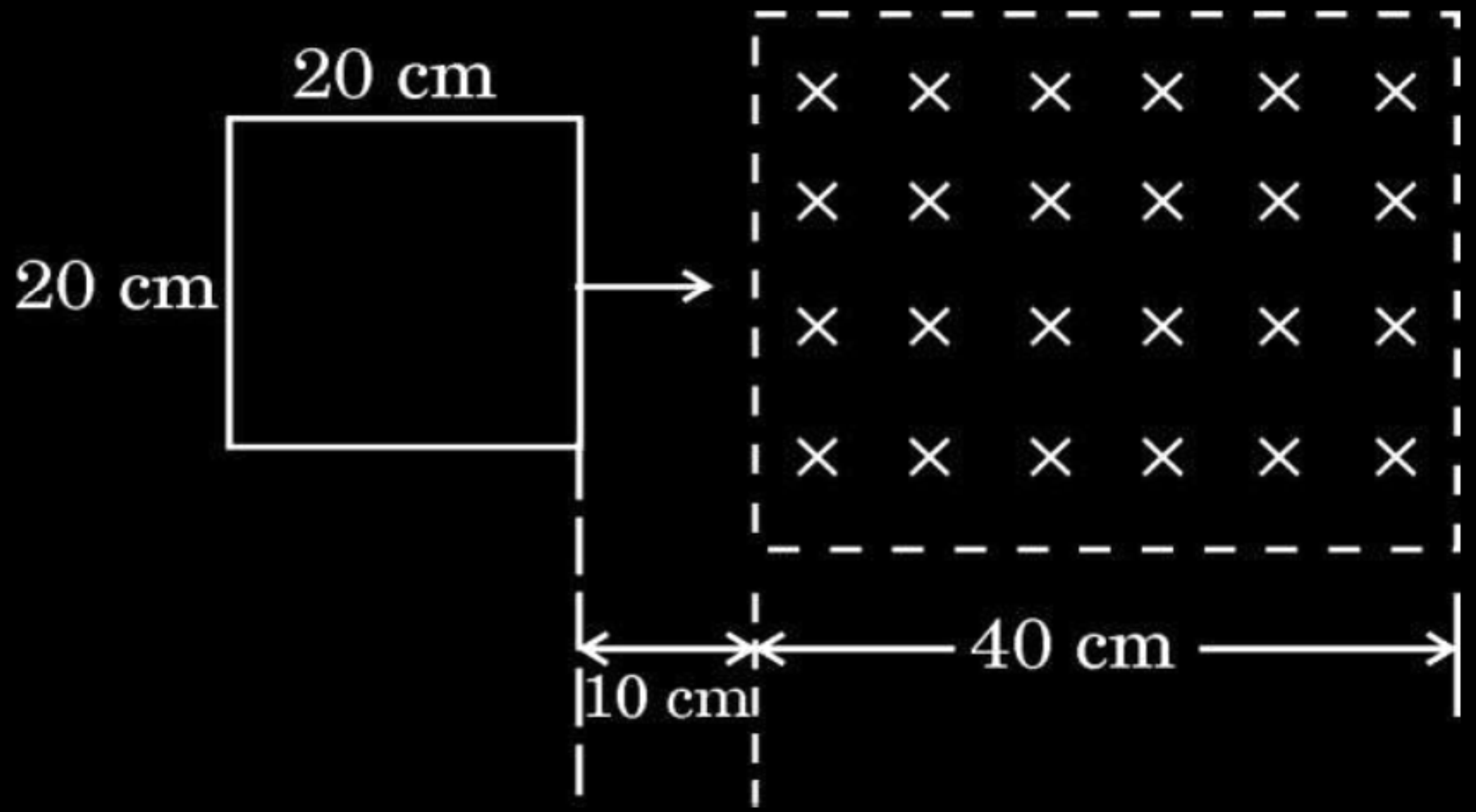
(a) current induced in the loop specifying its direction, and

(b) power required to move the arm.

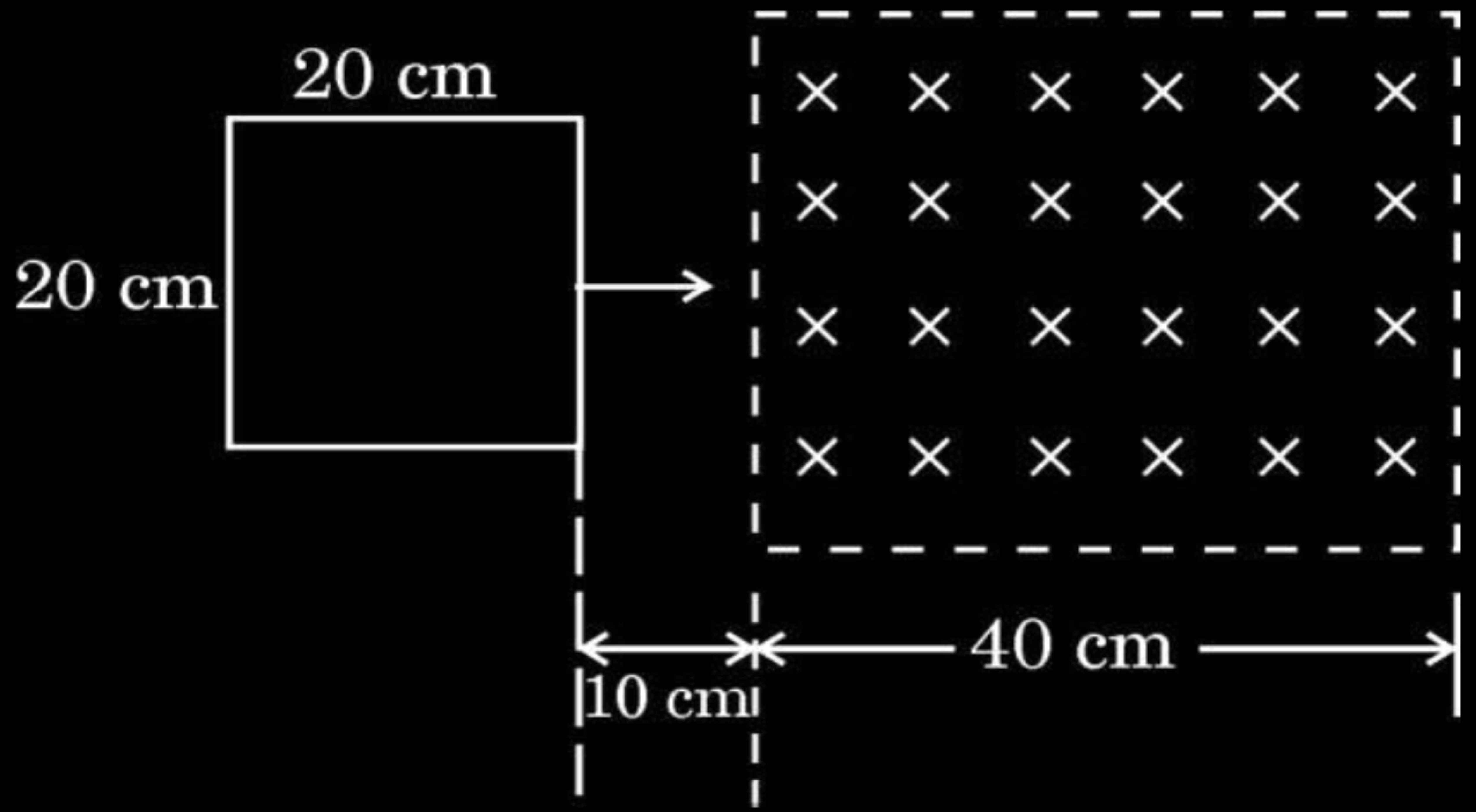


(a)(i) Magnetic flux enclosed by the loop MNPQ $\phi_B = Blx$ Since x is changing with time, the rate of change of flux ϕ_B will induce an e.m.f. given by $\varepsilon = \frac{-d\phi_B}{dt} = -\frac{d}{dt}(Blx)$	½
$\lvert\varepsilon\rvert = Blv$	½
Induced current in the loop $I = \frac{\varepsilon}{r} = \frac{Blv}{r}$	½
(ii) Direction - anticlockwise or along NMQP	½
(b) Power required $P = F\,v = IBlv$	½
$P = \frac{B^2 l^2 v^2}{r}$	½

(b) A square loop of side 20 cm starts moving at t = 0 with a velocity of 5 cm/s towards a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop. *3*

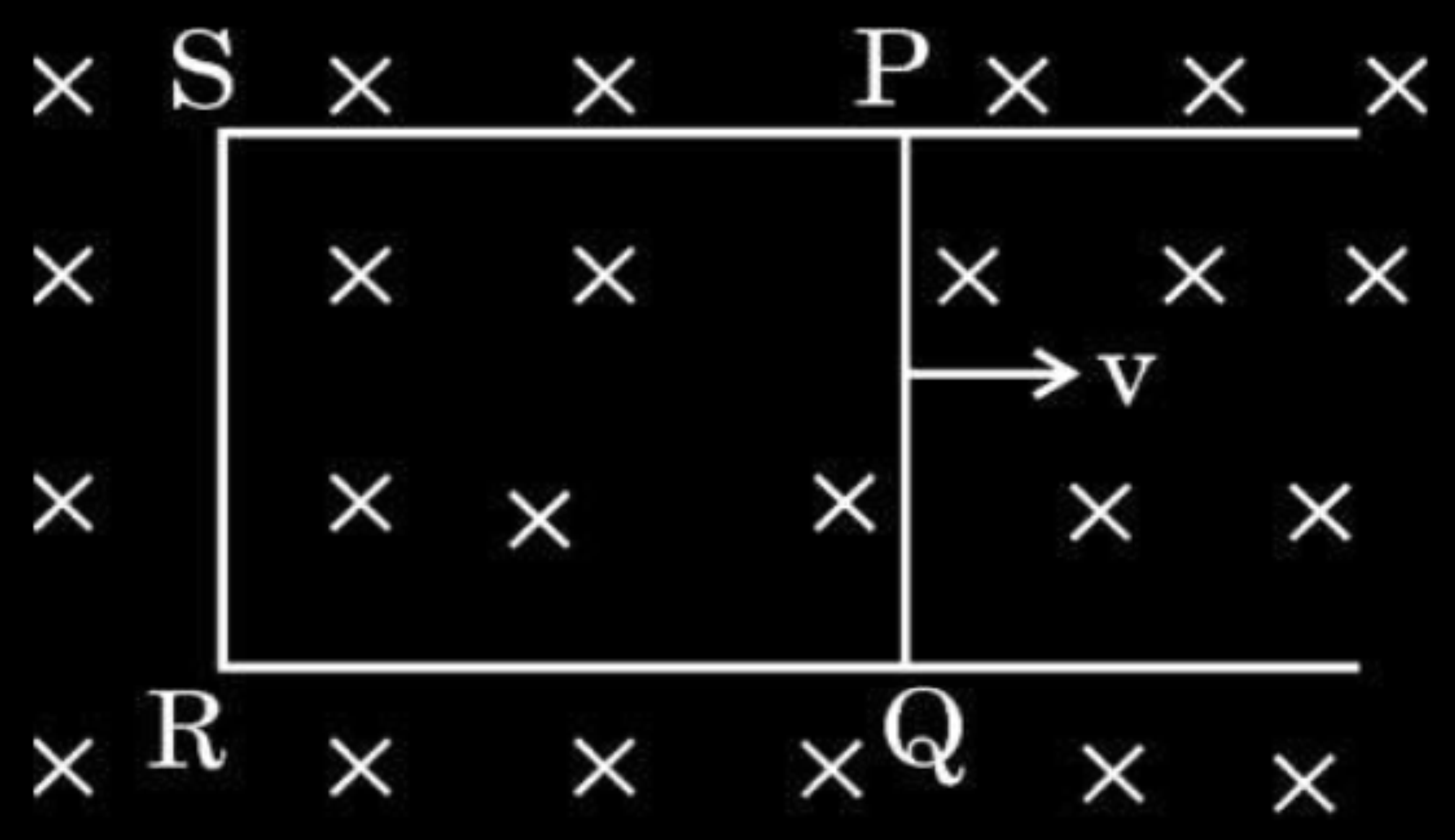


(b) A square loop of side 20 cm starts moving at t = 0 with a velocity of 5 cm/s towards a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop. *3*



(b) $t = \frac{l}{v}$ Induced e.m.f. during time interval $2\,s < t \leq 6\,s$ & $10\,s \leq t \leq 14\,s$	½ + ½

28. The figure shows a rectangular conductor PQRS in which the arm PQ of length 10 cm and resistance 0·4 Ω is free to move. It is kept in a uniform magnetic field B = 0·2 T acting perpendicular into the plane of PQRS. If arm PQ is moved with a velocity v of 5 cm/s as shown, find : *3*



(a) the current induced in the loop, and

(b) the power required to move the arm.

(Resistances of arms PS, SR and RQ are negligible.)

(a)	
$i = \dfrac{Blv}{R}$	½
$= \dfrac{0.2\times(10\times10^{-2})\times(5\times10^{-2})}{0.4}$	½
$i = 2.5\times10^{-3}\,A$	½
(b) $P = i^2 R$	½
$= \left(2.5\times10^{-3}\right)^2 \times 0.4$	½
$P = 2.5\times10^{-6}\,W$	½

29. A rectangular loop of sides 25 cm and 20 cm is lying in x-y plane. It is subjected to a magnetic field $\vec{B} = (5t^2 + 2t + 10)\hat{k}$, where B is in Tesla and t is in seconds. If the resistance of the loop is 4 Ω, find the emf induced and the induced current in the loop at t = 5 s. *3*

$\phi_B = \vec{B} \cdot \vec{A} = BA\cos 0^o$	½
$= (5t^2 + 2t + 10) \times (25\times10^{-2} \times 20\times10^{-2})$	
$\lvert\varepsilon\rvert = \left\lvert -\dfrac{d\phi_B}{dt} \right\rvert$	
$= 5\times10^{-2} \times (10t + 2)$	½
At $t = 5$ sec;	
$\varepsilon = (5\times10^{-2}) \times (52)$	½
$\varepsilon = 2.6$ V	½
$I = \dfrac{\varepsilon}{R}$	½
$= \dfrac{2.6}{4}$	
$I = 0.65$ A	½

1. Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current :

(a) increases in A and decreases in B.

(b) decreases in both A and B.

(c) increases in both A and B.

(d) remains same in both A and B.

1. Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current :

(a) increases in A and decreases in B.

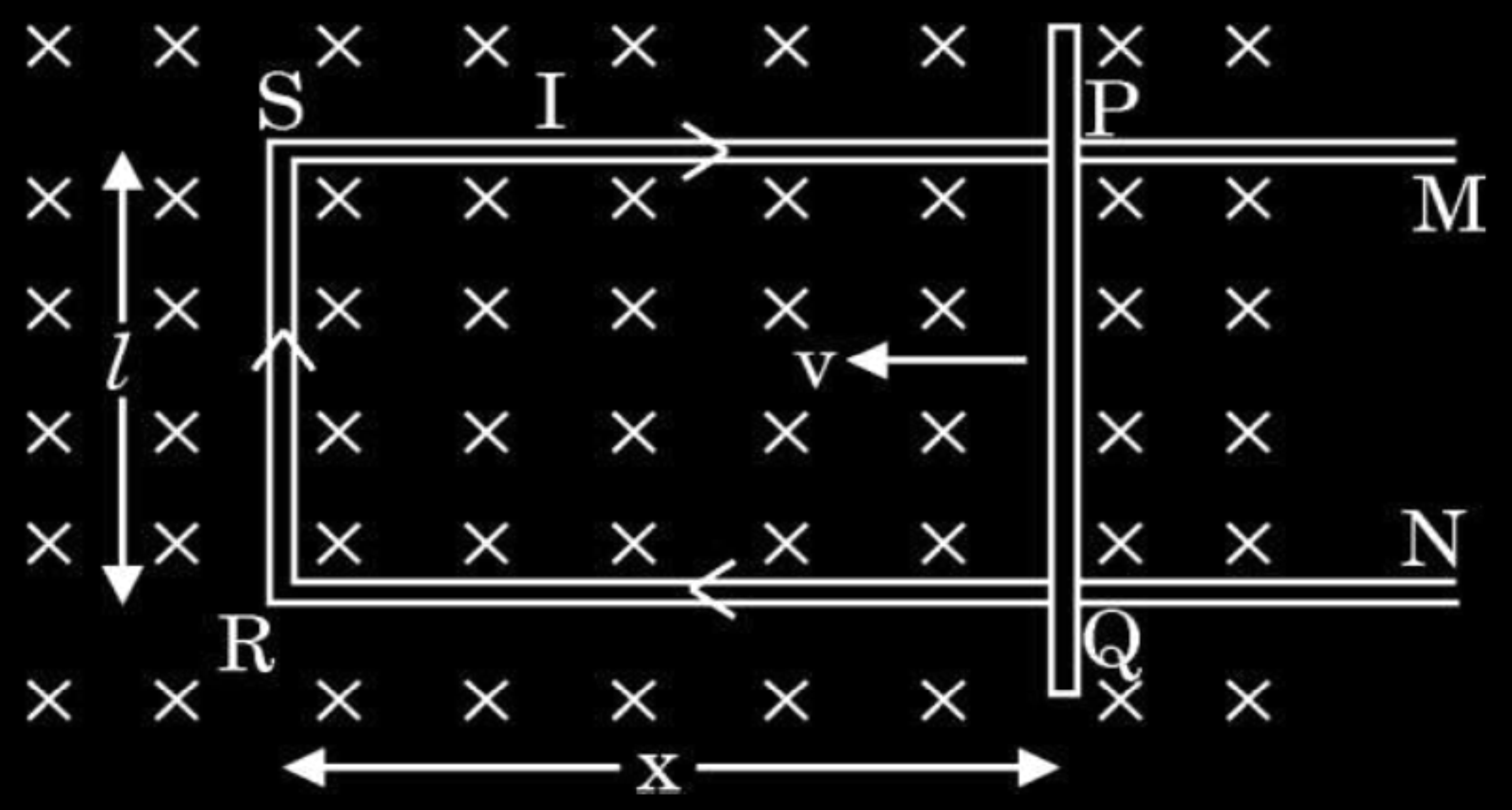
(b) decreases in both A and B.

(c) increases in both A and B.

(d) remains same in both A and B.

(b) Decreases in both A and B.

7. Figure shows a rectangular conductor PSRQ in which movable arm PQ has a resistance 'r' and resistance of PSRQ is negligible. The magnitude of emf induced when PQ is moved with a velocity $\vec{v}$ does ***not*** depend on :



(a) magnetic field ($\vec{B}$) (b) velocity ($\vec{v}$)

(c) resistance (r) (d) length of PQ

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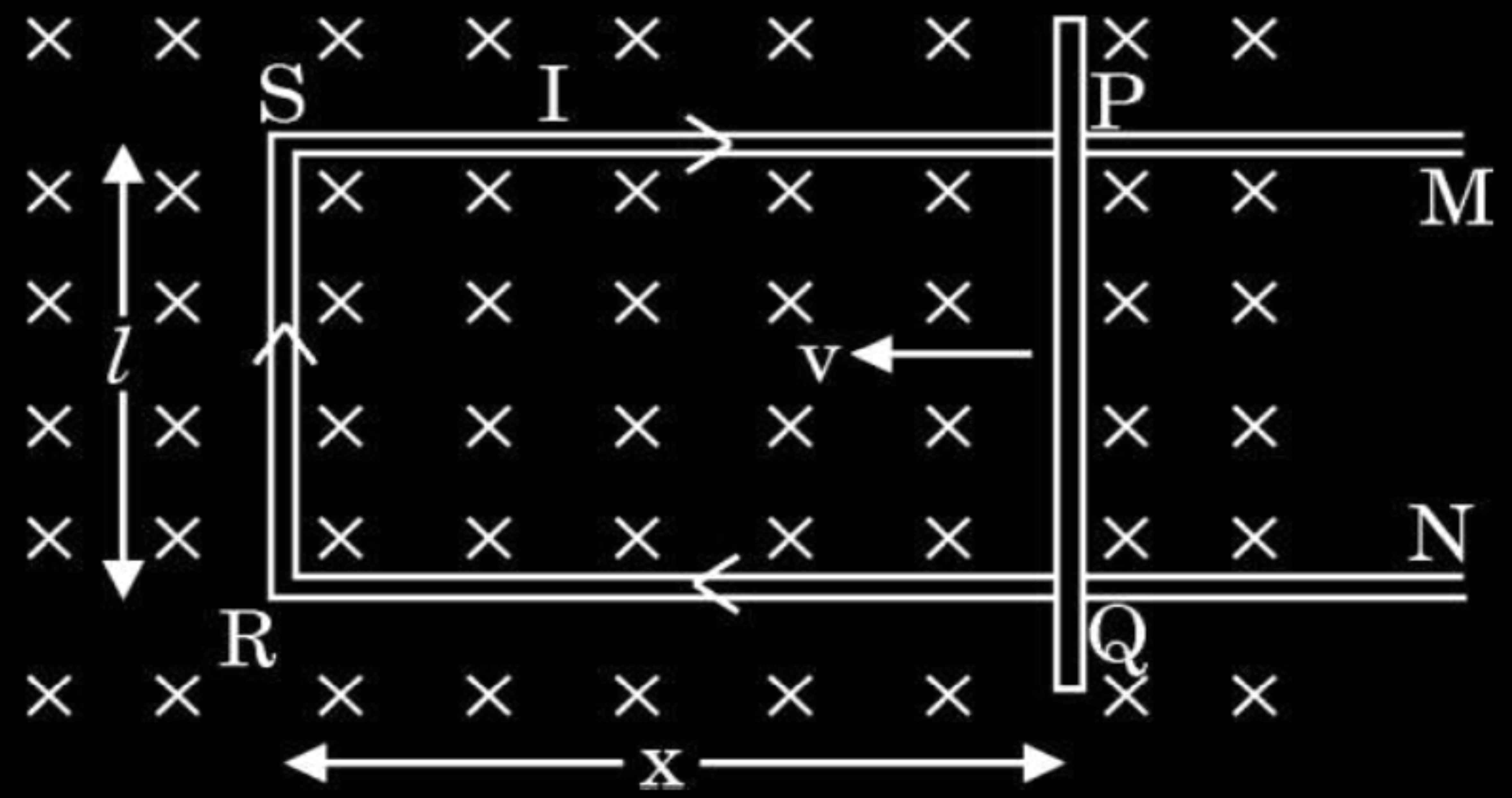


(a) magnetic field ($\vec{B}$)

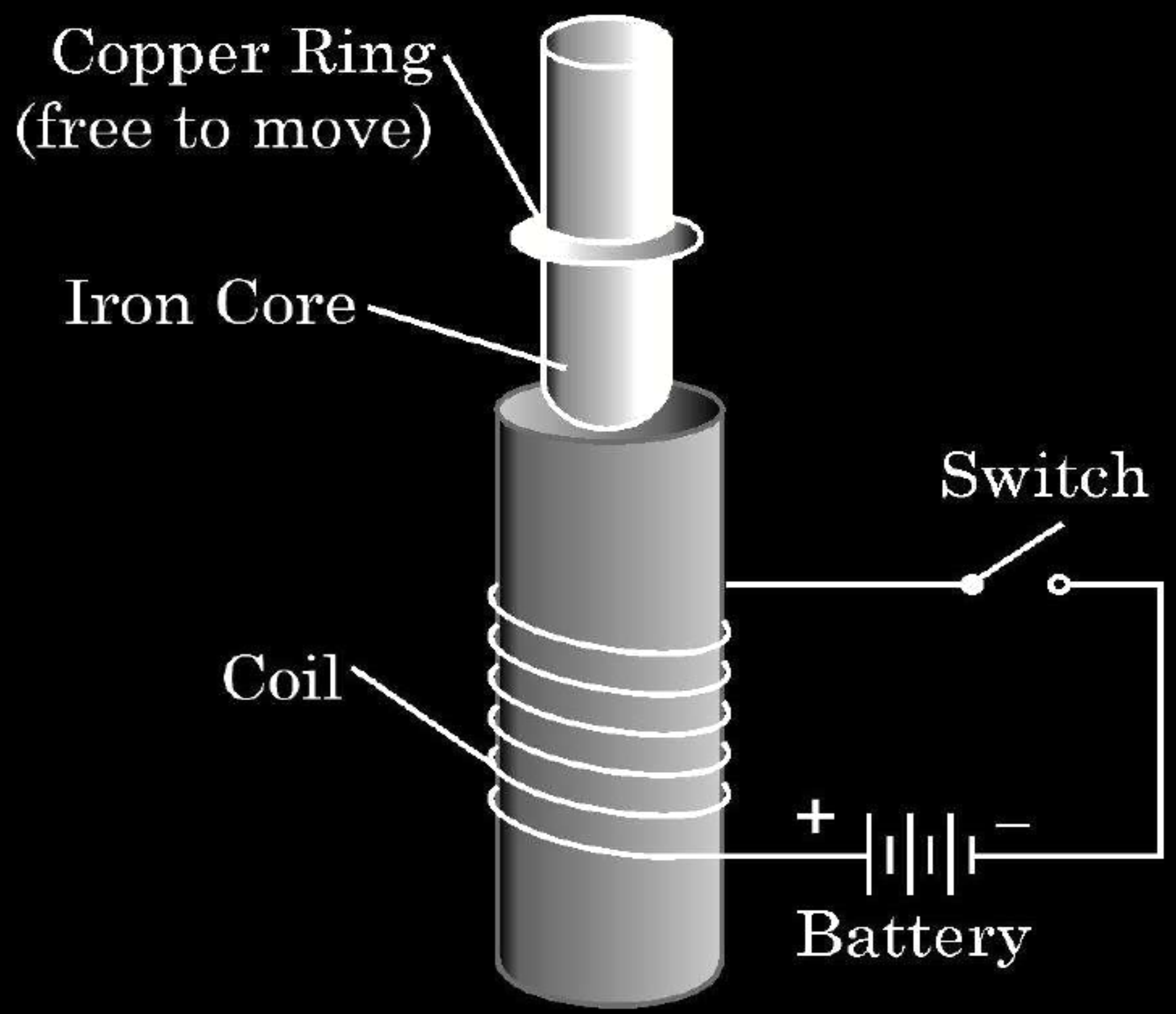
(b) velocity ($\vec{v}$)

(c) resistance (r)

(d) length of PQ

(c) Resistance (r)

34. (a) Consider the experimental set up shown in the figure. This jumping ring experiment is an outstanding demonstration of some simple laws of Physics. A conducting non-magnetic ring is placed over the vertical core of a solenoid. When current is passed through the solenoid, the ring is thrown off.



Answer the following questions :

(i) Explain the reason of jumping of the ring when the switch is closed in the circuit.

(ii) What will happen if the terminals of the battery are reversed and the switch is closed ? Explain.

(iii) Explain the two laws that help us understand this phenomenon. **4**

(i) The direction of induced current in the ring is such that the polarity developed in the ring is same as that of the polarity on the face of the coil , hence it will jump up due to repulsive force.	1
(ii) The polarity of the induced current in the ring will get reversed on changing the terminals of the battery, so the ring will jump again.	1
(iii) **Lenz's law** It states that the polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it.	1
Faraday's law of EMI Whenever there is change in magnetic flux through a coil, an emf is induced. The magnitude of the induced emf in a coil is equal to the time rate of change of magnetic flux through the coil.	1

4. A square shaped coil of side 10 cm, having 100 turns is placed perpendicular to a magnetic field which is increasing at 1 T/s. The induced emf in the coil is **1**

(A) 0.1 V (B) 0.5 V

(C) 0.75 V (D) 1.0 V

4. A square shaped coil of side 10 cm, having 100 turns is placed perpendicular to a magnetic field which is increasing at 1 T/s. The induced emf in the coil is **1**

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(D) 1. 0 V

28. State Lenz's law. Show that it is consistent with the principle of conservation of energy. *3*

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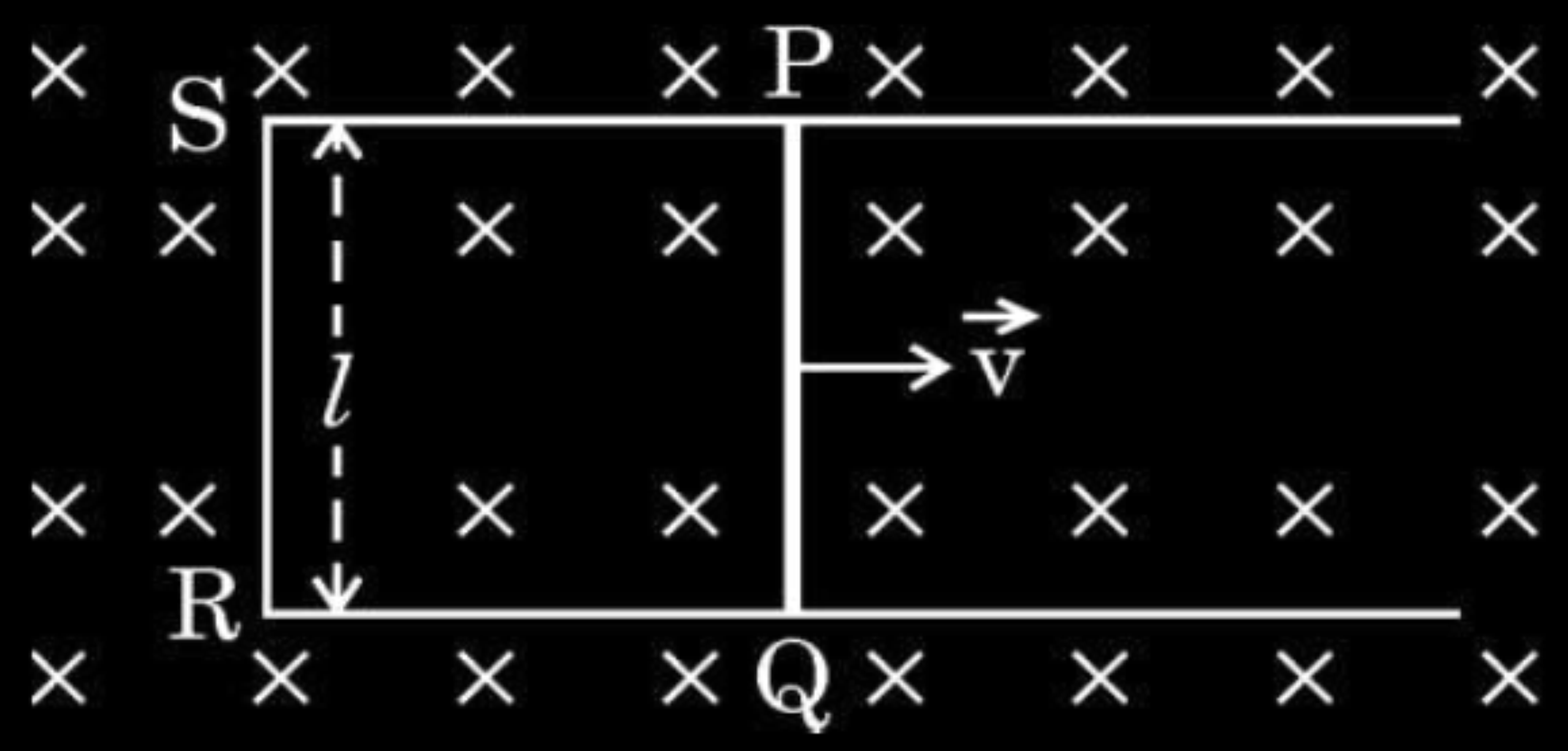
Lenz's Law The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.	1
The e.m.f. induced opposes the causes of its change. So in order to change the magnetic flux work is always needed to be done. This work done gets converted to induced e.m.f. Hence this is consistent with conservation of energy.	2

26. The figure shows a rectangular conductor PQRS with a movable arm PQ, kept in a uniform magnetic field $\vec{B}$ pointing into the page.



(i) PQ is moved towards the right with a velocity $\vec{v}$. Obtain the expression for the emf developed across PQ.

(ii) If r is the resistance of PQRS, find the force required to move PQ with constant velocity $\vec{v}$. *3*

(i) $\phi_B = B\,l\,x$	½
$\varepsilon = -\dfrac{d\phi_B}{dt}$	½
$= -\dfrac{d}{dt}(B\,l\,x)$	
$= -B\,l\,\dfrac{dx}{dt}$	
$\varepsilon = -B\,l\,v$	½
(ii) $i = \dfrac{\varepsilon}{r}$	½
$\vec{F} = i\,(\vec{l} \times \vec{B})$	½
$F = \dfrac{\varepsilon}{r}\; i\,B = \dfrac{B\,l\,v}{r}(l\,B)$	
$F = \dfrac{B^2\,l^2\,v}{r}$	½

30. Answer the following, giving reasons : *3*

(a) Induced emf does not always produce induced current.

(b) The motion of a copper plate is damped when it is allowed to oscillate between pole pieces of a strong magnet.

(c) No power is consumed in an ac circuit containing a pure inductor.

30. Answer the following, giving reasons : *3*

(a) Induced emf does not always produce induced current.

(b) The motion of a copper plate is damped when it is allowed to oscillate between pole pieces of a strong magnet.

(c) No power is consumed in an ac circuit containing a pure inductor.

(a) Even in the open loop e.m.f. can be induced but since circuit is open no current can flow.	1
(b) When copper plate oscillates between poles of a strong magnet eddy currents are generated in the plate opposing the change in flux.	1
(c) Power consumed by a pure inductor oscillates. For first half of the cycle it is positive and for the next half it is negative. So for a complete cycle it is zero.	1

26. A horizontal straight metallic rod of length 4 m is held at some height above the surface of Earth, in east-west direction. If it is allowed to fall from rest, find the :

(a) emf induced in the rod 2 s after it starts falling,

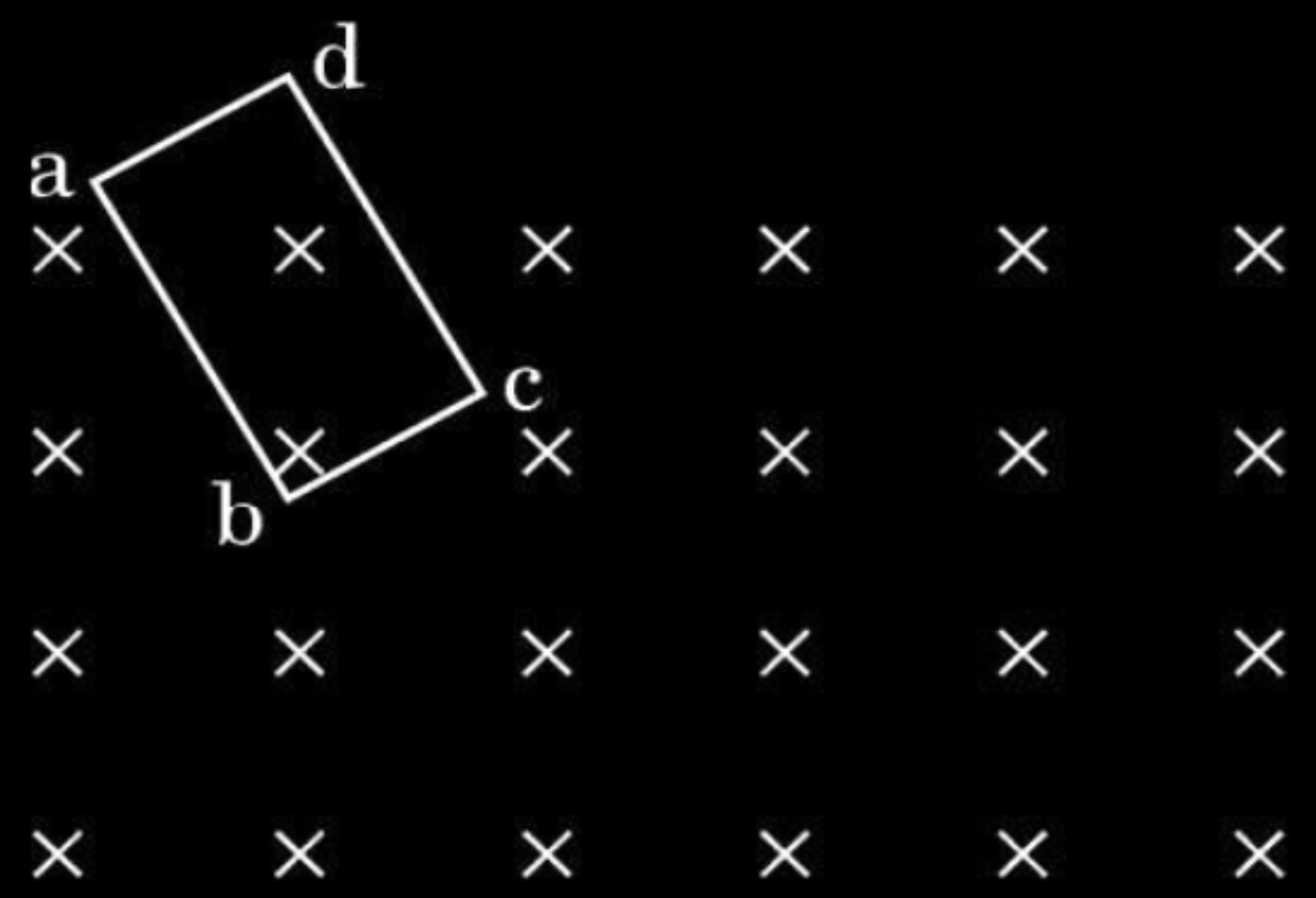
(b) polarity of the emf induced, and

(c) the end of the rod which is at the higher potential.

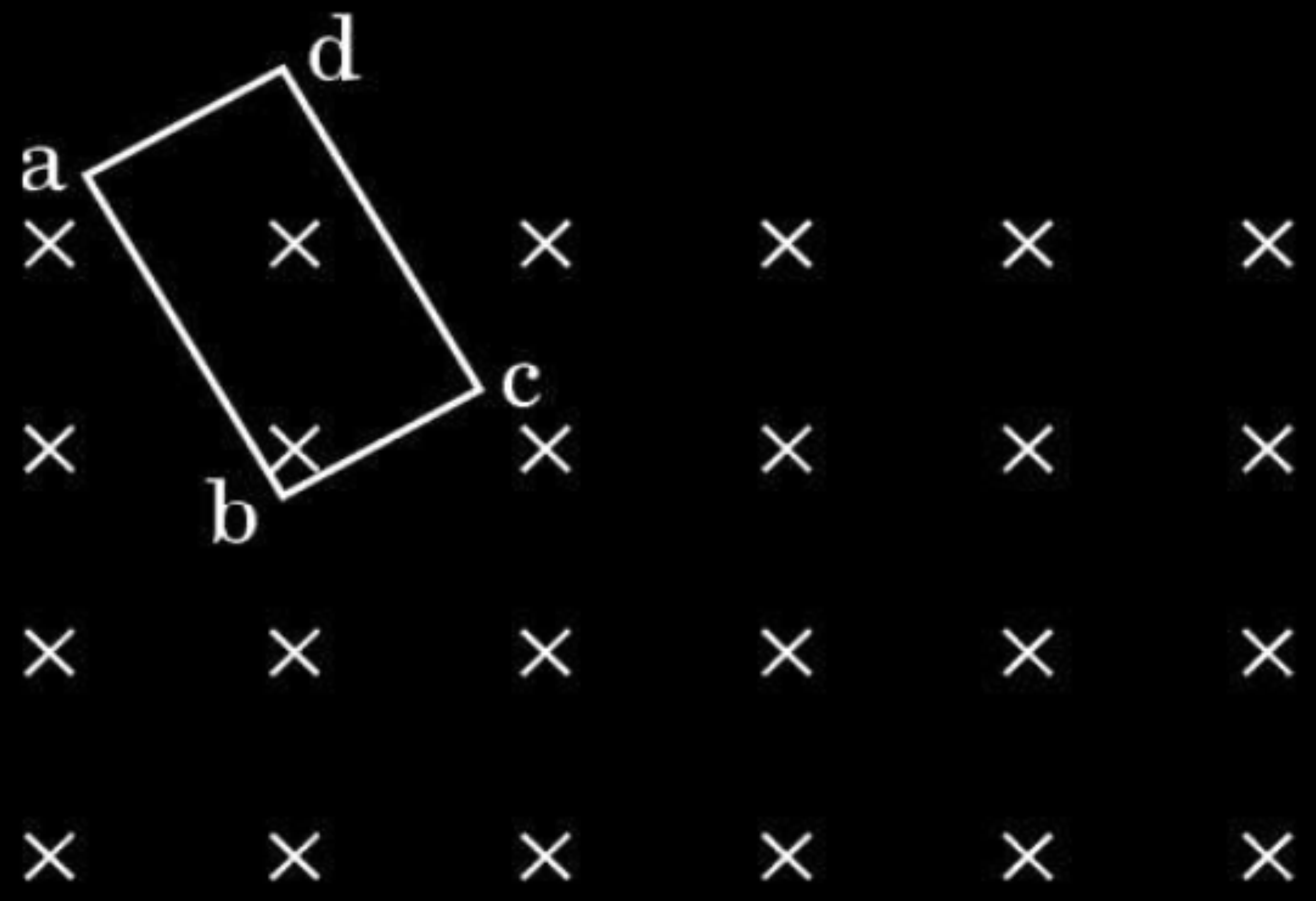
The horizontal component of the Earth's magnetic field at the place is $0{\cdot}3 \times 10^{-4}$ Wb/m^2 and take $g = 10$ m/s^2. *3*

(a)	
$\varepsilon = Blv$	½
$B = 0.3\times10^{-4}Wbm^{-2}$	
$v = 20ms^{-1}$	
$l = 4m$	½
$v = 0+10\times2$	
$v = 20m/s$	
$\varepsilon = 0.3\times10^{-4}\times20\times4$	½
$= 24\times10^{-4}V = 2.4mV$	½
(b) West to east	½
(c) East end of the rod is at higher potential.	½

26. (a) State Lenz's law. Determine the direction of the induced current when a rectangular conducting loop abcd in xy-plane is moved into a region of magnetic field which is directed along z-axis. *3*



26. (a) State Lenz's law. Determine the direction of the induced current when a rectangular conducting loop abcd in xy-plane is moved into a region of magnetic field which is directed along z-axis. *3*



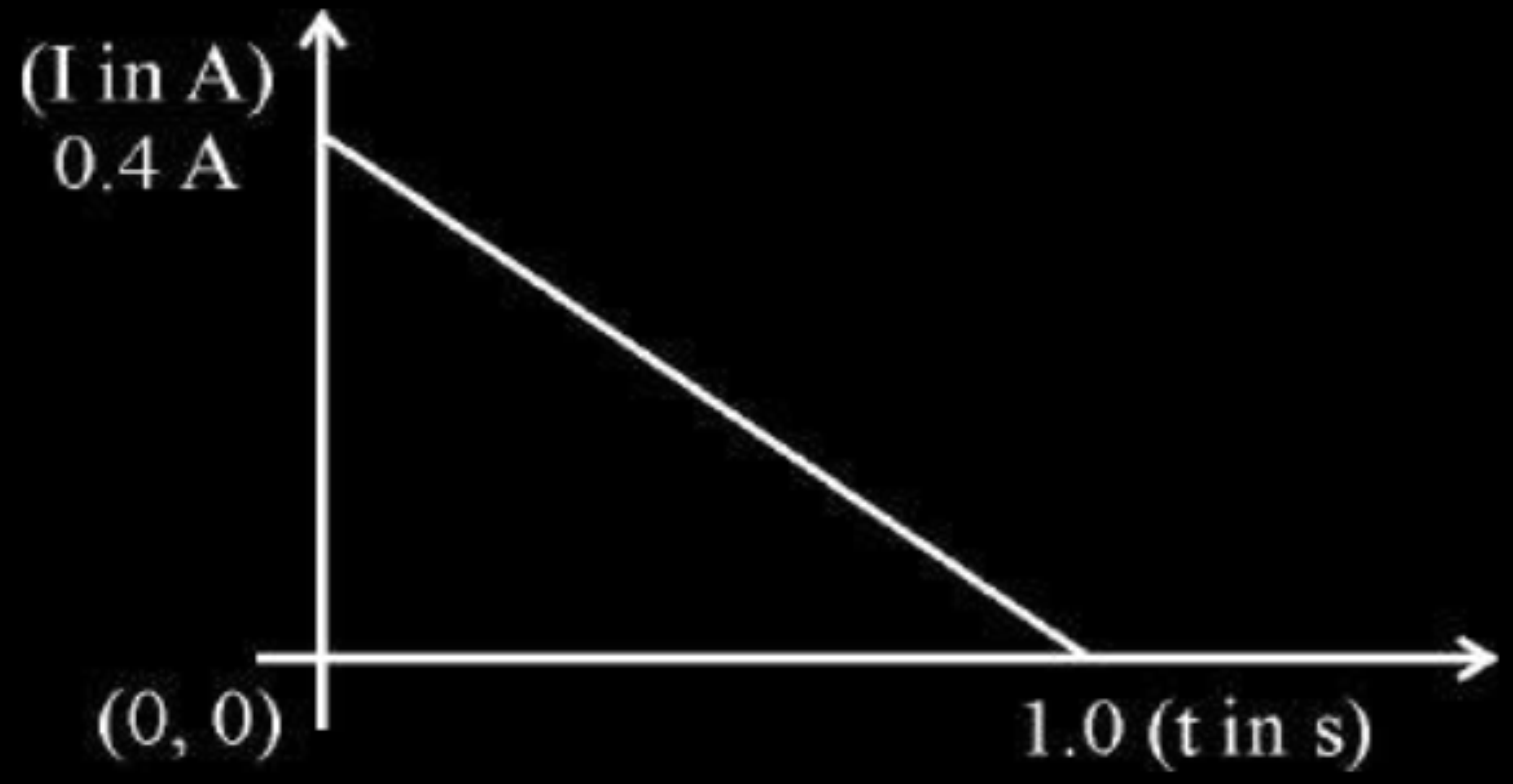
(a) Lenz's law " the polarity of induced e.m.f is such that it tends to produce a current which opposes the change in magnetic flux that produced it"  Direction of induced current is abcd/anticlockwise	2 1

(b) Two identical circular loops, one of copper and the other of aluminium are rotated about their diameters with the same angular speed in a magnetic field directed perpendicular to their axes of rotation. Compare (i) the emf induced, and (ii) the current in the two loops. Justify your answers. *3*

(b) Two identical circular loops, one of copper and the other of aluminium are rotated about their diameters with the same angular speed in a magnetic field directed perpendicular to their axes of rotation. Compare (i) the emf induced, and (ii) the current in the two loops. Justify your answers. *3*

(b) (i)Same e.m.f induced in both cases because they are identical and moving with same angular speed in same magnetic field.	1½
(ii) Induced current is more in copper loops, as its resistance is lesser than that of aluminum.	1½

34. When a conducting loop of resistance 10 Ω and area 10 cm^2 is removed from an external magnetic field acting normally, the variation of induced current in the loop with time is shown in the figure.



Find the

(i) total charge passed through the loop.

(ii) change in magnetic flux through the loop.

(iii) magnitude of the magnetic field applied. **3**

(i) Total charge passed through the loop (Q)	
$Q =$ area under the I-t graph	½
$= \frac{1}{2} \times 0.4 \times 1$ coulomb $= 0.2C$	½
(ii) Change in magnetic flux	
Total charge passing $= \left(\frac{\text{change in magnetic flux}}{R} \right)$	½
Change in magnetic flux $= [R \times 0.2C]$	
$= [10 \times 0.2]$ Wb	
$= 2$Wb	½
(iii) Magnitude of magnetic field applied	
Let B be the magnitude of the magnetic field applied	
Initial magnetic flux $= \mathrm{B} \times (10 \times 10^{-4})\ Wb$	
Final magnetic flux $=$ zero	
Change in magnetic flux $= (\mathrm{B} \times 10^{-3} - 0) = 2$	½
$\Rightarrow B = 2 \times 10^{3}\ Wb / m^{2}$	½
(**Note:** Award two marks to a student who only calculates charge and not able to calculate correctly the remaining two parts of this question)	

(a) A conductor of length 'l' is rotated about one of its ends at a constant angular speed 'ω' in a plane perpendicular to a uniform magnetic field B. Plot graphs to show variations of the emf induced across the ends of the conductor with (i) angular speed ω and (ii) length of the conductor l.

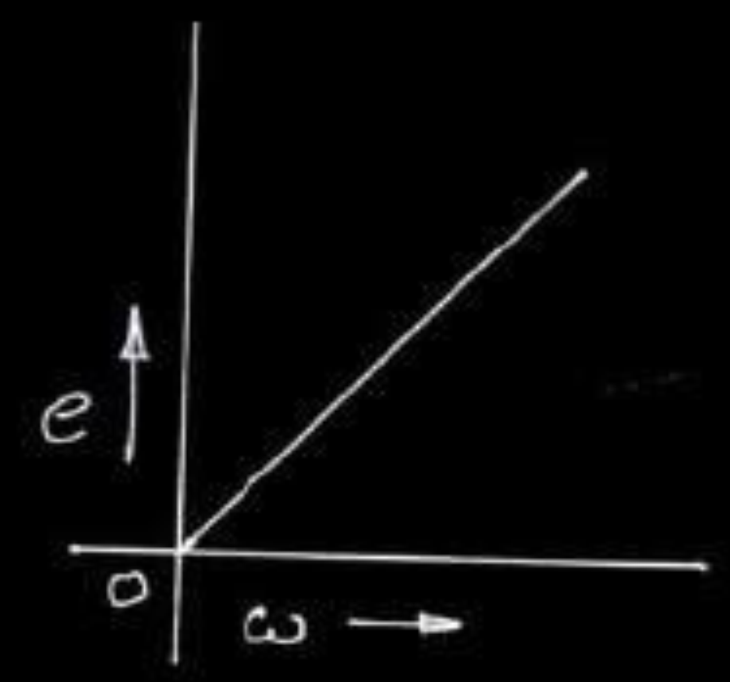
(b) Two concentric circular loops of radius 1 cm and 20 cm are placed coaxially.

(i) Find mutual inductance of the arrangement.

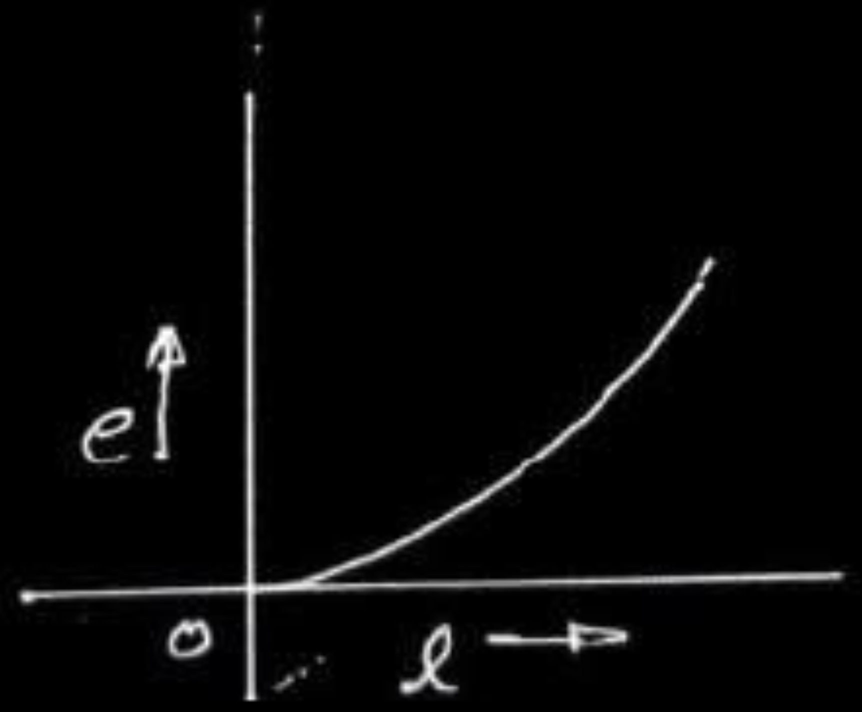
(ii) If the current passed through the outer loop is changed at a rate of 5 A/ms, find the emf induced in the inner loop. Assume the magnetic field on the inner loop to be uniform. **5** e.m.f. induced in the rod $= \left| -\frac{d\phi}{dt} \right|$ ½

$\therefore e = \frac{1}{2} Bl^2\omega$ ½

(i)



(ii)



½+½

(i) Imagine a current I to flow through the larger coil.

Magnetic flux linked with linked with the smaller coil $= MI$

$= B_{centre} \times \pi \times 10^{-4}\,\text{Wb}$

$= \frac{\mu_0 I}{2 \times 20 \times 10^{-2}} \times \pi \times 10^{-4}\,\text{Wb}$

$\Rightarrow M = \frac{\mu_0 \pi}{4} \times 10^{-3}\,\text{H}$

$= \frac{4\pi \times 10^{-7} \pi \times 10^{-3}}{4} H$

$= 9.9 \times 10^{-10} H$

$= 10^{-9} H$ ½

(ii)

$e.m.f.\ induced = -M \frac{dI}{dt}$ ½

$= -10^{-9} \times \frac{5}{10^{-3}} V$ ½

$= -5 \times 10^{-6} V$ ½

Alternatively

$e.m.f.\ induced = -\frac{d\phi}{dt}$

$= -\frac{d}{dt}(B_c A_i) = -A_i \frac{dB_c}{dt}$ ½

$= -A_i \frac{d}{dt}\left(\frac{\mu_0 I}{2R}\right)$

$= -\frac{A_i \mu_0}{2R} \frac{dI}{dt}$

$= \frac{-\pi \times 10^{-4} \times 4\pi \times 10^{-7} \times 5}{2 \times (20 \times 10^{-2}) \times 10^{-3}} V$ ½

$= -\frac{20\pi^2 \times 10^{-6}}{2 \times 20} V$

$= -5 \times 10^{-6} V$ ½

17. Two identical coils, one of copper and the other of aluminium are rotated with the same angular speed in an external magnetic field. In which of the two coils will the induced current be more ? *1*

17. Two identical coils, one of copper and the other of aluminium are rotated with the same angular speed in an external magnetic field. In which of the two coils will the induced current be more ? *1*

Copper

What happens when a block of metal is kept in a varying magnetic field ? *1*

What happens when a block of metal is kept in a varying magnetic field ? *1*

Eddy currents are produced in metal block / block gets heated

17. A conducting rod of length l is kept parallel to a uniform magnetic field $\vec{B}$. It is moved along the magnetic field with a velocity $\vec{v}$. What is the value of emf induced in the conductor ? *1*

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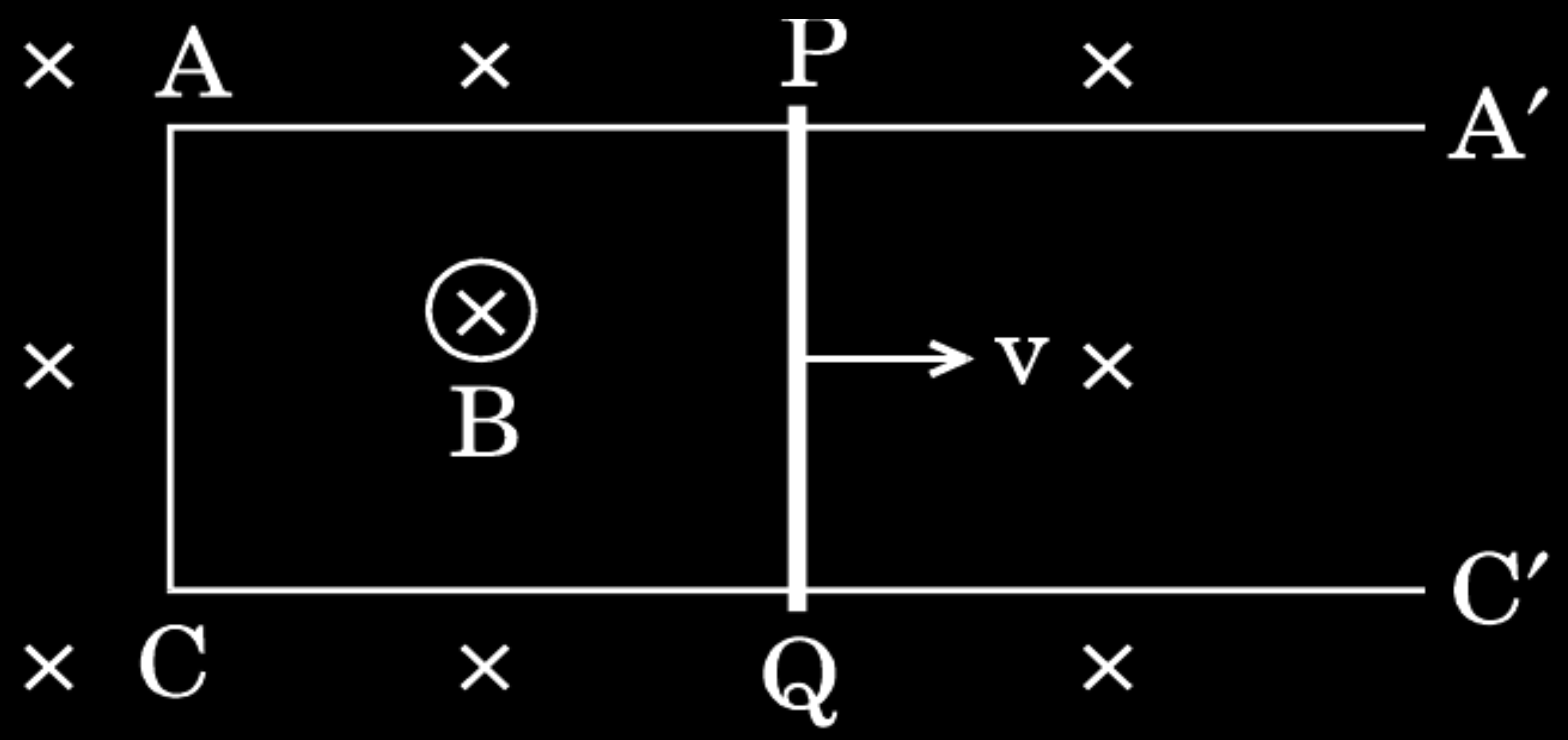
| Zero

(b) A conducting rod PQ of length 20 cm and resistance 0·1 Ω rests on two smooth parallel rails of negligible resistance AA′ and CC′. It can slide on the rails and the arrangement is positioned between the poles of a permanent magnet producing uniform magnetic field B = 0·4 T. The rails, the rod and the magnetic field are in three mutually perpendicular directions as shown in the figure. If the ends A and C of the rails are short circuited, find the

(i) external force required to move the rod with uniform velocity v = 10 cm/s, and

(ii) power required to do so. *5*



(b) (i) $$F=BIl$$ $$I=\frac{E}{R}=\frac{B\upsilon l}{R}$$ $$F=\frac{B^2\upsilon l^2}{R}$$ $$=\frac{0.4\times 0.4\times 0.1\times 0.2\times 0.2}{0.1}$$ $= 6.4 \times 10^{-3}$ N $$P=F.\upsilon=6.4\times 10^{-3}\times 0.1$$ $$=.64\times 10^{-3}W$$	 ½ ½ ½ ½

CBSE 2019

(a) A metallic rod of length 'l' and resistance 'R' is rotated with a frequency 'v' with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 'l', about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field 'B' parallel to the axis is present everywhere.

(i) Derive the expression for the induced emf and the current in the rod.

(ii) Due to the presence of current in the rod and of the magnetic field, find the expression for the magnitude and direction of the force acting on this rod.

(iii) Hence, obtain an expression for the power required to rotate the rod.

(b) A copper coil is taken out of a magnetic field with a fixed velocity. Will it be easy to remove it from the same field if its ohmic resistance is increased ?

(i)

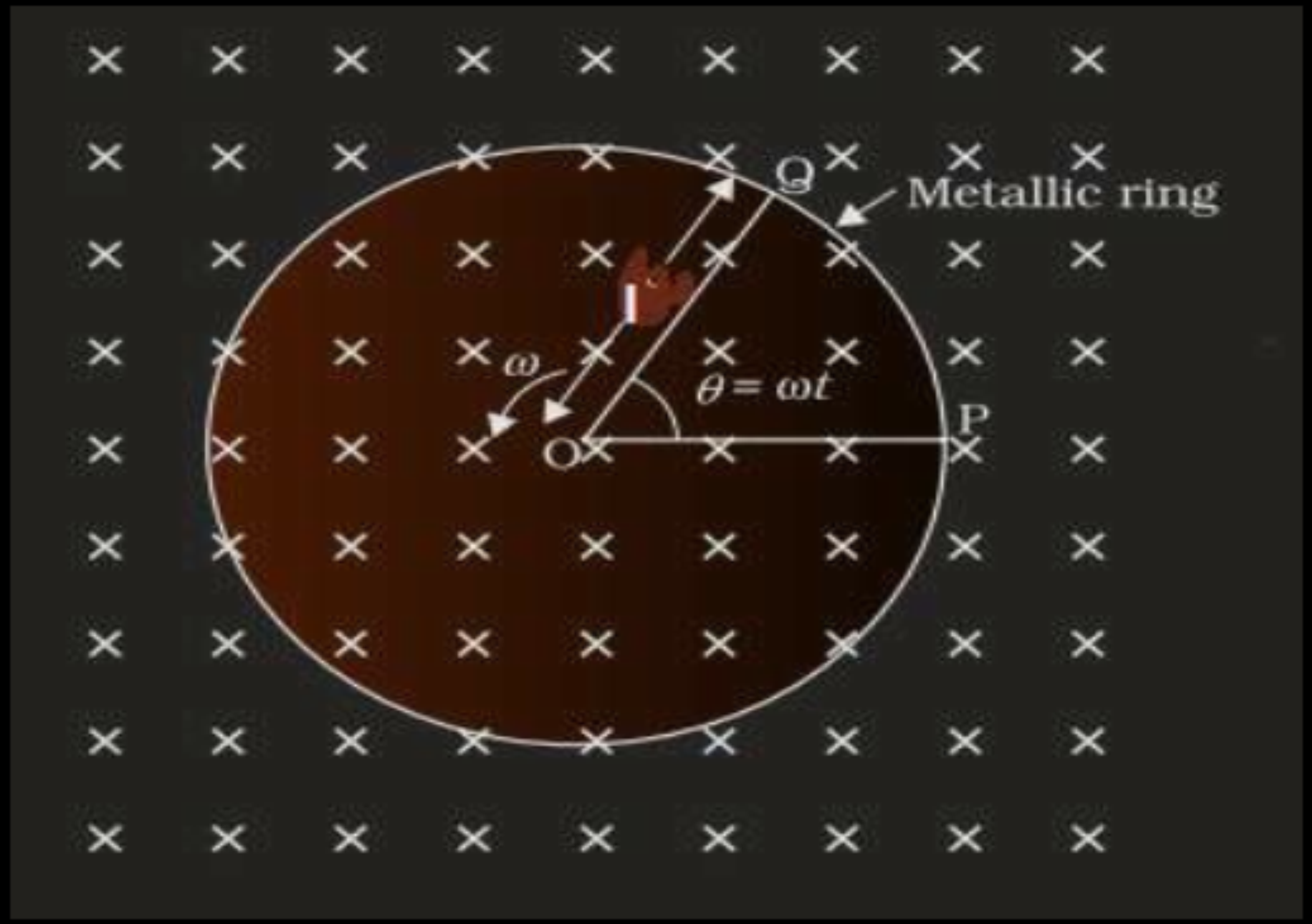
$$\varepsilon' = Blv$$

$$\varepsilon = Bl\frac{v}{2}$$

$$\varepsilon = Bl\left(\frac{l\omega}{2}\right)$$

$$\varepsilon = \frac{bl^2\omega}{2}$$ [Student may use any method to arrive at this result] 1

$$I = \frac{\varepsilon}{R} = \frac{bl^2\omega}{2R}$$ 1



(ii) F= I (l x b) ½

$$F = \frac{bl^3\omega B}{2R}$$ ½

Direction of force is perpendicular to both $\vec{l}$ and $\vec{B}$ ½

iii)

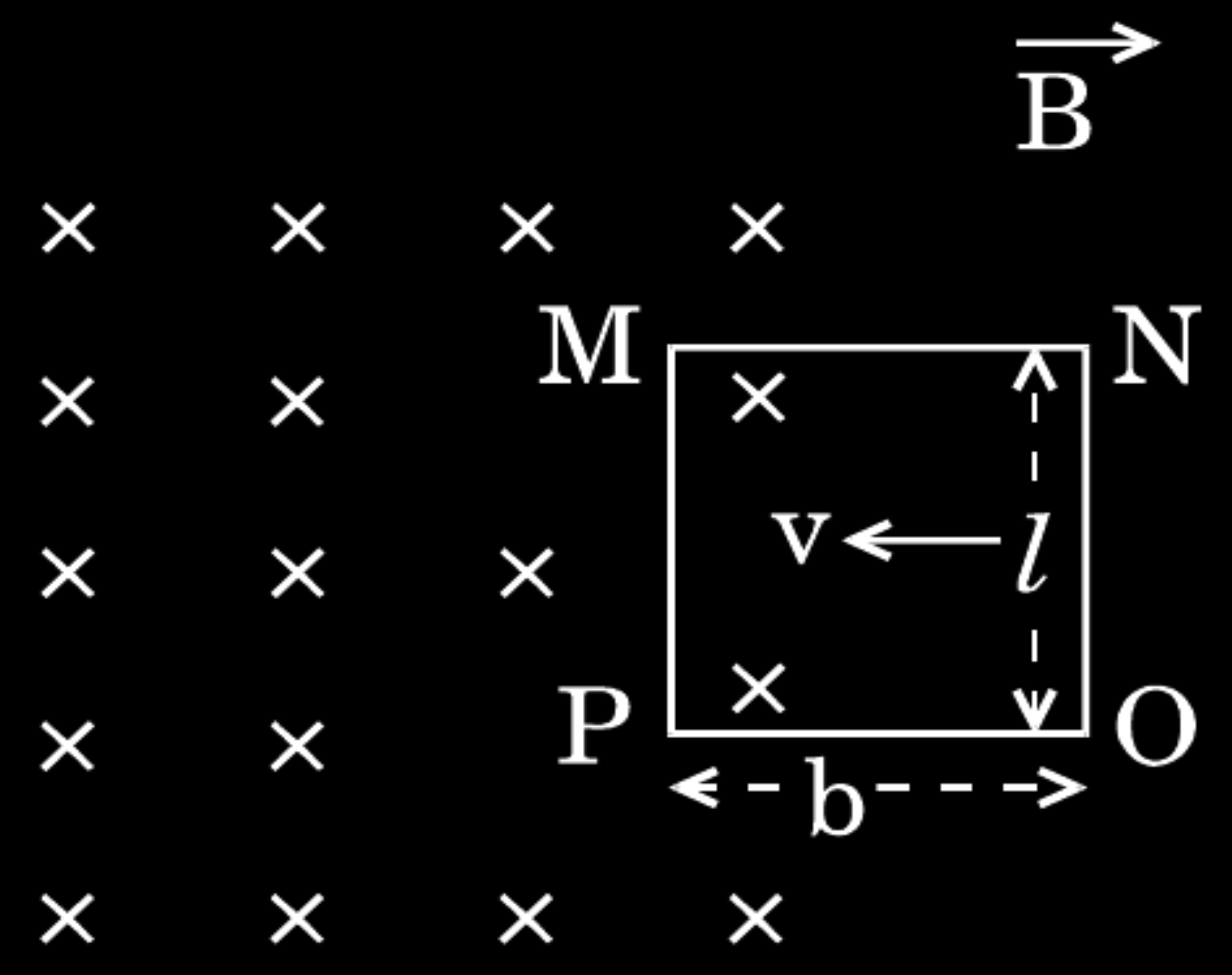
$$Power = i^2R = \left(\frac{bl^2\omega}{2R}\right)^2 R$$ ½

$$= \frac{b^2l^4\omega^2}{4R}$$ ½

Since induced current will reduce, it will be a little easier to remove the coil ½

[Even if student writes induced current decreases award ½ mark]

The figure shows a rectangular conducting frame MNOP of resistance R placed partly in a perpendicular magnetic field $\vec{B}$ and moved with velocity $\vec{v}$ as shown in the figure.



Obtain the expressions for the

(a) force acting on the arm 'ON' and its direction, and

(b) power required to move the frame to get a steady emf induced between the arms MN and PO.

a)	The induced emf in the moving conductor MNOP	
	$e = Blv$	½
	The induced current, $i = \frac{e}{R} = \frac{Blv}{R}$	
	Force on the arm 'ON', $F = Bil$	½
	$= \frac{B^2l^2v}{R}$	½
	The force is directed in the direction opposite to velocity of rod (v)	½
	Note : Award the last half mark if the student write $F = 0$ as $B = 0$ in the position shown	
b)	Power $P = F \times v$	½
	$= \frac{B^2l^2v}{R}$	½
	Note : Award the last half mark if the student write $P = 0$ as $B = 0$ in the position shown	

A small flat search coil of area 5 cm^2 with 140 closely wound turns is placed between the poles of a powerful magnet producing magnetic field $0{\cdot}09$ T and then quickly removed out of the field region. Calculate

(a) change of magnetic flux through the coil, and

(b) emf induced in the coil.

A small flat search coil of area $5\ \text{cm}^2$ with 140 closely wound turns is placed between the poles of a powerful magnet producing magnetic field 0·09 T and then quickly removed out of the field region. Calculate

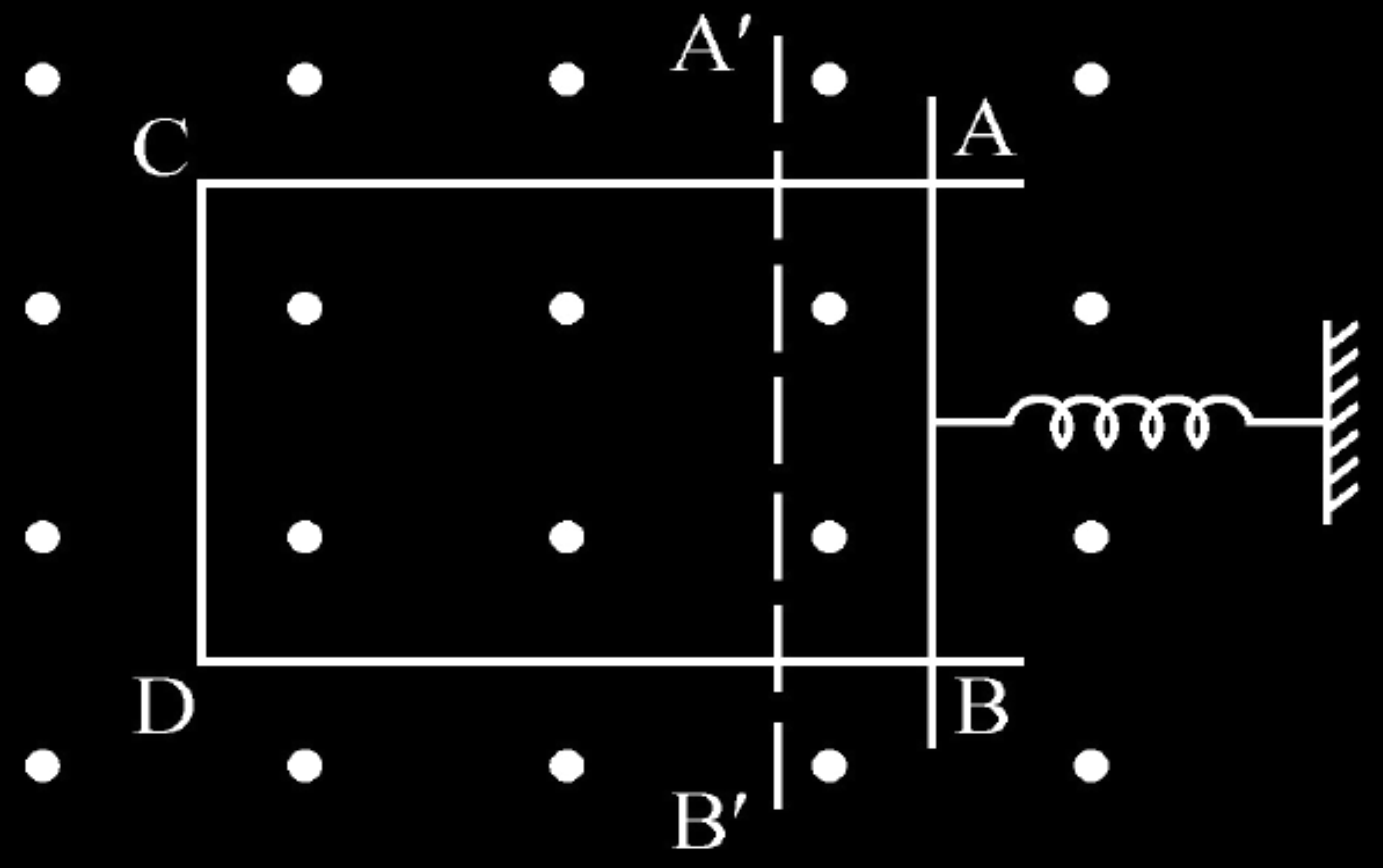
(a) change of magnetic flux through the coil, and

(b) emf induced in the coil.

i) $\Delta\phi = \phi_2 - \phi_1 = 0 - NBA\cos\theta$ $= 140 \times 0.09 \times 5 \times 10^{-4}\cos 0 = 63 \times 10^{-4}$ Wb	½
Alternatively If student assumes that the coil was initially kept with its plane parallel to the field, i.e. $\varphi = 90°$, $\Delta\phi = (0 - 0) = 0$ Wb award 1 mark	1
ii) $e = \dfrac{-\Delta\phi}{\Delta t} = \dfrac{-63\times10^{-4}}{\Delta t}$ V Alternatively, if the student takes $\Delta\phi = 0$, then e = o, [Note: Award this 1 mark, If a student writes that induced emf cannot be calculated as value of time interval Δt it is not given.]	½+½

CBSE 2018

A rectangular frame of wire is placed in a uniform magnetic field directed outwards, normal to the paper. AB is connected to a spring which is stretched to A′B′ and then released at time t = 0. Explain qualitatively how induced e.m.f. in the coil would vary with time. (Neglect damping of oscillations of spring)



CBSE 2017

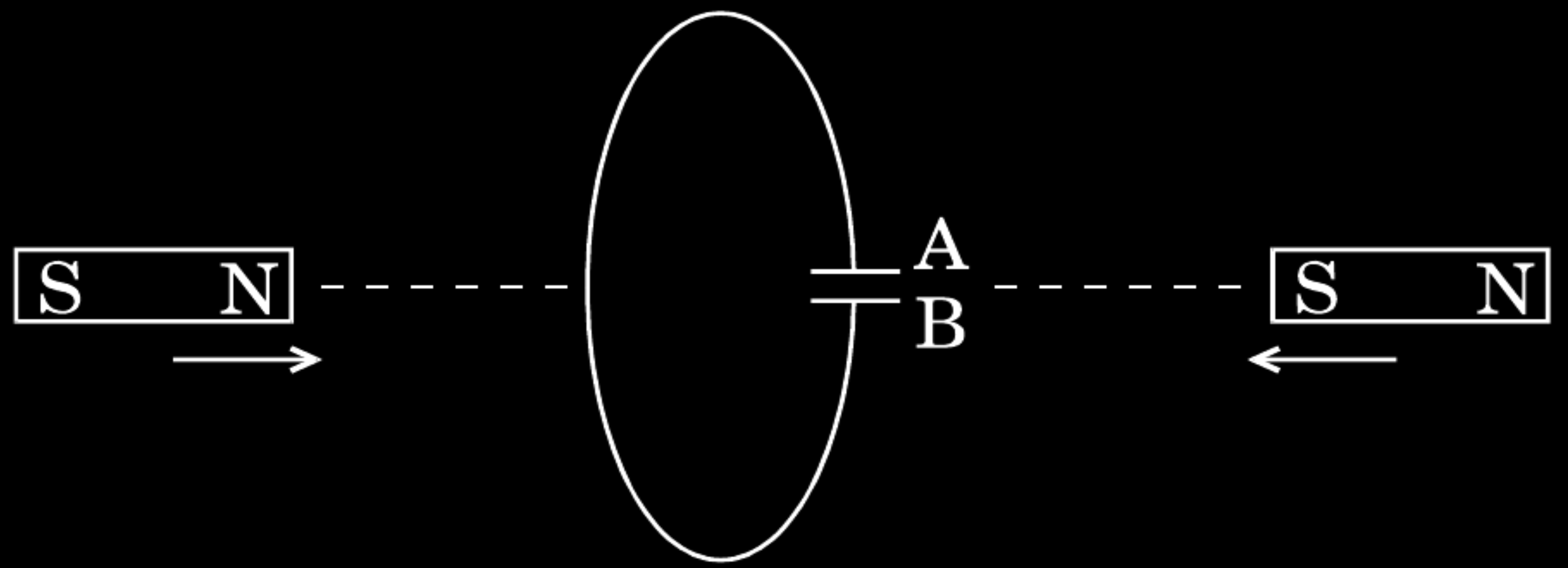
A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop ? Justify.

CBSE 2017

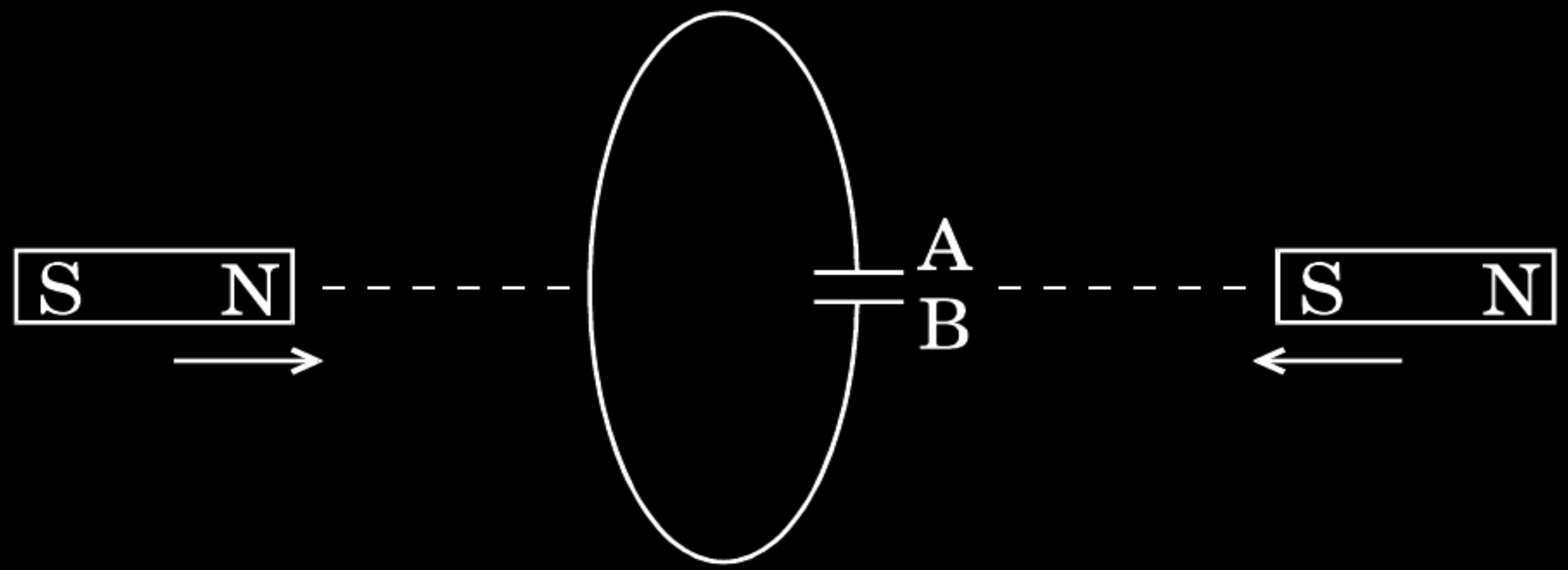
A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop ? Justify.

No,	½
As the magnetic field due to current carrying wire will be in the plane of the circular loop, so magnetic flux will remain zero.	½
Alternatively [Magnetic flux does not change with the change of current.]	

Predict the polarity of the capacitor in the situation described below :



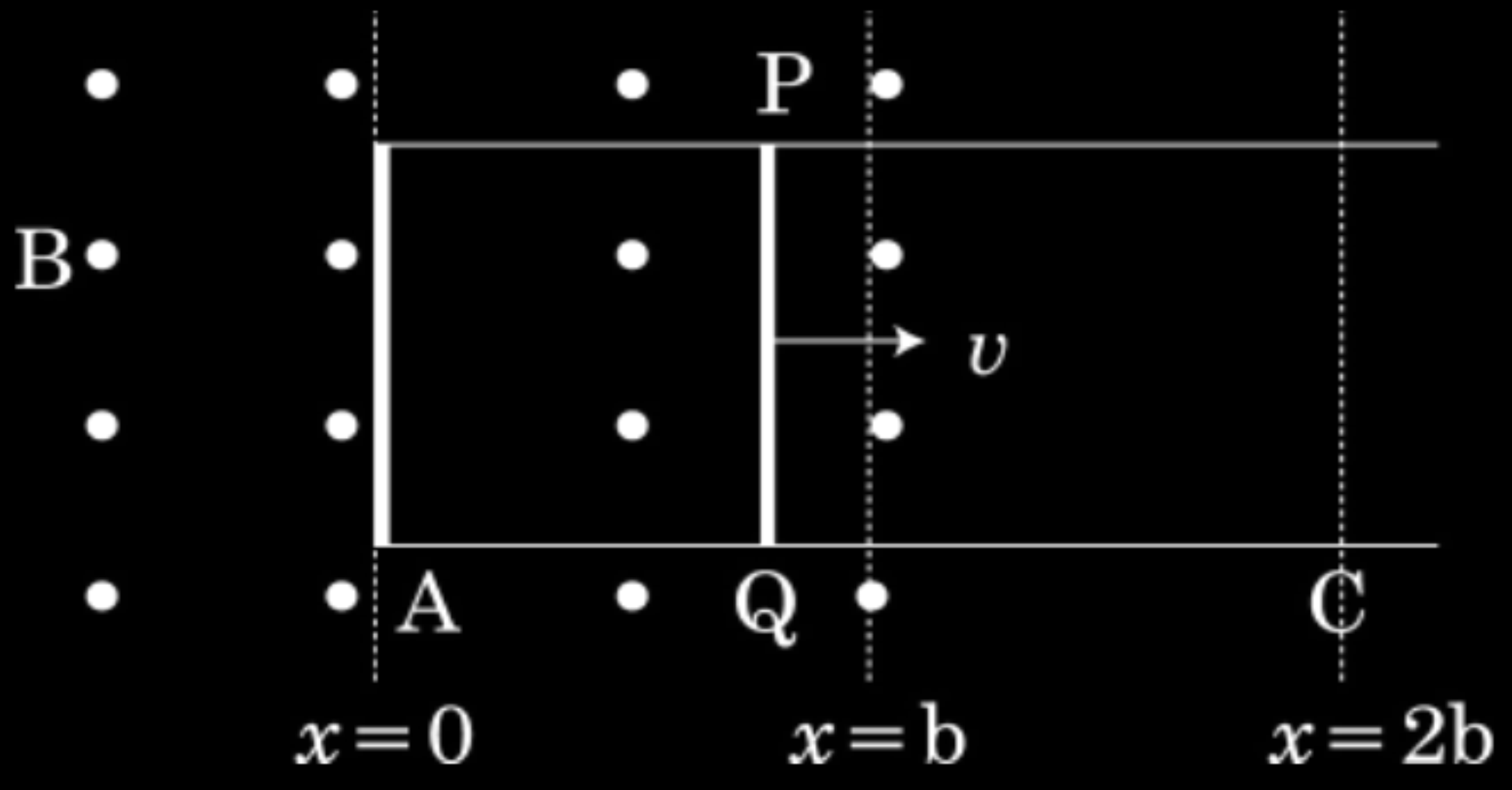
Predict the polarity of the capacitor in the situation described below :

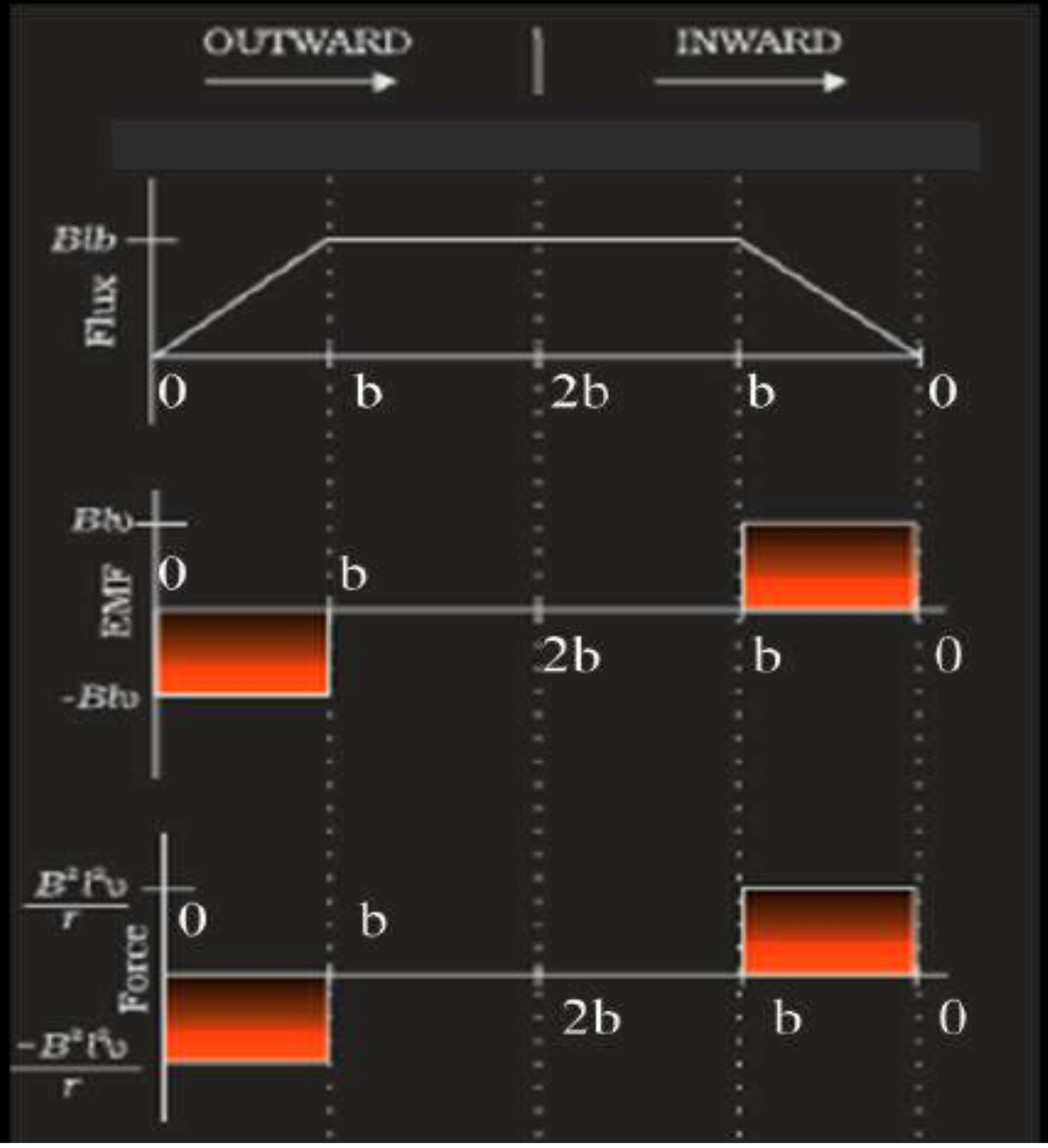


A is positive and	½
B is negative	½
(Also accept: A is negative and B is positive)	

(a) When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects. Identify the phenomenon causing this deflection and write the factors on which the amount and direction of the deflection depends. State the laws describing this phenomenon.

(b) Sketch the change in flux, emf and force when a conducting rod PQ of resistance R and length l moves freely to and fro between A and C with speed v on a rectangular conductor placed in uniform magnetic field as shown in the figure.



(a) The phenomenon involved is electromagnetic induction (EMI)	½
For the deflection:	
Amount depends upon the speed of movement of the magnet.	½
Direction depends on the sense (towards, or away) of the movement of the magnet.	½
The law describing the phenomenon is :	
The magnitude of the induced emf, in a circuit, is equal to the time rate of change of the magnetic flux through the circuit.	½
(Note: Also accept if a student writes: whenever magnetic flux linked with a conductor changes, an induced emf is setup in the conductor.)	
(Alternatively, $\epsilon = -\frac{d\phi_B}{dt}$)	
(b) 	1 1 1